Earth Sciences

Vrije Universiteit Amsterdam - Faculty of Science - Earth Sciences - 2025-2026

Earth Scientists study the processes and cycles within and between the four spheres, atmosphere, hydrosphere, biosphere and the geosphere by quantifying their magnitudes and rates in space and time. What is the influence of humankind on the Earth System in relation to natural processes? What can fluxes in the carbon cycle in the geological past, and in the present, tell us about the carbon cycle of the future?

Students are trained in fundamental knowledge and skills in the context of addressing the complex problems of our planet in today's society. The three specializations, Geology and Geochemistry, Earth and Climate, and Global Environmental Change and Policy offer a variety of courses on e.g. tectonics, landscape dynamics, climate modelling and environmental policy. The first year consists mainly of compulsory courses, practicals and an excursion in one of the two years. The second year offers a variety of electives focused on advanced and more specialized topics and themes, such as water related risks, humanity and climate, mantle properties and geothermal energy. The year is finished with the Master Thesis, which can include an internship.

The study guide has been published subject to changes resulting from this proposed decision. As soon as the Executive Board has taken a final decision with regards to the reorganisation, the relevant changes will be made to the study guide. This text will then be updated.

Info

Level	Master
Language	English
Duration	2 years
Type	Full Time
Credits	120 EC
Faculty	Faculty of Science

Teaching and examination regulation 2025-2026
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Earth Sciences specialisation Education

Description

The two-year master programme Education (E) basically consists of one year of further education and specialisation in Earth Sciences and one year of specific teacher training.

The programme of Earth Sciences in the first year (minimum of 60 credits) should at least contain an Earth Sciences Research Project (AM_1265 or AM_1267 24 credits; or AM_1266 15 credits) and Sociale geografie II (AM_1051, 12 credits). The remainder of the programme (total 24 or 33 EC) is to be filled with courses which can be chosen from the various master's degree programmes of the master Earth Sciences. We recommend students to use the remainder of the programme to broaden their knowledge by selecting courses from different sub-disciplines such as climatology, hydrology, sustainability, etc.

The educational programme in the second year is taught in Dutch by the ‘Teacher Academy' (Lerarenacademie) of the Faculty of Psychology and Education. It consists of 60 credits of compulsory modules. You will learn how to transfer your knowledge and motivate and inspire students in Geography (Aardrijkskunde).

The teacher training programme at VU Amsterdam is built around 20 themes (core practices).Courses are partly shared with master students from other faculties. You will apply these teaching practices directly in the classroom, as you will be working in a school for more than 50% of your study programme. With this specialisation, you will obtain a first-degree teaching qualification (eerstegraads lesbevoegdheid). This means that in two years, you will be qualified to teach both lower and upper secondary vocational education (HAVO/VWO) and pre-university education (VMBO) in the Netherlands. All teachers in the STEM disciplines are also qualified to teach the STEM elective NLT (Nature, Life and Technology).

Course registration

The course sociale geografie 1 (AB_450099) or a comparable course in the field of Human Geography. For more information contact Iris Pauw: I.Pauw@vu.nl

Programme Sections

- Education specific content
- Earth Sciences specific content

Education specific content

Programme Sections

- Teaching Geography

Teaching Geography

Description

Heb jij liefde voor het vak aardrijkskunde? Ben je nieuwsgierig naar de leerprocessen bij jongeren en wil je hen helpen de complexe wereld waarin we leven beter te begrijpen? Dan is de lerarenopleiding Aardrijkskunde aan de VU iets voor jou. Als leraar Aardrijkskunde leer je leerlingen kritisch te kijken naar hun eigen leefomgeving en de wereld om hen heen. Die wereld is - net als zijzelf - voortdurend in verandering. Met behulp van zowel de fysische als sociale kant van geografie leren leerlingen patronen en processen herkennen en begrijpen. Het begrijpen van de link tussen de mens en de aarde is nu urgenter dan ooit. Met de juiste kennis over actuele thema’s zoals de energietransitie en migratie, staan leerlingen steviger in de maatschappij. Als docent Aardrijkskunde is elke dag anders en biedt elke les weer nieuwe uitdagingen en kansen.

Course registration

Voldoende vakinhoudelijke kennis van sociale geografie en fysische geografie.

De aard en omvang van eventuele vak-deficiënties worden vastgesteld bij aanmelding, conform landelijke eisen. Afhankelijk van de omvang van de deficiëntie wordt de toelaatbaarheid bepaald. Mogelijk (bij een beperkte omvang) kan de student tijdens de lerarenopleiding de deficiënties wegwerken.

Programme Sections

- Leraar VHO februari start
- Leraar VHO September start

Leraar VHO februari start

Courses

Course Name	Period	Credits	Code
Pedagogische en Didactische verdieping Semester 1	P1+2+3	9EC	O_MPEDIVER_F
Peergroup 2 Semester 1	P1+2+3	0EC	O_MLPEERG_2F
Practice 3	P1+2+3	15EC	O_MLPRAK_3F
Didactics 1	P4	6EC	O_MLDIDAC_1F
Practice 1	P4	6EC	O_MLPRAK_1F
Peergroup 1 Semester 2	P4+5+6	0EC	O_MLPEERG_1F
Didactiek 2 Semester 2	P5+6	9EC	O_MLDIDAC2F
Practice 2	P5+6	9EC	O_MLPRAK_2F

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

Leraar VHO September start

Courses

Course Name	Period	Credits	Code
Didactics 1	P1	6EC	O_MLDIDAC_1
Practice 1	P1	6EC	O_MLPRAK_1
Peergroup 1 Semester 1	P1+2+3	0EC	O_MLPEERGR_1
Didactiek 2 Semester 1	P2+3	9EC	O_MLDIDAC2
Practice 2	P2+3	9EC	O_MLPRAK_2
Pedagogische en Didactische Verdieping Semester 2	P4+5+6	9EC	O_MPEDIVER
Peergroup 2 Semester 2	P4+5+6	0EC	O_MLPEERGR_2
Practice 3	P4+5+6	15EC	O_MLPRAK_3

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

Earth Sciences specific content

Courses

Course Name	Period	Credits	Code
Human Geography II	Ac. Year (September)	12EC	AM_1051

Programme Sections

- Earth Sciences electives
- Research project (choose one)

Earth Sciences electives

Courses

Course Name	Period	Credits	Code
Human Geography II	Ac. Year (September)	12EC	AM_1051
Causes and Consequences of Environmental Change	P1	6EC	AM_1049
Climate Systems	P1	6EC	AM_1124
Ecohydrology	P1	6EC	AM_450014
Science and Communication	P1	6EC	AM_470587
Sustainable Energy Systems	P1	6EC	AM_468018
Advanced Planetary Science	P2	6EC	AM_450273
Marine Geology and Paleoclimatology	P2	6EC	AM_450330
Science Journalism	P2	6EC	AM_471014
Capita Selecta Master Earth Sciences	P3	6EC	AM_1174
Climate Modelling	P3	6EC	AM_450004
Environmental Remote Sensing	P3	6EC	AM_450145
Global Biogeochemical Cycles	P4	6EC	AM_450332
Humanity and Climate	P4	6EC	AM_1057
Magmatic Processes	P4	6EC	AM_450189
Practical: Paleoclimate Change and Environmental Impacts	P4	6EC	AM_1144
Reflection Seismic for Geologists	P4	6EC	AM_450170
Metropole Ecology	P5	6EC	AM_1199

Research project (choose one)

Courses

Course Name	Period	Credits	Code
Research Project Earth & Climate	Ac. Year (September)	24EC	AM_1265
Research Project Geology and Geochemistry	Ac. Year (September)	24EC	AM_1267

Earth Sciences specialisation Earth and Climate

Programme Sections

- Earth Sciences specialisation Earth & Climate year 1
- Earth Sciences specialisation Earth & Climate year 2

Earth Sciences specialisation Earth & Climate year 1

Programme Sections

- Earth Sciences specialisation Earth & Climate year 1 constrained electives
- Earth Sciences specialisation Earth & Climate year 1 constrained electives
- Earth Sciences specialisation Earth & Climate year 1 compulsory courses

Earth Sciences specialisation Earth & Climate year 1 constrained electives

Courses

Course Name	Period	Credits	Code
Basin Tectonics and Subsurface Resources	P1	6EC	AM_1298
Climate Systems	P1	6EC	AM_1124
Advanced Spatial Analyses	P2	6EC	AM_1197
Marine Geology and Paleoclimatology	P2	6EC	AM_450330
Orogenesis	P2	6EC	AM_450190
Source to Sink	P2	6EC	AM_1302
Climate Modelling	P3	6EC	AM_450004
Environmental Remote Sensing	P3	6EC	AM_450145
Climate Dynamics and Processes	P4	6EC	AM_1230
Global Biogeochemical Cycles	P4	6EC	AM_450332
Imaging and Assessing Landscapes	P4	6EC	AM_1183
Practical: Paleoclimate Change and Environmental Impacts	P4	6EC	AM_1144
Reflection Seismic for Geologists	P4	6EC	AM_450170

Earth Sciences specialisation Earth & Climate year 1 constrained electives

Courses

Course Name	Period	Credits	Code
Capita Selecta Earth Sciences	Ac. Year (September)	15EC	AM_1322
Research Proposal Master Earth Sciences	P5+6	15EC	AM_1321

Earth Sciences specialisation Earth & Climate year 1 compulsory courses

Courses

Course Name	Period	Credits	Code
Landscape Dynamics	P1	6EC	AM_450331
Designing Interdisciplinary Research	P5	3EC	AM_1264

Earth Sciences specialisation Earth & Climate year 2

Programme Sections

- Earth Sciences specialisation Earth & Climate year 2 electives
- Earth Sciences specialisation Earth & Climate year 2 compulsory courses

Earth Sciences specialisation Earth & Climate year 2 electives

Courses

Course Name	Period	Credits	Code
Catchment Hydrology	P1	6EC	AM_1278
Ecohydrology	P1	6EC	AM_450014
Mantle Properties	P1	6EC	AM_1211
Science and Communication	P1	6EC	AM_470587
Sustainable Energy Systems	P1	6EC	AM_468018
Advanced Planetary Science	P2	6EC	AM_450273
Biological Oceanography	P2	6EC	AMU_0021
Catchment Biogeochemistry	P2	6EC	AM_1279
Climate Hydrological Processes	P2	6EC	AM_1196
Plastic and Chemical Pollution	P2	6EC	AM_1259
Science Journalism	P2	6EC	AM_471014
Capita Selecta Master Earth Sciences	P3	6EC	AM_1174
Energy and Climate Governance	P3	6EC	AM_1155
Governance of Ecosystem Services	P3	6EC	AM_468025
Multidisciplinary Uncertainties and Risks	P3	6EC	AM_1249
Water Politics and Governance	P3	6EC	AM_1192
Groundwater Processes	P4	6EC	AM_1164
Humanity and Climate	P4	6EC	AM_1057
Sustainable Mineral Resources	P4	6EC	AM_1289
Geology & Geochemistry Field Excursion	P5	3EC	AM_450229

Earth Sciences specialisation Earth & Climate year 2 compulsory courses

Courses

Course Name	Period	Credits	Code
Research Project Earth & Climate	Ac. Year (September)	24EC	AM_1265
Thesis Project Earth & Climate	Ac. Year (September)	30EC	AM_1228

Earth Sciences specialisation Global Environmental Change and Policy

Description

This specialisation integrates the disciplinary insights from Earth Sciences, Environmental Economics and Environmental Policy and Governance to arrive at interdisciplinary solutions to the complex sustainability problems of tomorrow. In this interdisciplinary specialisation, you will learn how to design solutions to the complex sustainability problems faced by our planet. From a strong grounding in Earth Sciences, we explore integrated approaches towards solving global sustainability challenges by adding the insights of environmental economics and environmental policy studies. Examples of main themes addressed in the programme are:

- Climate change processes, societal impacts of climate change, and policy solutions;
- Transition to renewable energy and a low carbon economy;
- Land use change and options for limiting ecosystem degradation and biodiversity losses;
- Sustainable management of water resources and flood risk.

Programme Sections

- Earth Sciences specialisation Global Environmental Change and Policy year 1
- Earth Sciences specialisation Global Environmental Change and Policy year 2

Earth Sciences specialisation Global Environmental Change and Policy year 1

Description

The first year lays the groundwork in the three disciplines and methods of interdisciplinary research, including

fieldwork. From a strong grounding in Earth Sciences, we explore integrated approaches towards solving global sustainability challenges by adding the insights of environmental economics and environmental policy studies. Climate change is a central and cross-cutting theme throughout the GECP programme.

Programme Sections

- Earth Sciences specialisation Global Environmental Change and Policy year 1 constrained electives
- Earth Sciences specialisation Global Environmental Change and Policy year 1 compulsory courses

Earth Sciences specialisation Global Environmental Change and Policy year 1 constrained electives

Description

Only in period 2, students are required to choose a constrained elective course. Students pick at least one of two economics courses. Students with a strong background in economics are encouraged to take Economics of Environmental Policy Instrument Design. Please note that this course is taught at the School of Business and Economics (SBE) of the Vrije Universiteit. The Environmental Economics for GEC&P course is taught in collaboration with the Environmental Policy for GEC&P course.

Courses

Course Name	Period	Credits	Code
Economics of Environmental Policy	P2	6EC	E_STR_EEP
Environmental Economics for GEC&P	P2	6EC	AM_1232

Earth Sciences specialisation Global Environmental Change and Policy year 1 compulsory courses

Courses

Course Name	Period	Credits	Code
Challenges and Solutions for GEC&P	P1	6EC	AM_1234
Climate Systems	P1	6EC	AM_1124
Environmental Policy for GEC&P	P2	6EC	AM_1241
Land Use Change and Ecosystems	P3	6EC	AM_1235
Earth science observation research methods	P4	6EC	AM_1285
Methods of Global Environmental Change	P4	6EC	AM_1236
Designing Interdisciplinary Research	P5	3EC	AM_1264
Research Proposal Master Earth Sciences	P5+6	15EC	AM_1321

Earth Sciences specialisation Global Environmental Change and Policy year 2

Description

In the second year, students deepen their studies in four distinct study paths: (1) energy, (2) ecosystems, (3) water, or (4) science communication. The year consists of three compulsory courses (Climate Impacts & Policy, Advanced Spatial Analysis, and the Master Thesis Project) that are taken by students from all four study paths and a variety of electives as part of the study paths (energy, ecosystems, water, and science communication). It is possible to mix courses from different study paths.

Programme Sections

- Earth Sciences specialisation Global Environmental Change and Policy year 2 constrained electives
- Earth Sciences specialisation Global Environmental Change and Policy year 2 compulsory courses

Earth Sciences specialisation Global Environmental Change and Policy year 2 constrained electives

Programme Sections

- Earth Sciences specialisation Global Environmental Change and Policy Energy constrained choice
- Earth Sciences specialisation Global Environmental Change and Policy Ecosystems constrained choice

- Earth Sciences specialisation Global Environmental Change and Policy Water constrained choice
- Earth Sciences specialisation Global Environmental Change and Policy other constrained choice

Earth Sciences specialisation Global Environmental Change and Policy Energy constrained choice

Description

A total of 18 credits can be obtained within each study path. Below is an overview provided of the recommended courses in the energy study path.

Courses

Course Name	Period	Credits	Code
Sustainable Energy Systems	P1	6EC	AM_468018
Energy and Climate Governance	P3	6EC	AM_1155

Earth Sciences specialisation Global Environmental Change and Policy Ecosystems constrained choice

Description

A total of 18 credits can be obtained within each study path. Below is an overview provided of the recommended courses in the ecosystems study path.

Courses

Course Name	Period	Credits	Code
Ecohydrology	P1	6EC	AM_450014
Ecological Data Analysis	P1	6EC	AM_1225
Catchment Biogeochemistry	P2	6EC	AM_1279
Plastic and Chemical Pollution	P2	6EC	AM_1259
Environmental Remote Sensing	P3	6EC	AM_450145
Governance of Ecosystem Services	P3	6EC	AM_468025

Earth Sciences specialisation Global Environmental Change and Policy Water constrained choice

Description

A total of 18 credits can be obtained within each study path. Below is an overview provided of the recommended courses in the water study path.

Courses

Course Name	Period	Credits	Code
Catchment Hydrology	P1	6EC	AM_1278
Ecohydrology	P1	6EC	AM_450014
Water Risks	P1	6EC	AM_1210
Catchment Biogeochemistry	P2	6EC	AM_1279
Climate Hydrological Processes	P2	6EC	AM_1196
Plastic and Chemical Pollution	P2	6EC	AM_1259
Integrated Modeling in Hydrology	P3	6EC	AM_1165
Water Politics and Governance	P3	6EC	AM_1192

Earth Sciences specialisation Global Environmental Change and Policy other constrained choice

Description

A total of 18 credits can be obtained within each study path. Below is an overview provided of the recommended courses in the other (communication) study path.

Courses

Course Name	Period	Credits	Code
Science and Communication	P1	6EC	AM_470587
Science Journalism	P2	6EC	AM_471014
Capita Selecta Master Earth Sciences	P3	6EC	AM_1174
Multidisciplinary Uncertainties and Risks	P3	6EC	AM_1249

Earth Sciences specialisation Global Environmental Change and Policy year 2 compulsory courses

Courses

Course Name	Period	Credits	Code
Climate Impacts & Policy	P1	6EC	AM_1240
Advanced Spatial Analyses	P2	6EC	AM_1197
Master Thesis GEC&P	P4+5+6	30EC	AM_1286

Earth Sciences specialisation Geology and Geochemistry

Description

This specialization focuses on processes operating on the Earth's crust and mantle, and other planets. You will learn about geodynamics, tectonics, sedimentary and structural geology, stratigraphy, petrology, isotope geochemistry, planetary evolution, and basin analysis; and apply geological, (isotope) geochemistry and radio-isotope dating techniques.

Programme Sections

- Earth Sciences specialisation Geology & Geochemistry year 1
- Earth Sciences specialisation Geology & Geochemistry year 2

Earth Sciences specialisation Geology & Geochemistry year 1

Programme Sections

- Earth Sciences specialisation Geology & Geochemistry year 1 constrained choice
- Earth Sciences specialisation Geology & Geochemistry year 1 electives
- Earth Sciences specialisation Geology and Geochemistry year 1 compulsory courses

Earth Sciences specialisation Geology & Geochemistry year 1 constrained choice

Courses

Course Name	Period	Credits	Code
Capita Selecta Earth Sciences	Ac. Year (September)	15EC	AM_1322
Research Proposal Master Earth Sciences	P5+6	15EC	AM_1321

Earth Sciences specialisation Geology & Geochemistry year 1 electives

Courses

Course Name	Period	Credits	Code
Capita Selecta Master Earth Sciences	P3	6EC	AM_1174
Magmatic Processes	P4	6EC	AM_450189
Reflection Seismic for Geologists	P4	6EC	AM_450170
Sustainable Mineral Resources	P4	6EC	AM_1289
Applied Stratigraphy (UU)	P4+5	7.5EC	AM_1301

Earth Sciences specialisation Geology and Geochemistry year 1 compulsory courses

Description

The specialisation Geology and Geochemistry focuses on the geology and geochemistry of the Earth. In doing so it integrates tectonics, petrology, geochemistry, sedimentary geology, and planetary science. The student acquires a broad background by following a set of compulsory courses in Year 1 before focusing on a research topic in one or more of the aforementioned fields.

Courses

Course Name	Period	Credits	Code
Basin Tectonics and Subsurface Resources	P1	6EC	AM_1298
Mantle Properties	P1	6EC	AM_1211
Orogenesis	P2	6EC	AM_450190
Source to Sink	P2	6EC	AM_1302
Multidisciplinary Uncertainties and Risks	P3	6EC	AM_1249
Designing Interdisciplinary Research	P5	3EC	AM_1264
Geology & Geochemistry Field Excursion	P5	3EC	AM_450229

Earth Sciences specialisation Geology & Geochemistry year 2

Programme Sections

- Earth Sciences specialisation Geology & Geochemistry year 2 electives
- Earth Sciences specialisation Geology and Geochemistry year 2 compulsory courses

Earth Sciences specialisation Geology & Geochemistry year 2 electives

Courses

Course Name	Period	Credits	Code
Advanced Isotope Geochemistry and Geochronology	P1	6EC	AM_1287
Catchment Hydrology	P1	6EC	AM_1278
Ecohydrology	P1	6EC	AM_450014
Science and Communication	P1	6EC	AM_470587
Sustainable Energy Systems	P1	6EC	AM_468018
Water Risks	P1	6EC	AM_1210
Advanced Planetary Science	P2	6EC	AM_450273
Catchment Biogeochemistry	P2	6EC	AM_1279
Climate Hydrological Processes	P2	6EC	AM_1196
Energy and Climate Change; Science, Policy and Economics	P2	6EC	X_428568
Science Journalism	P2	6EC	AM_471014
Capita Selecta Master Earth Sciences	P3	6EC	AM_1174
Energy and Climate Governance	P3	6EC	AM_1155
Physical volcanology	P3	6EC	AM_1297
Humanity and Climate	P4	6EC	AM_1057
Magmatic Processes	P4	6EC	AM_450189
Reflection Seismic for Geologists	P4	6EC	AM_450170
Sustainable Mineral Resources	P4	6EC	AM_1289
Applied Stratigraphy (UU)	P4+5	7.5EC	AM_1301

Earth Sciences specialisation Geology and Geochemistry year 2 compulsory courses

Description

The specialisation Geology and Geochemistry focuses on the geology and geochemistry of the Earth. In doing so it integrates tectonics, petrology, geochemistry, sedimentary geology, and planetary science. The student has acquired a broad background by following a set of compulsory courses in Year 1 and will focus in Year 2 on a set of

optional courses, finishing the Research Project started in Year 1, and the Master Thesis

Courses

Course Name	Period	Credits	Code
Master Thesis Geology and Geochemistry	Ac. Year (September)	30EC	AM_1186
Research Project Geology and Geochemistry	Ac. Year (September)	24EC	AM_1267
Geology & Geochemistry Field Excursion	P5	3EC	AM_450229

Courses

Advanced Isotope Geochemistry and Geochronology

Course Code	AM_1287
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. L.M. Kootker
Examiner	dr. L.M. Kootker
Teaching Staff	dr. L.M. Kootker, prof. dr. K.F. Kuiper
Teaching method(s)	Seminar, Excursion, Lecture

Course Objective

Students who attend this course will gain knowledge and understanding about two complimentary subjects: i) new applications of state of the art geochemistry to diverse aspects of Earth-Forensic and archaeological sciences with interaction with industry and ii) the current developments in geochronology as applied in the standard Geological Timescale from the Pre-Cambrium to Recent. Analytical and methodological approaches to constrain rates and duration of geological processes in sedimentary, magmatic and metamorphic systems are discussed.

You will develop the skills to

- Use the acquired knowledge to analyse, compare and explain distinct features of processes in which geochemical and geochronological tools are required.
- Read and critically assess significant literature about these subjects
- Actively participate in group discussions

Course Content

- Assessment of scientific literature of state-of-the art geochemical and geochronological methods
- new applications applied to forensic and cultural heritage research
- the mineralogical evolution of Earth
- The use and application of low- , medium and high-temperature thermochronometers in different geological systems
- Geochronological tools applied to sea mount chains and mantle plumes
- Assessment and development of the standard Geological Timescale.
- Dating tools for the Quaternary

Additional Information Teaching Methods

Lectures 12 * 3 hours 45 mins including visit to the Rijksmuseum. Self study 12 * 4 hours

Method of Assessment

2 essays, 2 presentations and one poster

Entry Requirements

BSc Earth Science

Literature

Selection of literature for individual essays and presentation projects to be announced on Canvas.

Additional Information Target Audience

2nd year MSc Earth Science

Additional Information

A variety of lecturers from the VU and industry

Recommended background knowledge

broad Earth Science BSc, chemistry back ground advantageous.

Advanced Planetary Science

Course Code	AM_450273
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. W. van Westrenen
Examiner	prof. dr. W. van Westrenen
Teaching Staff	prof. dr. W. van Westrenen
Teaching method(s)	Seminar

Course Objective

The aim of this course is to discuss in depth recent advances in planetary science in general, and geological/geochemical aspects of planetary science in particular.

Course Content

A series of lectures will examine in detail several key problems and active areas of research in planetary and exoplanetary science linked to specific space missions, specific rocky bodies in the solar system, and specific techniques (analytical, experimental or theoretical).

Additional Information Teaching Methods

The course will consist of several blocks of sessions concerning particular topics. Students will participate in flip the classroom sessions on specific subtopics, and in depth discussion of recent literature will form an integral part of the course. The course schedule will consist of lecture-based sessions and discussion sessions. The course counts for 6 EC = 160 SBU, which are divided between the different components of this course in the following way (1) 14 * 3 hour lectures / discussion sessions = 115 SBU, (2) Background reading and preparation of lecture = 36 SBU

Method of Assessment

The final mark for this course consists of the following components:
Written exam (35%),
presentation (20%),
exercises to be done during the course (40%).

Literature

Recent literature references will be provided during the course.

Recommended background knowledge

Background in, or at least deep interest in, geology / geochemistry is required. The course builds upon the Introduction to Planetary Science course in the VU BSc Earth Sciences programme but it is possible to take the course without having done the Introduction to Planetary Science course.

Advanced Spatial Analyses

Course Code	AM_1197
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.A. de Bruijn
Examiner	dr. J.A. de Bruijn
Teaching Staff	dr. J.A. de Bruijn, dr. N.A.J. Schutgens
Teaching method(s)	Study Group, Computer lab, Written partial exam, Lecture

Course Objective

The objective of this course is to become familiar with more advanced approaches for the analysis of (large) spatial and temporal datasets in the context of earth and environmental sciences, for example related to (addressing) climate change.

Course Content

In this course you will acquire the necessary skills to conduct spatial-temporal analysis in the context of earth and environmental sciences. The course starts with improving more fundamental programming in Python, and gradually focuses on tools that are useful for spatial-temporal analyses: reading and writing spatial and time-series data, libraries for handling spatial data and time series data, data analysis, and visualization of results. You will learn these skills using several practical cases, such as making climate stripes, simulating wildfires, optimization of an energy-grid using weather data, and using machine learning to detect flood events in satellite imagery. The gained knowledge in each exercise should be transferable across the domain of earth sciences.

Additional Information Teaching Methods

Each week will consist of one or two lectures and multiple computer labs (tutorials) where students can work on the assignments. Teachers and/or teaching assistants will be available to assist.

Method of Assessment

The course requires 8 weeks of part-time (50%) study, and students are expected to spend approximately 20 hours a week on this course.

Entry Requirements

This course assumes a basic knowledge of GIS and GIS-based spatial analysis (such as the course GIS and Digital Spatial Data (AB_1076), or GIS en Aardobservatie AB_1271 or equivalent), as well as basic knowledge of Python. This requirement is met by all students who completed their BSc in Earth Sciences, or Earth, Economics and Sustainability at VU University. MSc students in Econometrics and Operations are not expected to experience issues when they do not have this background, especially when they are experienced in other programming languages.

Additional guidance will be provided to brush up Python knowledge for students who need it.

Literature

Selected literature will be provided on Canvas.

Additional Information Target Audience

MSc students in Earth Sciences, MSc students in Hydrology, MSc students in Econometrics and Operations Research with specialization Climate Econometrics, and other relevant studies.

Additional Information

It is advisable for students to have their own laptop for computer exercises.

Explanation Canvas

We will use canvas as well as a custom website to provide information and assignments throughout the course. Assignments will be submitted through canvas.

AI in het Onderwijs

Course Code	O_MAIIHO
Credits	3
Period	P1, P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	prof. dr. ir. F.F.J. Hermans
Examiner	dr. H.B. Westbroek
Teaching Staff	prof. dr. ir. F.F.J. Hermans, dr. S.M. Calor
Teaching method(s)	Seminar, Lecture

Course Objective

Het doel van dit vak is op basis van kennis over AI en over onderwijs theorieën een helder beeld schetsen van hoe je je als docent en school wilt verhouden tot generatieve AG.

Course Content

Studenten leren in dit vak:

- Hoe generatieve AI werkt
- Wat de impact van AI op onderwijs kan zijn, vanuit een aantal verschillende theorieën en invalshoeken
- Wat de risico's van AI zijn, bijvoorbeeld mbt bias en deskilling
- Reflecteren op het gebruik van AI in de eigen onderwijspraktijk

Belangrijk: Voor dit vak is geen technische voorkennis vereist en het is voor alle studenten van de lerarenopleiding (en aanverwante studies) geschikt, zeker niet louter voor de beta's!

Additional Information Teaching Methods

Het vak bestaat uit:

- 5 verplichte hoorcolleges (10 uur)
- Huiswerkopdrachten maken (25 uur)
- Lezen literatuur (25 uur)
- Uitwerken beleidsplan (24 uur)

Method of Assessment

Studenten sluiten dit vak af door het schrijven van een beleidsplan over het gebruik van AI binnen hun vak of school.

Literature

Zie Canvas

Applied Stratigraphy (UU)

Course Code	AM_1301
Credits	7.5
Period	P4+5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. W. van Westrenen
Examiner	prof. dr. W. van Westrenen
Teaching Staff	

Teaching method(s)	Seminar, Excursion

Course Objective

On completion of this course, students should be better able to:

1. Use geological and geophysical datasets to evaluate the stratigraphy of basins in the subsurface;
2. Predict the occurrence and distribution of source and reservoir facies and their fluids in the subsurface; and
3. Identify tectonic, climatic, and eustatic controls on basin stratigraphy that explain the occurrence and distribution of source and reservoir facies in the subsurface.

In terms of skills, on completion of this course, students should be better able to:

- Describe rocks in cores and thin sections;
- Use seismic sections and wireline logs to interpret the stratigraphic architecture of basins in the subsurface; and
- Study source rocks and reservoirs in the field.

Course Content

This course is part of the stream 'Subsurface Resourcing' in the Geology and Geochemistry specialisation. This stream contains a series of elective courses that prepares for activities in the subsurface (geothermal, CO2 sequestration, hydrocarbon exploration and production). Many examples of successful techniques, originally developed for the oil and gas industry, need also to be applied elsewhere in order to achieve safe and accurate operations.

The aim of this course is to bridge the methodological and scale gap between outcrop-based geology and subsurface investigations. Students who enroll in this course should have learned, so far in their academic careers, how to study rocks exposed at the surface of the Earth and how to collect and interpret field data. In this course, students will learn how to study basins in the subsurface and how to collect and interpret subsurface data. This is arguably a more challenging occupation, for subsurface datasets are more incomplete and the economic implications of subsurface exploration immense. Students will learn how—and why—facies and fluids occur in basins in the subsurface and how the geologic history of a region affects the occurrence and distribution of those fluids and the facies that contain them. The content and learning outcomes of this course are particularly relevant for those wishing to pursue a career in applied geology (e.g., exploration geologist or seismic interpreter).

Additional Information Teaching Methods

Lectures, project work, student presentation, and field study.

Method of Assessment

Class report (60%): Students are required to hand in their class notes for a session selected by the course coordinator. Students will be given the opportunity to annotate their class notes and complement them with information and insights from the literature.

Field trip guide (40%): Students will work in pairs and collaborate on the research and writing of a guide for the field trip to England. Each pair will choose one of the localities to be visited during the field trip and prepare a succinct description and interpretation of its geology. The interpretation should explore the nature of the control on structure and/or stratigraphy. Students should place their observations and interpretations in the broader context of the geology of the North Sea. Before the trip, students will be given the opportunity to receive formative feedback on a draft version of their guide. After the trip, students will prepare an illustrated final version of their guide.

Entry Requirements

This course is only accessible to MSc students.

Literature

The lecturers will make extensive literature lists available. In addition students can refer to:

- Miall, A. D. 2016. Stratigraphy: A Modern Synthesis. Cham: Springer.
- Woodcock, N., and Strachan, R., eds. 2012. Geological history of Britainand Ireland. 2nd ed. Chichester: Wiley.
- Wong, Th., D. A. J. Batjes, and J. de Jager, eds. 2007. Geology of theNetherlands. Amsterdam: KNAW.

Additional Information

Lecturers: Dr. João Trabucho (Utrecht University), and specialists from the industry (EBN, Shell, TNO). Field trip occurs in spring.

Recommended background knowledge

Students should have followed the Basin Tectonics and subsurface geo-resources compulsory course.

Basin Tectonics and Subsurface Resources

Course Code	AM_1298
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. A. Beniest
Examiner	dr. A. Beniest
Teaching Staff	dr. A. Beniest, dr. A. Beniest
Teaching method(s)	Seminar, Computer lab, Practical, Written partial exam, Lecture

Course Objective

The first learning objective of the course is to develop skills to analyse and review the tectonic evolution and infill of sedimentary basins. The second learning objective is using and evaluating this information as boundary conditions for the exploration and development of subsurface resources and storage, including geothermal energy, water, heat storage and carbon capture storage. At the end of this course, the student will have a good impression on the technical and societal impact of the exploration and production of geo-resources.

To achieve this objective, the course involves training in:

- Integrating different data sets and techniques to understand the tectonic and sedimentary processes controlling the evolution of sedimentary basins.
- Formulating hypotheses, and critically evaluating sedimentary basin data to test them.
- Combining class material and independent, critical reading of literature.
- Working in small interdisciplinary groups.
- Communicating effectively in oral and written form.

Course Content

This course consists of two parts:
Part I: Basin tectonics
Part II: Subsurface resources

Part I: The main topics addressed in this part of the course are:

- The tectonic processes that guide the structural-stratigraphic evolution of sedimentary basins
- The structural and sedimentary characteristics of sedimentary basins in different tectonic settings: compressional, extensional and strike-slip

This part of the course starts with lectures and in-class exercises on the different types of basins in different tectonic settings. After two weeks, this theory will be tested with an exam. This is followed by a case study using a modelling program and well-data to investigate the subsidence of a basin in the Alps. Lastly, the sedimentary infill of Dutch basins and how they can potentially be used during the Energy Transition, will be discussed during a workshop using cores from the Dutch offshore.

The students will learn about the tectonic processes that lead to basin formation in an interactive way, including lectures, practical exercises done in class and a computer and/or core practical. Societal relevance of Dutch basins for the Energy Transition in the Netherlands will be explored from different angles.

Part II:

The subsurface resourcing part contains a series of lectures that prepares for activities in the Dutch subsurface (geothermal, CO2 sequestration, hydrocarbon exploration and production). Many examples of successful techniques, originally developed for the oil and gas industry, need also to be applied elsewhere to achieve safe and accurate operations involving the subsurface.

During this part of the course a number of issues, technical and otherwise, that impact subsurface resourcing are reviewed. Emphasis is placed on the need to study subsurface from the regional to the local scale and to integrate data and concepts from many different disciplines. The main objective of the course is to teach students to appreciate the overall application of basin studies to the evaluation of subsurface resources with a strong link between basin tectonics and stratigraphy at all scales, and the importance of taking an integrated view through the appreciation on both technical and geological challenges in the sub-surface.

The course starts with a general introduction to the general usage of the subsurface and expected future energy supply and demand scenarios. Second, the lectures will evolve around technical challenges for various subsurface actives, including salt storage, carbon storage, geothermal energy and heat storage. Throughout the lecture series, the students will work in small teams on a 3-week case study to perform a risk-assessment to real-life cases of subsurface exploration activity in the Dutch subsurface.

Additional Information Teaching Methods

Part I:

- Lectures with practical exercises that complement the theory
- Computer practical and/or core practical
- Guest lectures
- Reading and writing.

Part II:

- Guest lectures from industry experts
- Case study activity: risk assessment of the Dutch subsurface
- Practical examples worked by students.

Method of Assessment

Part I:

- Final written exam: 25%
- Report based on computer and/or core practicals: 25%
- Presence at the core practical: pass/fail

Part II:

- Case study report 30%
- Case study presentation 20%

Note that:

- A minimum grade of 5.5/10 for both parts is required to pass.
- An average final grade of 5.5 is required to pass

Literature

Part I:

Main books covering the course content:

- Allen, P.A. and Allen, J.R. (2004). Basin Analysis. Blackwell Publishing. ISBN: 978-0-470-67376-847
- Allen, P.A. (2017) Sediment Routing Systems. Cambridge University Press. ISBN: 9781107091993
- A reading list of published papers will also be provided.

Part II:

Powerpoint presentation material is posted on Canvas.

Additional Information

Part I: This course focuses on the tectonic formation and evolution of sedimentary basins and their stratigraphic infill. By the end of this course, students should be able to reconstruct the tectonic evolution of sedimentary basins from various types of data including well-data, seismic data and field data. The societal aspect of the usage of the Dutch subsurface for resources during the Energy transition will be covered by guest lectures and workshops. This part of the course is taught by Anouk Beniest (VU Amsterdam).

Part II: This part of the course is followed jointly by MSc Earth Science students from Utrecht University and Vrije Universiteit Amsterdam. Students are taken on a rollercoaster of integrated geologic concepts and swept in a short

time from place to place across the globe to look at regional geology from an explorers' perspective. Mental alertness and the flexibility to follow these rapid changes are therefore essential to gaining maximum benefit. This part of the course is led by Dineke Wiersma (GasUnie).

Explanation Canvas

Part I and Part II:

- all lectures and practical material will be made available through Canvas

Biological Oceanography

Course Code	AMU_0021
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.E. Vonk
Examiner	dr. J.E. Vonk
Teaching Staff	
Teaching method(s)	

Course Objective

- Identify main biological oceanographic processes (4) and describe their interdependence
- Specify and differentiate between biological processes involved in the biological pump
- Survey relevant literature about a given research topic within biological oceanography and summarize in an abstract
- Develop and formulate a follow-up research question / objective on a given topic within biological oceanography and describe how the research should be conducted
- Present a lecture aimed at explaining a research topic to a scientific audience and proposing a continuation research project
- Explain the importance of the biology of the ocean for global change developments
- Critically discuss, oppose and defend scientific issues within the field of biological oceanography

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130120

Additional Information Teaching Methods

- Lecture
- Seminar
- Computer lab session/practical training
- Presentation/symposium
- Self-study
- Supervision/feedback meeting
- Fieldwork/excursion
- Working independently on e.g. a project or thesis

Method of Assessment

- Exam
- Abstract and formulation research objective
- Oral presentation

Literature

(**not** obligatory) the book: Oceanography and Marine Biology. An Introduction to Marine Science. (D.W. Townsend. Sinauer Associates, ISBN 9780878936021) may serve as introduction to the course.

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Capita Selecta Earth Sciences

Course Code	AM_1322
Credits	15
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.E. Vonk
Examiner	dr. M.C. de Ruiter
Teaching Staff	dr. M.C. de Ruiter
Teaching method(s)	

Course Objective

In this course, students perform fieldwork to complement their thesis research. This allows students to expand fieldwork skills, and adapt the planning of thesis research to fieldwork that may for example be dependent on seasonal conditions or to the timing of international (ship) expeditions.

Important: this course is formally scheduled in P5+P6 but can be at anytime, e.g. during the late summer.

Course Content

Students plan, prepare, perform, and report fieldwork related to their thesis research.

Additional Information Teaching Methods

One or more face-to-face meetings planned on an individual basis between student and thesis supervisor.

Method of Assessment

A dataset, fieldwork report, or other method agreed upon between thesis supervisor and course coordinator.

Entry Requirements

The course specifically aims to expand fieldwork (and/or extended labwork) opportunities to support thesis research. Therefore, the course requires a thesis research plan and a fieldwork plan upon admission. This is to be agreed upon between student, thesis supervisor, and course coordinator.

Literature

Targeted to own research topic

Additional Information Target Audience

The course specifically aims to expand fieldwork (and/or extended labwork) opportunities to support thesis research. Therefore, the course requires a thesis research plan and a fieldwork plan upon admission.

Additional Information

The course can be taken at any time during an academic year and is not limited to any specific period within the academic year.

Capita Selecta Master Earth Sciences

Course Code	AM_1174
Credits	6
Period	P3

Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. W. van Westrenen
Examiner	prof. dr. W. van Westrenen
Teaching Staff	
Teaching method(s)	Seminar

Course Objective

In this course students perform an in depth literature review of a specific topic of their choosing in Earth Science research in the broadest sense; or to write a detailed report of an extracurricular activity such as (1) organisation of / participation in an international expedition or workshop; (2) learning how to use software that is compulsory to enter specific internships with companies.

Course Content

Students prepare an in depth literature review or report of a specialist topic in Earth Sciences that is not treated in detail in regular compulsory or optional courses in the MSc Earth Sciences curriculum.
Students are coupled to a Department of Earth Science's staff member of their choosing. Student and staff member agree on the outline of the topic to be discussed and precise scope of the review or report, culminating in a written review report that is graded by the supervising staff member.

Additional Information Teaching Methods

One or more face to face meetings planned on an individual basis between student and staff member.

Method of Assessment

A written literature review report.

Entry Requirements

The course is open to all MSc Earth Science students

Literature

Extensive literature review.

Additional Information Target Audience

This course is primarily aimed at 2nd year MSc Earth Sciences students.

Additional Information

The literature review or report writing can be undertaken at any time during an academic year and is not limited to any specific period within the academic year.

Catchment Biogeochemistry

Course Code	AM_1279
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.E. Vonk
Examiner	dr. J.E. Vonk
Teaching Staff	dr. J.E. Vonk, dr. F.M. Béen
Teaching method(s)	Lecture

Course Objective

The main objective of Catchment Biogeochemistry is to provide students with a basic understanding of the landscape controls on water biogeochemistry. Streams and rivers serve as integrators of their surroundings and are therefore a diagnostic tool to assess the properties and ongoing (natural or anthropogenic) changes of the land that they drain.
More specifically, the student will at the end of the course:

- have acquired a qualitative and quantitative understanding of biogeochemical processes in the soil that influence the biogeochemical composition of water;
- understand how the biogeochemical composition of water changes within the hydrological cycle of a catchment: from precipitation, to groundwater to surface water;
- understand, describe and utilize techniques for assessing biogeochemical constituents in water, and be able to interpret these data;
- be familiar with different types of pollutants affecting water quality, and understand how pollutants enter, move through, and interact with the hydrological system.
- be able to independently interpret, visualize, and discuss existing catchment biogeochemistry data by means of development and presentation of a conference poster
- be able to comprehend and present scientific articles on catchment biogeochemistry, and be able to criticise and discuss its content

Course Content

The biogeochemistry and quality of water on Earth is a function of processes that occur in the human and natural environment. Water of sufficient quality is required for, for example, agricultural and domestic usage, as well as for healthy ecosystems. Knowledge on water biogeochemistry is needed to study and interpret the quality of ground and surface water.

The following course topics are included: water sampling, analysis and assessment of water and its biogeochemical constituents, isotopic composition of water and its constituents, processes that act upon biogeochemical constituents in water (e.g. weathering, degradation), carbonate chemistry, dynamics and transport of pollutants.

Additional Information Teaching Methods

The course consists of lectures, supplemented with practicals (assignment), group discussions, and a lab afternoon.

Method of Assessment

Written examination (40%),
assignment (40%),
discussions (20%).
One has to pass all components (higher than 5.5), no compensation possible.

Additional Information Target Audience

Hydrology MSc students and other Earth Science-related MSc programs

Catchment Hydrology

Course Code	AM_1278
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. W.R. Berghuijs MSc
Examiner	dr. W.R. Berghuijs MSc
Teaching Staff	dr. W.R. Berghuijs MSc
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

The objective of this course is to provide students with scientific theory, tools and methods for understanding and evaluating the surface water response of catchments to incoming precipitation. This requires knowledge about processes regulating the flow of water on the land surface and in river channels, the techniques for quantification

of surface water flows and statistical methods for predicting extreme runoff events. In addition, experience with surface water flow modelling for predicting the behavior of rivers under different land use or climate conditions will be acquired.

The course contributes to the Knowledge and Understanding and Application of Knowledge and Understanding final attainment levels of the MSc Hydrology Program. Knowledge and understanding is obtained through studying theory as provided in the textbook, during the oral lectures and through self-study of scientific papers on rainfall-runoff response topics. Knowledge and understanding is applied to a rainfall-runoff model which simulations will be evaluated with measured data.

Course Content

The course consists of three main topics. We start with runoff processes leading to stormflow. This is followed by lectures on different hydrological modelling practices, and goodness-of-fit criteria. Finally, an overview of hydrodynamic and hydraulic theory that governs flow in open channels is given as well as an overview of discharge measuring techniques.

Topics are hillslope hydrology, hydrograph analysis, statistical methods to describe and quantify spatial and temporal variation in catchment runoff and reservoir and flow routing and scaling in hydrology. The spectrum of available models for runoff modelling, from classical lumped models to data-demanding distributed, physically-based hydrological models, will also be discussed. Finally, theory and understanding will be applied in a series of modelling exercises applying a rainfall – runoff model to simulate runoff.

Additional Information Teaching Methods

The tuition consists of eleven classroom lectures and several computer modelling workshop sessions. The number of contact hours is in the order of 42.

Method of Assessment

The assessment is through a written exam (70%), an assessment of a selection of exercises (10%), and the HBV modeling assignment report (20%). Grades of all assessments should be at least a 5.5.

Entry Requirements

The student should be familiar with the subjects of the BSc course Introduction to Hydrology (AB_1074)

Literature

Main Textbook:

- S.L. Dingman, 2015. Physical hydrology, 3rd edition. ISBN: 978-1-4786-1118-9

Articles and other material (subject to updates)

- Seibert, J. & Vis, M. J. P. Teaching hydrological modeling with a user-friendly catchment-runoff-model software package. (Links to an external site.)
- Hydrology and Earth System Sciences, 2012, 16, 3315-3325 (Links to an external site.)
- Selection of Reader "Hydrology of catchments, rivers and deltas. Download Hydrology of catchments, rivers and deltas ", TUDelft, 2016
- Kirchner, J. W. Catchments as simple dynamical systems: Catchment characterization, rainfall-runoff modeling, and doing hydrology backward. (Links to an external site.)Water Resources Research, 2009, 45, W02429

Additional Information Target Audience

First-year M.Sc. Hydrology students, students from Earth Sciences, Earth and Economy or Natural Sciences M.Sc. programmes.

Additional Information

The course is open for participation to students from alternative M.Sc. programmes at the VU University Amsterdam, or from other universities.
If you are a professional and wish to attend this course you can also participate on a contract basis. In both cases please do contact the course coordinator to find out if you fulfill the background knowledge requirements and for enrolment procedures.

Custom Course Registration

The course is open for participation to students from alternative M.Sc. programs at the VU University Amsterdam, or from other universities. If you are a professional and wish to attend this course you can also participate on a contract basis. In both cases please do contact the course coordinator to find out if you fulfill the background knowledge requirements and for enrollment procedures.

Recommended background knowledge

The student should have good background knowledge of mathematics and physics at BSc level and have basic computer skills.
In addition, the student should have basic knowledge of Earth Science, as provided by the System Earth course (AB_450067).

Causes and Consequences of Environmental Change

Course Code	AM_1049
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ing. P.G.B. Chopin
Examiner	dr. ing. P.G.B. Chopin
Teaching Staff	dr. ir. C.J.E. Schulp, A. Steinmann, prof. dr. P.H. Pattberg, dr. ing. P.G.B. Chopin, dr. F.M. Béen, dr. E.J.M. Boere MSc, prof. dr. ir. P.H. Verburg
Teaching method(s)	Study Group, Seminar, Computer lab, Lecture

Course Objective

The goal of this course is to develop a common knowledge base about the causes and consequences of environmental changes and about how different disciplines perceive and address environmental problems. The course provides the opportunity to develop skills expected of master's graduates, notably literature search, efficient reading and synthesis, group work and writing skills and for grasping sustainability issues.

By the end of this course, students should be able to:

- Link the different components of socio-ecological systems: demands in terms of resources, biological processes, impacts and response driving the causes and consequences of environmental change
- Capture the links between society, economics and environmental and how these interactions drive the fate of a given system
- Characterize the type of relationship among these elements by using the different system concepts that will be viewed in the course: system boundaries, reinforcing and balancing loops, non-linearity, threshold effects, uncertainty
- Judge the relevance of indicators used to describe the systems in a real-life setting linked to decision-making in environmental processes analysis
- Reflect on concept around social sciences, e.g. societal or individual responsibility and how these concepts can impact the responses to environmental changes
- Demonstrate a capacity in collaborating in an interdisciplinary environment and contribute to a shared goal
- Develop skills in handling databases, presenting information in a synthetic and well framed way, writing in an efficient way

Course Content

Current human activities including agriculture, fishing, mining among others alter the core functions of global as well as local ecosystems. Anthropogenic climate change and the accelerating loss of biodiversity are pressing drivers of the planet change and are receiving increasing attention. However, many others local to global environmental impacts are also happening in parallel such as the degradation of ecosystems services, disturbance of the nitrogen cycles notable in link with the food chain, intensive land use change and various pollutants affecting natural ecosystems and human health. More sustainable development pathways are urgently needed. Identifying such pathways require an interdisciplinary and system understanding of the drivers and impacts of human activities to search for solution. This requires the involvement of numerous academic disciplines including natural and social sciences and appropriate assessment of current system functioning via system thinking and assessment of change of these system with appropriate indicators.

Causes and consequences of environmental change presents the breadth of environmental problems and negative impacts, using different discipline to examine causes, effects and possible solutions. It is a compulsory course in

Environmental Resource Management master program as well as by students from other curricula from the social environmental or earth sciences, as an extra curricula course. The course serves to develop a system understanding of environmental, social and economic impacts using i) System Dynamic approaches (e.g., causal loop diagram building in groups), ii) scenario of climate change mitigation, iii) indicators-based assessment in groups and iv) debate on the necessity of economic growth through 4 key practical exercises proposed during the course. Those practical exercises are fed by multiple guest lectures targeting specifically planetary boundaries (biosphere integrity, climate change, land system change, biogeochemical cycles, freshwater use, pollution) and society and population as a driver of negative impacts and potential solutions through for instance appropriate governance.

Additional Information Teaching Methods

The course is worth 6 ECTS credits which corresponds to 168 hours of work per student. The course comprises two sets of activities. The first takes place in classes, where information is presented through lectures, guest presentations, workshops and debates. The second includes assignments where student's ability to achieve the course's objectives is tested and students get hands-on experience with some of the methods discussed during the course. Assessment involves these written assignments and the exam. Feedback opportunities are included in class activities, question hours and presentations are organized for the assignments.

Approximate time allocation:

- Class: about 44 hours (HC=16, CPR=8, WC=20)
- Reading and exam preparation: about 64 hours
- Assignments and presentation: about 60 hours

Method of Assessment

Type of assessment:

- Assignments (A) worth 40% of the final grade (minimum grade to pass the course: 5.5)
- Written exam (E) worth 60% of the final grade (minimum grade to pass the course: 5.5). One resit possibility in December/January.
- It is compulsory to attend the assignment-related working lectures, activities and practical exercises.

Literature

The course builds on a series of scientific articles, which will be announced in the course guide, as well as selected book chapters which include:

Harris, J.M. and Roach, B. (2014) Environmental and Natural Resource Economics: A Contemporary Approach. 3rd Edition. Chapter 9 (Lecture Population, Affluence, Technology) and Chapters 1+7 (Lecture Economic instruments). Required text for ERM Course on Environmental Economics (Period 2).

Additional Information Target Audience

The course is part of the MSc programme 'Environment and Resource Management' but is open to all MSc students.

Additional Information

The course is coordinated by Dr. Pierre Chopin, and lectured by:

- Dr. Pierre Chopin (p.g.b.chopin@vu.nl)
- Mrs. Adna Steinmann (a.steinmann@vu.nl)
- Dr. Nynke Schulp (nynke.schulp@vu.nl)
- Dr. Esther Boere (e.j.m.boere@vu.nl)
- Prof. Dr. Philipp Pattberg (philipp.pattberg@vu.nl)
- Prof. Dr. Jeroen van den Bergh
- Dr. Harry Aiking (h.aiking@vu.nl)
- Dr. Frederic Béen (f.m.been@vu.nl)

Challenges and Solutions for GEC&P

Course Code	AM_1234
Credits	6
Period	P1

Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.C. Schoderer
Examiner	dr. M.C. Schoderer
Teaching Staff	prof. dr. P.J.H. van Beukering, dr. M.C. Schoderer
Teaching method(s)	Computer lab, Study Group, Excursion, Lecture

Course Objective

After completing this course the student is able to:

- Understand the scope of the master spec. Global Environmental Change and Policy
- Understand the position of personal background in the scope of the master spec. Global Environmental Change and Policy
- Explain the relation between global environmental change, policy and local practice in a field situation with stakeholders
- Conduct a DPSIR analysis and write an implementation plan with a multidisciplinary team
- Pitch a solution to a scientific problem for a group of stakeholders

Course Content

The course is an introduction to the master specialization program Global Environmental Change and Policy. It is building a bridge between the different scientific backgrounds of the students and the scope of the master specialization. The scope of the program is explained, with focus on global environmental change (the challenge) and policy (solutions). In more detail, the course will a) introduce key concepts from the field of environmental sciences; b) introduce the DPSIR approach; c) provide brief introductions to the specific fields of environmental policy analysis and environmental economics; and d) expose students to stakeholders and organizations in the field via excursions, including civil society, business and government.

Additional Information Teaching Methods

Lectures and fieldwork

Method of Assessment

DPSIR assignment (70%)
Presentation (30)%

Summary of field excursion (pass/fail)

Literature

Literature is made available via the Canvas page.

Explanation Canvas

The course has a dedicated Canvas page.

Climate Dynamics and Processes

Course Code	AM_1230
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. D.M.V.A.P. Roche
Examiner	prof. dr. D.M.V.A.P. Roche
Teaching Staff	prof. dr. D.M.V.A.P. Roche, dr. A.G.C.A. Meesters, B. Jong
Teaching method(s)	Seminar, Lecture

Course Objective

By the end of the course, students should have developed an in-depth knowledge of the base functioning of the atmosphere and ocean components of the climate system, from a physical systems point of view. Amongst the questions that will be developed are: what are the drivers of atmospheric convection? What is the effect of the rotation of the planet? What is the physical background of known climatological features such as planetary waves, monsoons, etc. Why are the oceanic boundary currents located on the west side of ocean basins? How is the deep ocean ventilated? The course knows a fair amount of theoretical and mathematical analysis.

Course Content

The climate on Earth is mainly dependent on the redistribution of heat from the sun through the atmosphere and, through the coupling of the two bodies, the oceans. Both atmosphere and ocean are governed by the equations of thin fluid layers on a rotating sphere: as a consequence, numerous aspects of the atmospheric and ocean dynamics can be derived from specific approximations of the well-known Navier-Stokes equations. Those form the core of the course and specific simplifications will be used to describe well-known phenomena occurring in both the atmosphere and the ocean that form the basis for the understanding of the climate on Earth. We will illustrate several aspects with state of the art examples, such as the weakening of the jet stream.

Additional Information Teaching Methods

Lectures, group discussion and computer exercises. There willl be six lectures and 4 workshops and a student presentations. The workshops will consist of Python/Matlab exercises. The students are required to develop a specific project for analysis and present this towards the end of the project.

Method of Assessment

Compulsory participation in computer exercises, oral presentation and written exam. The final mark is based on 20% being the oral presentation mark 20% exercises and 60% the final written exam.

Entry Requirements

Climate Science (BSc) and Climate Modelling (Msc)

Literature

G.K. Vallis: Essentials of Atmospheric and ocean dynamics, Cambridge University Press, 2019. DOI:10.1017/987107588431. / ISBN 978-1-107-69279-4 (Paperback)

Additional Information Target Audience

MSc Earth Science students with an interest in physics of climate

Recommended background knowledge

Mathematics and Physics (BSc), some experience in Python/Matlab programming is also useful

Climate Hydrological Processes

Course Code	AM_1196
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. D. Coumou
Examiner	prof. dr. D. Coumou
Teaching Staff	
Teaching method(s)	Seminar, Computer lab

Course Objective

Present-day climate change is already increasing the intensity and/or frequency of extreme weather around the

globe. A key challenge for environmental scientists today is to assess the likely societal impacts from future changes in hydro-meteorological extremes, with heavy rainfall leading to flooding and persistent dry conditions leading to drought-related impacts. To do such impact assessments, scientists have to rely on global climate models in combination with local hydro-meteorological impact models. This process is associated with substantial uncertainty for example because those models operate at different spatial resolution. To be able to properly interpret future impact studies, one needs to understand the most relevant physical processes of the climate system and how those processes are represented in climate models. This will be the focus of this course.

This course aims to provide students a solid understanding of the climate system including the most important underlying physical processes, its overall characteristics and its variability. The focus will be towards the intimate links between the climate system, the hydrological cycle and the large-scale atmospheric circulation. We will analyse which type of weather extremes have already increased in intensity and what kind of changes are to be expected for the near future.

Key goals for students to reach at the end of the course are:

1. Student can describe in own words how present-day climate change affects the hydrological cycle and how this impacts society.
2. Student can give a clear vision of the risks of climate change in his/her lifetime.
3. Student can describe in own words the concepts of radiative transfer, the greenhouse effect and the role of water and clouds on global climate
4. Student can describe in own words the fundamental drivers of tropical and extra-tropical circulation and its implications for weather
5. Student can describe in own words the importance of non-linear dynamics in the climate and how this can lead to abrupt transitions
6. Student has created a basic climate model of a 'Daisy' world and analyzed its dynamics
7. Student can critically analyze and interpret climate-related reporting in the media
8. Student can apply the new knowledge in class in presentations, discussions and debates.

Course Content

This course consists of several sessions consisting of (1) lectures by the professors with interactive discussion; (2) some practical assignments/coding, and (3) student presentations & discussions. You will team up in pairs of two (maybe three) students to develop a Daisy world climate model and do a series of experiments with it. Three afternoons will be devoted to this python coding where students will develop a basic climate model and perform some experiments. Also students will present in larger groups (~4/5) a recent scientific paper (or set of papers) which discusses the state-of-the-art of a particular lecture-topic. During these presentation sessions, all students are expected to participate in discussions.

Additional Information Teaching Methods

Class lectures (20hr), Computer practicals (12hr)

Method of Assessment

The course will be assessed through a written closed book examination based on the compulsory readings and the lectures. Also questions related to insights learned during the computer exercises can be asked.

There may be some quantitative question where a small calculation has to be made, so please bring a calculator. This closed book examination counts for 80% of the final grade. Furthermore, students will present a scientific paper in groups and discuss this in class. The presentation in class will together constitute 20% of the final grade.

The computer exercises are not graded but they are, however, a compulsory part of the course and need to be handed in via Canvas, for students to obtain a 'pass'.

Students will not receive a final grade if not all computer exercises have been handed in.

Literature

Apart from attending the sessions, you are expected to study the readings associated with the lectures (‘compulsory reading’) before attending them. This helps you both to actively engage in discussions during our meetings, and also makes it easier to prepare for the final exam at the end of the course.

The readings are provided as chapters and papers and aim to give you a broad understanding of the climate system as befits a master’s program. However, you will be examined only on some elements of the course; in other words you do not have to memorize entire pieces of text. The structure of the lectures in class will help you identify the key questions on which you will be examined.

Recommended background knowledge

In the computer practicum Daisy World, you will work together in groups of 2 or 3 to build a simple simulation of Daisy World in python. If you do not have any experience in python, please complete the modules up to and including ‘Functions’ from the ‘Learn the Basics’ chapter on https://www.learnpython.org/

Please do so before the first practicum (normally in week 3 of the course)

Climate Impacts & Policy

Course Code	AM_1240
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. W.J.W. Botzen
Examiner	prof. dr. W.J.W. Botzen
Teaching Staff	prof. dr. D. Coumou, N.A.M. van Maanen, S. Nirandjan, dr. E.E. Koks, C. Ferman Carral, prof. dr. W.J.W. Botzen, dr. V.L. Lupi
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

After completing this module, the student is able to:

- Understand processes in climate and economic systems and their interactions which cause climate impacts on society;
- Understand and apply methods for assessing impacts from extreme weather events and climate change in order to guide policies that limit these impacts;
- Be able to interpret, analyse, and critically reflect on the results gained from applying these methods.

Course Content

This interdisciplinary course combines knowledge from climate sciences, analysis of risks from extreme weather events and climate economics in order to assess climate change impacts. This knowledge forms the basis for guiding the design of policies that limit these impacts on society, including climate change adaptation and mitigation policies at the global and local level.

At the start of the course, processes of climate and economic systems and their interactions that lead to climate change impacts on society are discussed. Moreover, students learn methods for integrative climate impact analysis, such as climate models for assessing relations between climate change and extreme weather events, catastrophe model approaches which are used for assessing flood, and drought risks, and integrated assessment models of climate and the economy. Students work on assignments in which these methods are applied and results are used for the evaluation and design of policies that limit climate impacts through global climate policy agreements and local natural disaster risk management strategies, like flood protection.

Additional Information Teaching Methods

6 EC = 160 hours in total. The course will be taught through a combination of lectures, practical classes, and (computer) exercises. The methods will be applied to environmental problems, and students will reflect on the results. Moreover, for about half of the course students will work on their models, analyses and reporting themselves.

Method of Assessment

The assessment will consist of written assignments (40%) and a closed book examination (60%).

Entry Requirements

Admitted to MSc Earth Sciences, specialization Global Environmental Change and Policy, the MSc track Climate Econometrics, or the MSc track Climate Finance.

Literature

See the study manual on Canvas.

Additional Information Target Audience

Additional Information Target Audience

Second year students of MSc Earth Sciences, specialization Global Environmental Change and Policy , the MSc track Climate Econometrics, and the MSc track Climate Finance.

Climate Modelling

Course Code	AM_450004
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. D.M.V.A.P. Roche
Examiner	prof. dr. D.M.V.A.P. Roche
Teaching Staff	dr. P. Bakker, prof. dr. D.M.V.A.P. Roche
Teaching method(s)	Seminar, Computer lab

Course Objective

The objective of this course is to provide an overview of numerical climate models and their applications, with a focus on Earth Science studies.

Course Content

Geological archives show convincingly that the climate system experiences variability on a wide range of time-scales. For Quaternary studies, climate variations at the following time-scales are most important: glacials-interglacials, millennia and centuries-decades. This course focuses at the mechanisms behind these variations, thereby using climate models as a tool, i.e. numerical computer models in which the dynamics of the climate system are calculated. The combination of these models and geological data will be treated extensively. The course consists of lectures giving an overview of climate models and their application (different types for different time-scales), computer practicals and discussion meetings, in which students discuss the recent literature in detail. In this way the course considers case studies for the different time-scales and deals with recent developments in climate modelling. The following two questions are central to the course:

1. What is the driving mechanism behind climate change at a particular time-scale?
2. How can we optimise the combination of climate models and geological data in order to increase our understanding of climate evolution?

Additional Information Teaching Methods

Lectures, discussion meetings and computer exercises.

Method of Assessment

Compulsory participation in discussion meetings, computer exercises, oral presentation and written exam.

Literature

- Text book: Goosse, H. (2015) Climate System Dynamics and Modelling. Cambridge University Press, 358 p., ISBN 978-1-107-44583-3
- Additional: Lecture notes and selected papers (made available through Canvas).

Additional Information

The course is open for participation to students from alternative M.Sc. programmes at VU University Amsterdam, or from other universities.
If you are a professional and wish to attend this course you can also participate on a contract basis. In both cases please do contact the course coordinator to find out if you fulfill the background knowledge requirements and for enrollment procedures.

Climate Systems

Course Code	AM_1124

Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. ir. S. Houweling
Examiner	prof. dr. ir. S. Houweling
Teaching Staff	prof. dr. D.M.V.A.P. Roche, prof. dr. ir. S. Houweling
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

Quantitative approaches are increasingly important to mechanistically understand the processes leading to and resulting from Climate Change. The aim of this course is to gain basic knowledge of the modern climate system as a prerequisite for studying the climate of the present and the past.

At the end of the course students will:

- know the main properties that characterize the atmospheric and oceanic state;
- understand the main processes driving the atmospheric and oceanic circulation;
- know important influences of climate change on atmospheric and oceanic dynamics;
- be able to critically evaluate and present scientific literature on the Earth’s climate system.

Course Content

The course gives an introduction of present-day atmospheric and oceanic processes that explain the main large-scale circulation patterns. Besides important basic parameters and processes, the main circulation patterns will be discussed together with their implications for the regional climate and climate change. The students will apply the gained knowledge by selecting a topic on climate processes in relation to the cryo-, bio-, geo-, or anthropo-sphere, that they will study in detail and present to others.

Additional Information Teaching Methods

Lectures, literature reading, oral presentations by the students and discussions on the contents and quality of the presentation.

Method of Assessment

Written exam about the theory (50% of the grade)
oral and written presentation of a topic (50% of the grade)

Entry Requirements

Basic knowledge of physics. Interest in climate change.

Literature

Lecture notes (powerpoints of the presentations by the teacher), selected papers and Ruddiman, W.F., 2013. Earth's Climate: past and future.W.H. Freeman and Company New York.

Additional Information Target Audience

Target audience is geoscience / environmental science students. Not suited for second year MSc Hydrology students

Custom Course Registration

Subscription via Canvas

Designing Interdisciplinary Research

Course Code	AM_1264

Credits	3

Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.C. Blondeel
Examiner	dr. M.C. Blondeel
Teaching Staff	dr. M.C. Blondeel
Teaching method(s)	Seminar, Lecture

Course Objective

- Understand and describe the differences between mono-, multi-, inter- and transdisciplinary research;
- Implement the necessary steps to arrive at a research proposal, including identifying adequate research questions, literature review, selecting methods and data, developing an analytical framework, as well as providing structured discussion and analysis;
- Begin to develop a (interdisciplinary) research proposal;
- Present findings to a diverse audience.

Course Content

With the ever increasing complexity of environmental and socio-economic problems from climate change to global food security there is more than ever the need for holistic research programs to drive solutions, including interdisciplinary research that combines the theories tools and methods from a variety of academic disciplines. The purpose of this course is to introduce students to the theories and methods of designing an interdisciplinary research project, to apply that knowledge further throughout the Earth Sciences programme. This module introduces students to the challenges and opportunities of interdisciplinary research. By discussing the differences between mono-disciplinary approaches, multi-, inter- and trans-disciplinarity, the module critically analyzes the theoretical underpinnings of interdisciplinarity in the context of global change research. In addition to providing a meta-theoretical discussion of doing research, the module also introduces the practical steps necessary to develop a sound research proposal, including how to identify a suitable research question, how to choose adequate methods, how to analyze data, as well as how to present the findings to varying audiences.

Additional Information Teaching Methods

Lectures and seminars

Method of Assessment

Submission and presentation of a poster that presents an interdisciplinary research proposal.

Literature

Articles and book chapters via Canvas.

Didactics 1

Course Code	O_MLDIDAC_1F
Credits	6
Period	P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	prof. dr. M. Meeter, dr. I. Pauw, dr. S. Donszelmann, D.S. Hoefakker-Laanstra, drs. M.A. Huijskes MA, dr. H.B. Westbroek, dr. C.D.P. van Oeveren, drs. A.J.C. Monquil, dr. O.P.M. van Buuren, drs. Y.G. Meindersma, C.L. Geraedts, dr. B. de Vries, drs. R. Cornelisse MSc, M. Beukering, drs. E.D. van Noort, drs. H.R. Goudsmit, drs. M.M. Bout, prof. dr. ir. F.F.J. Hermans, dr. S.M. Calor, drs. L.J. van Well-van Grootheest, N. el Abdellaoui MSc, drs. J.K.W. Riksen, drs. N.H. Ypenburg

Teaching method(s)	Seminar, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het handelen als leraar in simpele en overzichtelijke situaties (bijvoorbeeld op het niveau van één les met aandacht voor de klas als groep).

Course Content

De cursus Didactiek 1 is onderdeel van de eerste fase (fase I) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 1 (stage). Tijdens Didactiek 1 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

De cursus is geordend rondom zogeheten kernpraktijken die fundamenteel zijn voor het beroep van leraar. Bij Didactiek 1 staan de volgende kernpraktijken centraal: (1) Contact maken, (2) De les starten, (3) Krediet opbouwen en uitgeven, (4) De les voorbereiden, (5) Orde houden, en (6) Uitleggen, instructie geven en de les afsluiten. De reikwijdte van het didactisch denken en handelen is in deze eerste fase meestal nog beperkt tot één les. De ervaring leert dat deze zes kernpraktijken voor de meeste lio's vrijwel direct aan de orde komen in het eerste deel van de praktijkstage en daardoor een goede basis vormen.

Het fysieke onderwijs van Didactiek 1 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio's werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen.

Het onderdeel AD in de cursus Didactiek 1 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio's ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio's toegankelijk. Op deze manier kunnen lio's gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio's van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Didactiek 1 wordt afgesloten met een (geschreven) startproef, bestaande uit een lesontwerp, een verantwoording op basis van theorie, en een uitwerking van eigen leerdoelen.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten

kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld via Canvas. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Didactics 1

Course Code	O_MLDIDAC_1
Credits	6
Period	P1
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	drs. L.J. van Well-van Grootheest, drs. M.A. Huijskes MA, drs. J.K.W. Riksen, dr. O.P.M. van Buuren, drs. E.D. van Noort, drs. Y.G. Meindersma, dr. S.M. Calor, dr. I. Pauw, drs. A.J.C. Monquil, M. Beukering, drs. N.H. Ypenburg, dr. C.D.P. van Oeveren, dr. S. Donszelmann, drs. M.M. Bout, drs. H.R. Goudsmit, dr. H.B. Westbroek, C.L. Geraedts, N. el Abdellaoui MSc, drs. R. Cornelisse MSc, D.S. Hoefakker-Laanstra, dr. B. de Vries, prof. dr. ir. F.F.J. Hermans, prof. dr. M. Meeter
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het handelen als leraar in simpele en overzichtelijke situaties (bijvoorbeeld op het niveau van één les met aandacht voor de klas als groep).

Course Content

De cursus Didactiek 1 is onderdeel van de eerste fase (fase I) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 1 (stage). Tijdens Didactiek 1 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

De cursus is geordend rondom zogeheten kernpraktijken die fundamenteel zijn voor het beroep van leraar. Bij Didactiek 1 staan de volgende kernpraktijken centraal: (1) Contact maken, (2) De les starten, (3) Krediet opbouwen en uitgeven, (4) De les voorbereiden, (5) Orde houden, en (6) Uitleggen, instructie geven en de les afsluiten. De reikwijdte van het didactisch denken en handelen is in deze eerste fase meestal nog beperkt tot één les. De ervaring leert dat deze zes kernpraktijken voor de meeste lio's vrijwel direct aan de orde komen in het eerste deel van de praktijkstage en daardoor een goede basis vormen.

Het fysieke onderwijs van Didactiek 1 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio's werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de

kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen.

Het onderdeel AD in de cursus Didactiek 1 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio's ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio's toegankelijk. Op deze manier kunnen lio's gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio's van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Interactieve hoorcolleges, werkcolleges en zelfstudiemodules.

Method of Assessment

Didactiek 1 wordt afgesloten met een (geschreven) startproef, bestaande uit een lesontwerp, een verantwoording op basis van theorie, en een uitwerking van eigen leerdoelen.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld via Canvas. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Didactiek 2 Semester 1

Course Code	O_MLDIDAC2
Credits	9
Period	P2+3
Course Level	400

Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	C.L. Geraedts, prof. dr. M. Meeter, drs. M.M. Bout, drs. A.J.C. Monquil, drs. E.D. van Noort, drs. R. Cornelisse MSc, dr. H.B. Westbroek, dr. S. Donszelmann, dr. O.P.M. van Buuren, D.S. Hoefakker-Laanstra, N. el Abdellaoui MSc, dr. C.D.P. van Oeveren, drs. J.K.W. Riksen, dr. B. de Vries, prof. dr. ir. F.F.J. Hermans, drs. L.J. van Well-van Grootheest, drs. H.R. Goudsmit, drs. Y.G. Meindersma, M. Beukering, drs. M.A. Huijskes MA, drs. N.H. Ypenburg, dr. I. Pauw, dr. S.M. Calor
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van (korte) lessenreeksen.

Course Content

De cursus Didactiek 2 is onderdeel van de tweede fase (fase II) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 2 (stage). Tijdens Didactiek 2 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Didactiek 2 is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Didactiek 2 staan de volgende kernpraktijken centraal: (1) Zien wie er in je klas zit, (2) Leerprocessen bevorderen, (3) Leerprocessen zichtbaar maken, (4) Leiding geven, (5) Een veilig leerklimaat creëren, en (6) Leerprocessen toetsen. Ten opzichte van de cursus Didactiek 1 wordt de focus verlegd van de (individuele) les naar het leerproces van de leerling. De reikwijdte van het didactisch denken en handelen wordt daarmee ook groter: er wordt een begin gemaakt met het ontwerpen en uitvoeren van reeksen van lessen.

Het fysieke onderwijs van Didactiek 2 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio’s werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD in periode 2 genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. In periode 3 staan, na de peergroupsessies in de ochtend, een aantal grote onderwijsthema’s centraal.

Het onderdeel AD in de cursus Didactiek 2 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio’s van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Didactiek 2 wordt afgesloten met een (geschreven) basisproef, bestaande uit een ontwerp voor een reeks van drie lessen, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik.

Entry Requirements

Voorwaardelijk voor afronding van Didactiek 2: een voldoende beoordeling van Didactiek 1.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie

NB: Voor studenten die vóór 1 september 2023 gestart zijn geldt in studiejaar 2023/2024 het curriculum van 2022/2023. Dit betekent dat het vak Didactiek 2 een studielast van 6 EC (ipv 9EC) heeft (en ze zich voor die variant intekenen). Studenten die de colleges van Didactiek 2 (6 EC) nog niet (geheel) gevolgd hebben sluiten aan bij colleges van het nieuwe vak Didactiek 2 (9 EC). Het is tot en met augustus 2024 nog mogelijk om Didactiek 2 (6EC) uit het oude curriculum af te ronden en het examen onder de eindtermen van het oude curriculum af te leggen.

Didactiek 2 Semester 2

Course Code	O_MLDIDAC2F
Credits	9
Period	P5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	drs. Y.G. Meindersma, drs. E.D. van Noort, prof. dr. ir. F.F.J. Hermans, drs. N.H. Ypenburg, dr. O.P.M. van Buuren, dr. H.B. Westbroek, dr. I. Pauw, dr. S.M. Calor, drs. L.J. van Well-van Grootheest, N. el Abdellaoui MSc, dr. B. de Vries, drs. A.J.C. Monquil, M. Beukering, C.L. Geraedts, drs. R. Cornelisse MSc, drs. H.R. Goudsmit, dr. C.D.P. van Oeveren, drs. M.A. Huijskes MA, D.S. Hoefakker-Laanstra, drs. J.K.W. Riksen, dr. S. Donszelmann, drs. M.M. Bout, prof. dr. M. Meeter
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van (korte) lessenreeksen.

Course Content

De cursus Didactiek 2 is onderdeel van de tweede fase (fase II) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 2 (stage). Tijdens Didactiek 2 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Didactiek 2 is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Didactiek 2 staan de volgende kernpraktijken centraal: (1) Zien wie er in je klas zit, (2) Leerprocessen bevorderen, (3) Leerprocessen zichtbaar maken, (4) Leiding geven, (5) Een veilig leerklimaat creëren, en (6) Leerprocessen toetsen. Ten opzichte van de cursus Didactiek 1 wordt de focus verlegd van de (individuele) les naar het leerproces van de leerling. De reikwijdte van het didactisch denken en handelen wordt daarmee ook groter: er wordt een begin gemaakt met het ontwerpen en uitvoeren van reeksen van lessen.

Het fysieke onderwijs van Didactiek 2 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio’s werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD in periode 2 genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. In periode 3 staan, na de peergroupsessies in de ochtend, een aantal grote onderwijsthema’s centraal.

Het onderdeel AD in de cursus Didactiek 2 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio’s van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Didactiek 2 wordt afgesloten met een (geschreven) basisproef, bestaande uit een ontwerp voor een reeks van drie lessen, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik.

Entry Requirements

Voorwaardelijk voor afronding van Didactiek 2: een voldoende beoordeling van Didactiek 1.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie

NB: Voor studenten die vóór 1 september 2023 gestart zijn geldt in studiejaar 2023/2024 het curriculum van 2022/2023. Dit betekent dat het vak Didactiek 2 een studielast van 6 EC (ipv 9EC) heeft (en ze zich voor die variant intekenen). Studenten die de colleges van Didactiek 2 (6 EC) nog niet (geheel) gevolgd hebben sluiten aan bij colleges van het nieuwe vak Didactiek 2 (9 EC). Het is tot en met augustus 2024 nog mogelijk om Didactiek 2 (6EC) uit het oude curriculum af te ronden en het examen onder de eindtermen van het oude curriculum af te leggen.

Earth science observation research methods

Course Code	AM_1285
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M. Langer
Examiner	dr. M. Langer
Teaching Staff	dr. J. Kiesel, dr. S.S.N. Veraverbeke, S. Nirandjan, dr. M. Langer, dr. B. Andeweg
Teaching method(s)	Study Group

Course Objective

This course aims to prepare the students for the GECP methods course (AM_1236). The course will provide insights into different natural hazard types. In practical exercises the students will learn different techniques of geospatial data analysis with GIS and Python using remote sensing products, weather reanalysis, and other geospatial data. The goal is to generate hazard maps as they are used in risk analysis also in combination with field mapping.

In addition, the students will gain insights into field measuring techniques to assess the quality of remote sensing products and geospatial data such as DEMs, vegetation cover, and building footprints. The methods learned in this course can also be used to upscale variables collected through field measurements.

This course will enable students to process geospatial data on a regional and local scale to analyze and map different types of natural hazards and to identify exposed assets such as buildings and infrastructure.

Course Content

Contents:

- Observation techniques including data obtained using remote sensing and weather models applied to geological and meteorological hazards such as flooding, wildfires, landslides, earthquakes and volcanic eruptions.
- Analyzing such data and using these techniques to generate maps of hazards and exposure of assets such as buildings and other infrastructure.
- Application of simple field measurement techniques to assess the quality and validity of remote sensing products and other geospatial data such as digital elevation models.

At the end of the course students will:

- have obtained a basic understanding of different natural hazards (geological and meteorological);

- identify which hazards they will focus on during the AM_1236 fieldwork course;
- have knowledge of how to map those hazards using GIS and gained an understanding on the creation of assessing exposure;
- know field methods of assessing validity and quality of remote sensing and other geospatial data products

Additional Information Teaching Methods

TBD

Method of Assessment

Assessment is based on weekly assignments. These must be carried out in small groups. The tasks include the creation of risk maps, which must be presented in the form of short written reports.

Additional Information Target Audience

- Introductory lectures on different types of hazards
- Exercises on the mapping of each hazard using GIS and Python

Ecohydrology

Course Code	AM_450014
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K. Naudts
Examiner	dr. K. Naudts
Teaching Staff	dr. K. Naudts
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

Ecohydrology is a combination of ecology (study of how organisms interact with each other and with the natural environment) and hydrology (study of how water cycles in terrestrial environments). It focuses on the role of ecosystems in the water cycle of terrestrial landscapes. The objectives of the course is to provide understanding of the functioning of ecosystems in relation to water availability and the movement of water in terrestrial ecosystems under different climates. This ecohydrological knowledge forms the basis for supporting decisions on sustainable land use from a water resources point of view. It requires fundamental theoretical knowledge on plant physiology and on the exchange of water between the soil, vegetation and the atmosphere. As such, limitations to ecosystem functioning posed by water availability in relation to evaporation and transpiration by different plant communities is a central theme in this course.

Course Content

This course describes and discusses basic interactions between the vegetated land surface, the atmosphere and the hydrosphere. Basic questions dealt with include: what determines the broad vegetation patterns of the world, and how do these in turn determine the ecohydrological behaviour of different vegetation types? This requires understanding primary ecohydrological processes (rainfall interception, transpiration, soil moisture dynamics) and feedback mechanisms between the vegetation and the atmosphere as well as insight into the measurement, data analysis and modelling of these processes. Tropical and temperate deforestation impacts on catchment hydrological functioning and climate as well as desertification
processes are considered. Ecohydrological processes in boreal and tundra regions, as well as in montane cloud forests will be discussed in some detail. Emphasis throughout the course is on a combination of process understanding, interpretation of experimental results, and modelling. Finally, a workshop is included to become familiar with the basics of meteorological data processing, plant water flow and evaporation modelling.

Additional Information Teaching Methods

The tuition consists of nine classroom lectures, student presentation sessions, two workshops and a proposal writing assignment.

Method of Assessment

Written test on lecture notes and selected literature (65%), paper presentation assignment (15%), and a proposal writing assignment (20%).

Entry Requirements

The student should be familiar with the subjects of the BSc course Introduction to Hydrology (450024) as detailed in the Introduction to Hydrology (2012) .

Literature

Scientific papers and handouts are provided during the course via Canvas

Additional Information Target Audience

First-year MSc Hydrology students, students from alternative Earth Sciences, Earth and Economy or Natural Sciences MSc programmes

Additional Information

The course is open for participation to students from alternative M.Sc. programmes at the VU University Amsterdam, or from other universities.
If you are a professional and wish to attend this course you can also participate on a contract basis. In both cases please do contact the course coordinator to find out if you fulfill the background knowledge requirements and for enrollment procedures.

Recommended background knowledge

The student should have a good background knowledge of mathematics and physics at BSc level and basic computer skills

Ecological Data Analysis

Course Code	AM_1225
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.T. Weedon
Examiner	dr. J.T. Weedon
Teaching Staff	dr. J.T. Weedon
Teaching method(s)	Study Group, Seminar, Computer lab, Practical, Written partial exam, Lecture

Course Objective

The final attainment levels of this course, include that students:

- Are acquainted with possible experimental designs for ecological research and can select the most suitable design depending on
  experimental objectives and hypotheses
- Are acquainted with possible statistical analyses, understand the theory and the assumptions underlying the various analyses and can test the underlying assumptions
- Can select the most suitable statistical analysis depending on the design chosen and the statistical assumptions
- Can interpret the chain of hypotheses, design and analysis to validate hypotheses combining empirical data with statistical models

Course Content

A proper experimental design combined with a suitable statistical analysis is essential to -biological- science, even though it is considered by many as a necessary evil. In this course, the whole chain of hypothesis and design to analysis and interpretation is covered to allow students to apply a range of statistical techniques independently.

The application and implementation of the techniques (using the statistical programming language R) is the basis. Possible experimental designs are discussed in relation to specific biological questions and hypotheses. The

application of statistical analysis is treated in relation to these designs. Theory and especially the assumptions underlying the tests are treated to the extent that this information is necessary to apply the tests properly. Both -combinations of- regression and analysis of variance techniques and multivariate analysis techniques such as unconstrained and constrained ordination are dealt with.

Additional Information Teaching Methods

As application is central to this course, case studies, assignments and working with real biological data is the core of this course. Starting of with the research question, hypothesis and the lab/field/model situation a proper design and statistical analysis will be discussed. A specific case study is used to illustrate this chain of arguments. Theory, assumptions and tests are all treated in the context of these case studies and are coupled directly to the case study and subsequent assignments. The course is finalised with an assignment involving applying the learned theory and techniques, to a series of case studies. Knowledge of some of the main principles of applied statistics is also tested in a short-answer exam. This set-up translates into 30 contact hours for lectures, 35 contact hours for practicals.

Method of Assessment

Exam (50%)
Report on the final case studies (50%)

Entry Requirements

Methodology and statistics 1 and 2 or equivalent statistics courses (min. 12 EC). This implies that we require students to understand the interpretation of P-values, type I and type II errors and statistical hypotheses testing in general. In addition, students are required to have understanding of t-tests (paired and unpaired), linear regression and one-way ANOVAs.

Literature

There is no required textbook, however students are strongly recommended to have access to one of:

- Quinn, G.P. and M.J. Keough (2002), Experimental design and data analysis for biologists Cambridge University Press
- Discovering Statistics Using R (2012) A. Field, J. Miles & Z. Field SAGE Publications

The latter is particularly recommended for students without any previous experience with the R programming language.

In addition the following articles and books may be helpful for parts of the course:

- Dalgaard, P. (2008) Introductory Statistics with R
- Logan, M. (2010) Biostatistical Design and Analysis Using R: A Practical Guide. Wiley
- Borcard, D., F. Gillet and P. Legendre (2011) Numerical ecology in R. Springer
- Bolker, B.M., M.E. Brooks, C.J. Clark, S.W. Geange, J.R. Poulsen, M.H.H.
- Stevens and J.S. White (2009). Generalized linear mixed models: a practical guide for ecology and evolution. Trends Ecol. Evol. 24: 127-135
- Gurevitch, J. and L.V. Hedges (1999) Statistical issues in ecological meta-analyses. Ecology 80: 1142-1149

This literature is complemented by lecture handouts, explanations of the assignments, answers to the assignments,and additional notes provided on Canvas.

Additional Information Target Audience

The course is compulsory for MSc Ecology & Evolution students at the VU and for UvA students doing the Ecology and Evolution specialization of the master Biological Science. The course is also open for master students in Biology, Ecology or Earth Sciences and PhD students at the VU and UvA universities with a deficiency in experimental design and statistics.

Additional Information

All contact hours are at VU Amsterdam

Lecturers:
dr. J.T. Weedon (VU)
dr M. Bruijning (UvA)

Recommended background knowledge

Students who have doubts about their background knowledge in statistics (see "Entry requirements" above) are

strongly encouraged to contact the course coordinator (james.weedon@vu.nl) in August or September before the course to get advice for preparatory readings.

Economics of Environmental Policy

Course Code	E_STR_EEP
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	prof. dr. M.W. Hofkes
Examiner	dr. ir. J.H. Ansink
Teaching Staff	dr. M.J. Koetse, dr. ir. J.H. Ansink
Teaching method(s)	Study Group, Lecture

Course Objective

By the end of this course, you should be able to:

ACADEMIC AND RESEARCH SKILLS

- explain the main mechanisms of economic policy instruments for managing environmental problems
- explain the pros and cons of economic policy instruments under different conditions and in different contexts

BRIDGING THEORY AND PRACTICE

- apply this knowledge to solve contemporary environmental problems

SOCIAL PROFESSIONAL SKILLS

- make meaningful contributions to team work

Course Content

A sound economic analysis of environmental policies is imperative for achieving effective and efficient solutions to solve the important sustainability problems of our planet. Many of these problems arise because the consequences of everyday behaviour on the environment or natural resources are not borne by the individuals who choose how to behave. Market-based policy interventions harness the power of financial incentives to encourage firms and individuals to take into account the full societal impacts of their decisions.

This course begins with foundational concepts in environmental economics, including concepts such as externalities, public goods, and cost-benefit analysis. It introduces key market-based instruments but places stronger emphasis on the practical and political constraints that shape environmental policy. Students will explore how factors such as institutional capacity, political feasibility, lobbying pressure, and information limitations often prevent the adoption of textbook-optimal policies. The course critically examines how these constraints influence the real-world effectiveness, fairness, and efficiency of various policy options.

While covering instruments such as taxes, subsidies, emission trading systems, and individual transferable quotas, the course focuses on how and why governments choose among them in specific contexts. Criteria for policy selection—like administrative simplicity, equity, and resistance to manipulation—are analyzed alongside empirical case studies. Doing so, the course covers pressing environmental problems such as climate change, the depletion of fish stocks, water management, and air pollution.

Additional Information Teaching Methods

Each week of the course typically consists of two on campus lectures and one on campus interactive. In the lectures we assess the design and use of policy instruments both using traditional criteria (e.g. efficiency, effectiveness) as well as through context-specific constraints (e.g. institutions, transaction costs, public acceptance, distributional concerns etc.). In the interactives we practice with applying our knowledge on policy instruments to specific environmental problems in which these constraints take a central role. In week 6 of the course this application of knowledge is extended to an extensive case study which uses a similar approach. A weekly quiz is used to test understanding of the lecture materials.

Method of Assessment

- Quizzes (10%) (ACADEMIC AND RESEARCH SKILLS , BRIDGING THEORY AND PRACTICE)
- Interactives (active participation) (BRIDGING THEORY AND PRACTICE, SOCIAL PROFESSIONAL SKILLS)
- Group assignment (30%) (ACADEMIC AND RESEARCH SKILLS , BRIDGING THEORY AND PRACTICE, SOCIAL PROFESSIONAL SKILLS)
- Exam (60%) (ACADEMIC AND RESEARCH SKILLS , BRIDGING THEORY AND PRACTICE)

Entry Requirements

Bachelor's degree in Economics;
Bachelor's degree in Earth, Economics and Sustainability;

or

Bachelor's degree in Econometrics, Environmental Sciences, Transport Science, Engineering, or Geography with sufficient background in economics and statistics.

Literature

Title: Policy Instruments for Environmental and Natural Resource Management
Author: Sterner, T. and J. Coria
Edition: 2nd Edition
Publisher: New York: Routledge and RFF Press (2012)
ISBN: 978-1-61726-097-1

Link: https://www.routledge.com/Policy-Instruments-for-Environmental-and-Natural-Resource-Management/Sterner-Coria/p/book/9781617260988?srsltid=AfmBOopcYTq6avkj7_Ac2KP-qJRhk6Vyhlvrfz6YjLV_UJJ2jzxPTwPn

The textbook will be complemented with papers from academic journals.

Energy and Climate Change; Science, Policy and Economics

Course Code	X_428568
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.C. de Ruiter
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130161

Additional Information Teaching Methods

Method of Assessment

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Energy and Climate Governance

Course Code	AM_1155
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. P.H. Pattberg
Examiner	prof. dr. P.H. Pattberg
Teaching Staff	prof. dr. P.H. Pattberg
Teaching method(s)	Written partial exam, Lecture

Course Objective

After concluding this course, students should be able to:

1. define and explain key concepts of relevance to the climate change governance and energy issue;
2. understand the causes, impacts and effects of climate change and the key scientific controversies in the regime;
3. identify, explain and analyze the various policy options for mitigation and adaptation at different levels of governance with a specific focus on energy related options;
4. understand the key political challenges in the climate change regime, the common problems facing all countries, and the various and changing coalitions in the regime;
5. explain the long-term objective, the principles, the commitments of countries and other key elements of the Climate Change Convention, including the Paris Agreement;
6. to explain the role of energy in the climate change regime, and the various aspects of policy with respect to renewable energy transition;
7. to analyze global energy policy taking into account questions of energy security, energy access and affordability as well as sustainability;
8. to make a judgment about which principles, policy instruments and approaches are likely to be most efficient, equitable and/or effective in addressing the climate change problem, including energy policy.

Course Content

Global governance of human-induced climate change, including both mitigation and adaptation, is a hotly debated subject. Current (international and transnational) climate policy is the result of a complex and long-lasting negotiation process at multiple levels of governance. In this process, the science of the complex earth and climate system is closely linked to questions on the socio-economic effects of climate change, the options for global environmental governance as determined by the structure of international organizations, international economic and political relations and environmental law. Reaching the 1.5 degree temperature target of the 2015 Paris Agreement will require a fundamental transition of global energy systems. Therefore the course will investigate the relation between climate change and global energy policy. The relation between climate change and energy will also take the complexities of the energy trilemma (security, sustainability, equity) into account.

The course includes:

- an overview of the science of climate change, its impacts , uncertainties, mitigation, adaptation;
- discussion of climate change policy options at multiple levels of governance, including the international climate change regime, national and European policies, and transnational approaches;
- analysis of the political challenges in climate change and the positions of different countries and actors;
- assessment of the economics of climate change including analyzing the flexible mechanisms (Emission trading, Clean Development Mechanisms) and options for Post Kyoto measures;
- analysis of interrelated energy challenges (energy trilemma);
- analysis of the challenges for a transition to renewable energy at various scales;
- the relation between global energy policy (including its geo-political aspects) and the climate change regime.

Additional Information Teaching Methods

Seminar/‘Werkcollege’ (s)

Method of Assessment

Written exam (E);
Summary of COP (report)

Simulation (Sim; pass/fail);

Weight of each graded component: 100

Compensation: is it possible to compensate one component with another? NO

Mode of re-examination for the different components: re-sit for final paper

Literature

See Course manual on Canvas

Additional Information Target Audience

master students

Recommended background knowledge

basic knowledge of environmental policy and governance

Environmental Economics for GEC&P

Course Code	AM_1232
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. V.L. Lupi
Examiner	dr. V.L. Lupi
Teaching Staff	dr. ir. M.A. Poblete Cazenave, dr. V.L. Lupi
Teaching method(s)	Study Group, Computer lab, Lecture

Course Objective

After completing this module, the student is able to:

- Describe the fundamental nature of environmental problems from an economic perspective, in relation to notions like externalities, public goods and free riding.
- Derive optimal levels of pollution and resource use from a societal perspective.
- Describe what these economic fundamentals and optima imply for the feasibility of various solutions to existing environmental problems.
- Understand climate change as a global externality and its direct and indirect effects on the environment, societies and economy.
- List economic policy instruments that are available, and discuss their (dis)advantages with respect to predefined policy aims.
- Argue how international trade and the environment are interlinked, and discuss the role of trade in stimulating or hampering sustainable development.
- Describe what are critical and debatable assumptions of core policy insights within environmental economics (e.g. those related to economic costs-benefit analysis of climate policy, and those related to economic methods used).
- To understand the fundamental nature of water as a natural resource and water-related problems from an economic perspective, and to become familiar with the instruments and institutions of water resource management, like water pricing and water markets, as well as their efficiency and effectiveness.

Course Content

This course introduces students to the economic analysis of environmental problems, and aims to give an overview of the environmental policy instruments in different policy contexts, from the local up to the global perspective. A critical cause of environmental problems is that not all the costs (including environmental costs) caused by economic actors are borne by those responsible for generating them. Likewise, benefits from producing some goods and services do not always end up with those who bear the costs. This problem will be conceptualised in this course through the notion of externalities. There are various instruments and institutional arrangements for addressing externalities. Criteria for their selection and evaluation will be studied. Applications of environmental policies at various administrative levels (i.e. local, national, international), different economic sectors and regional

contexts will be discussed. This course tackles some of the most pressing environmental problems that our planet faces today, such as the depletion of biodiversity, climate change, and environmental effects of trade and globalization.

This course is suited for students with a broad range of disciplinary backgrounds. It provides some of the fundamental building blocks for the MSc track Global Environmental Change and Policy (GEC&P). It offers insight to the economic theory of market failure (i.e. negative externalities) and discusses appropriate policy responses both from the microeconomic principles and examples from practice (many of which are provided by guest lecturers). After this course, students will be able to structure public debates on environmental policy interventions and will be able to assess the economic implications (e.g. trade-offs) of implementing different policy instruments. We encourage students with a good background in economics to follow the course ‘Economics of Environmental Policy Instrument Design’ provided by the School of Business and Economics.

Additional Information Teaching Methods

6 EC = 160 hours in total. Lectures and literature study (96 hours) and preparing and making assignments (64 hours).

Method of Assessment

Closed-book exam (50%)
Written assignments (50%)

Entry Requirements

Admitted to MSc Earth Sciences, specialization Global Environmental Change and Policy

Literature

Harris, J.M. & Roach B. (2018). Environmental and Natural Resource Economics: A Contemporary Approach – Fourth Edition. Routledge: Taylor & Francis Group, London.

Additional Information Target Audience

First year students of MSc Earth Sciences, specialization Global Environmental Change and Policy

Environmental Policy for GEC&P

Course Code	AM_1241
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.C. Schoderer
Examiner	dr. M.C. Schoderer
Teaching Staff	dr. M.C. Schoderer
Teaching method(s)	Study Group, Lecture

Course Objective

After completing this course, the student is able to:

- Understand the concepts of governance and policy as they apply to challenges related to global environmental change;
- Distinguish between various instruments and approaches to environmental policy-making;
- Assess the effectiveness, legitimacy, and broader impacts of environmental policies;
- Perform an analysis of policy interventions and their logics;
- Present concrete science-based recommendations for decision-makers.

Course Content

Action to address environmental problems – i.e. environmental governance – nowadays does not only come from nation states but also from a plethora of other actors. In this course, you will learn what environmental policy and governance is and how some of the actors that provide for it operate. Specifically, you will learn to understand how

environmental governance and policy is developed, implemented, assessed and contested at multiple levels, from domestic environmental policies to international treaties and transnational governance arrangements. You will familiarize yourself with the various modes of governance, from public to private, and the different governance instruments, from command and control to information-sharing and networking. In more detail, the course is thematically divided to address (a) the emergence and basic concepts of environmental policy and governance; (b) the role of interstate actors in environmental policy and governance (hierarchies); (c) the role of market-based approaches in environmental policy and governance (markets); (d) the role of networked governance with a focus on non-state actors (networks).

Additional Information Teaching Methods

Lectures

Method of Assessment

The course has the following required deliverables/assessments:

- Research papers (60%)
- Video presentation (40%)

Students must pass all elements (5.5 or higher).

Literature

Readings will be made available via Canvas.

Additional Information Target Audience

Students in the GEC&P specialization of MSc Earth Sciences.

Environmental Remote Sensing

Course Code	AM_450145
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S.S.N. Veraverbeke
Examiner	dr. S.S.N. Veraverbeke
Teaching Staff	dr. S.S.N. Veraverbeke, prof. dr. ir. S. Houweling
Teaching method(s)	Seminar, Computer lab, Written partial exam, Lecture

Course Objective

The course objectives are twofold:

1. Gaining an in-depth understanding of the theory of several remote sensing research applications
2. Executing an independent remote sensing project

Course Content

Lectures, including guest lectures, will cover specialized remote sensing research topics, for example on photogrammetry, surface water, fires, soil moisture, aerosols, greenhouse gases, urban environments, etc. The course also includes a project assignment, in which students will execute an independent remote sensing project.

Additional Information Teaching Methods

Lectures, including guest lectures, supplemented with reading materials.
Individual project with consultation sessions.

Method of Assessment

Written exam and individual project assignment, including poster presentation.

Literature

Selected research papers.

Additional Information Target Audience

First-year MSc Hydrology students and students from MSc Earth Sciences or other Natural Sciences MSc programs.

Recommended background knowledge

It is highly recommended that students have followed the course AB_1271 GIS en Aardobservatie in the VU BSc programs, or a similar course at another university.
The materials of this BSc course are assumed to be known, including practical experience with GIS and object-oriented coding (preferably Python).

Geology & Geochemistry Field Excursion

Course Code	AM_450229
Credits	3
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. A. Beniest
Examiner	dr. A. Beniest
Teaching Staff	dr. J.M. Koornneef, dr. A. Beniest
Teaching method(s)	Fieldwork

Course Objective

The aim of this course is to familiarise students with the multidisciplinary aspects of studying geological processes, using different kinds of local observations that can be synthesised to generate larger scale insights. This approach is illustrated by looking at the geological evolution of Cyprus. Another objective of the course is to orient students in tackling phenomenological observations derived from particular natural laboratories. It is the aim to make students familiar with the principles of 'problem-based learning techniques' by making field observations.

Course Content

The excursion follows a transect through Cyprus, from the coast to the top of Mount Troodos. The excursion addresses a range of inter-related tectonic, petrological and sedimentary processes associated with the formation and emplacement of the Troodos ophiolite and overlying metamorphic and sedimentary rocks. Students will consider the processes involved directly in front of the outcrops. The students learn to understand the nature, structure and evolution of spreading ridges, transform faults, mass transfer, and associated petrological and sedimentological processes and geochemical cycles. Students train critical thinking and communication skills in group discussions.

Additional Information Teaching Methods

Approximately 11 day field programme

Method of Assessment

Information written down in the field note books will be evaluated and their overall evaluation will be part of the grade. This grade will be linked with active participation in the field surveys and discussions / presentations during the evening sessions (20%). A written examination will be scheduled during the trip or after our return to Amsterdam (80%).

Entry Requirements

Participation in the compulsory programme of the Specialization Geology and Geochemistry of the MSc Earth Sciences and having passed 18 ec worth of courses from this programme.

Students who have not followed the compulsory programme, but would still like to participate can contact the

responsible staff via email with a short motivation to request permission to participate in this course.

Literature

Excursion guide and possibly selected additional publications from the scientific literature.

Additional Information Target Audience

The target audience for this course are:

- Students in year 2 of the MSc Earth Sciences - Geology and Geochemistry. (Compulsory course)
- Students with a general interest in the ocean and ocean floor processes.

Additional Information

Rules concerning the deadline for subscription and having proper mountain equipment will be strictly enforced. Due to the heat on Cyprus, the starting date has been moved from August to October. We therefore have to fit the timing around other courses and exams. Hence the exact time of travel will vary each year. The final dates will be announced as early as possible, but before August1st.

In 2023, the excursion will go to Cyprus and will be led by Prof. Gareth Davies.

On the basis of past experience our best estimate for the student contribution to expenses for travel and subsistence for this course is approximately 540 euros per person but the amount varies depending on the number of participating students.

Custom Course Registration

Admission is granted to students who took the compulsory courses of year 1 of the G&G programme in Blocks 1-3) and have earned at least 18 ec for those courses. Participants should register in time (before June 1st) using VUnet and should notify the responsible staff by e-mail.

Students who have not followed the compulsory courses of the G&G program, but who would still like to participate, can request permission to participate via an email with a short motivation to the responsible staff.

Recommended background knowledge

Sedimentary Basins AM_450154 and Orogenesis AM_450190.

Global Biogeochemical Cycles

Course Code	AM_450332
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	W. Heffernan
Examiner	W. Heffernan
Teaching Staff	W. Heffernan, prof. dr. ir. S. Houweling, dr. J.E. Vonk
Teaching method(s)	Computer lab, Lecture

Course Objective

To better understand the importance of biogeochemical cycles in the Earth system, with specific attention to the role of biogeochemistry in climate change including for example ocean acidification, nitrogen and CO2 fertilisation, and impacts on society.

Course Content

Biogeochemical cycles are crucial for mankind and the Earth system. For example, they govern our food supply, regulate climate on long time scales based on the balance between weathering rates and volcanic eruptions, on short time scales via emissions of fossil fuels and climate-carbon feedbacks in oceans and permafrost regions, etc. The main subject of this course is the exchange of carbon, nitrogen, and phosphorus between the soil, water,

atmosphere, and biota on global and local scales in different climatic zones (tropics, temperate, boreal and arctic zone) and environments. Each week consists of two lectures focused around a theme which will be followed by a practical where you will work with actual data to better understand links between biogeochemical cycles and the Earth system. The themes include the global terrestrial and oceanic carbon cycle and its relation to climate; the terrestrial nitrogen cycle and links to food production and pollution as well as climate; forests including deforestation and forest fires; carbon and nitrogen cycles in Arctic ecosystems and wetlands with a special focus on permafrost dynamics; and oceanic cycles of nitrogen, carbon, and phosphorus. All these cycles are linked through the atmosphere which will be another theme during the course.

Additional Information Teaching Methods

14 Lectures, 7 practicals that include group work, discussion, and presentations

Method of Assessment

Evaluation of the practicals, student participation in group work, student engagement during class discussions, written exam, and possibly an essay

Literature

Lecture notes and scientific papers will be made available during the course.

Additional Information Target Audience

MSc students Earth Sciences, Hydrology, Environment and Resource management

Recommended background knowledge

We will use the Python scripting language during practicals and elemental knowledge of scripting is required (a manual will be provided), although some exercises can also be done using Excel

Governance of Ecosystem Services

Course Code	AM_468025
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M. Schaafsma
Examiner	dr. M. Schaafsma
Teaching Staff	dr. M. Schaafsma
Teaching method(s)	Seminar, Lecture

Course Objective

The aim of this course is for students to acquire profound knowledge and understanding of the governance of ecosystems and biodiversity, specifically to:

1. understand the concepts and key elements of ecosystems and biodiversity governance
2. analyse governance issues and evaluate the strengths and weaknesses of various policy instruments and organizational frameworks related to sustainable ecosystem services and biodiversity management in different contexts.

After following this course, students should be able to:

- discuss the characteristics of ecosystems and biodiversity and how these relate to governance mechanisms worldwide (LO1);
- discuss available policy instruments at different governance levels (LO2);
- analyse the legitimacy, transparency, accountability, inclusiveness, fairness, integration, capability, adaptability, and effectiveness of these governance options in providing ecosystems and biodiversity protection (LO3);
- discuss potential trade-offs between the protection of ecosystem services and biodiversity and other socio-economic interests at the local community, regional or national sector level (LO4).

Course Content

Current governance systems of ecosystem services and biodiversity have evolved through policies developed at local (e.g. community based forest management), national (e.g. protection of wetlands), supranational (e.g. EU legislation such as Natura2000) and global levels (e.g. Convention on Biological Diversity). Governing biodiversity and ecosystem services is complicated because of a number of factors, such as their public good characteristics, different value systems, the often unclear distribution of user and property rights, the trade-offs between resource use and other societal goals, distributional outcomes, lack of coordination between various actors, and high transaction costs of effective enforcement and control. Also, scientific uncertainty surrounding the relationship between biodiversity protection, ecosystems and ecosystem services as well as the complexity of how socio-ecological systems function and change, including the presence of thresholds, play a role. Yet, because of the crucial nature of ecosystem services and biodiversity in human survival, ‘good’ governance of ecosystems and biodiversity is highly important.

The course will focus on how ecosystems are currently managed, what policies and policy measures are in place, and what lessons have been learned so far in developing and implementing different governance structures and policy instruments at the local, national and international level to safeguard ecosystems and biodiversity in the long run.

The course will elaborate on the multi-dimensional aspects of the governance of ecosystems and biodiversity, illustrated for different types of ecosystems and governance structures, varying from local community-based approaches to Payments for Ecosystem Services (PES) and global agreements and coordination such as the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES).

Important governance questions that hence arise in practical ecosystem and biodiversity governance and management, and that will be discussed during the lectures, are:

- What are the key institutions and who are the actors? Who is included in governance processes? Who holds power?
- How do the actors interact? What are the processes? How are decisions made?
- What are the links between the different levels of decision-making (local, national, global)?
- What policy instruments are used? What type of interventions are discussed?
- How do the policies, instruments and/or interventions score on different governance evaluation criteria?

These questions also form the type of questions to consider for the paper.

Additional Information Teaching Methods

The course corresponds to 6 ECTS and consists of a combination of (guest) lectures and interactive sessions over a period of 4 weeks, required reading, a paper and presentation, and an exam. Various topics concerning the governance of ecosystem services will be discussed during the lectures, guest lectures and interactive sessions. Literature related to the (guest) lecturers’ presentations is provided on Canvas. The course involves multiple guest lecturers, who translate theories and concepts of ecosystem services management and governance into policies and actions. The guest lectures will be supported by literature on governance of ecosystems and theory lectures.

In addition to the (guest) lectures, students will work on a short scientific paper about a topic of their own choice. A list of suggested topics to choose from will be provided at the beginning of the course. To guide students in their writing, an interactive session is organized once per week (also on the first day of the course). The paper is developed and written during the first 3 weeks of the course and presented in the fourth week. The paper and the presentations are compulsory and will be graded.

Classes are scheduled in the first three weeks of the course and, together with the required reading, paper writing, presentation and exam preparation, are estimated to account for about 168 hours of students’ time. Attendance in all classes is mandatory, but will be checked irregularly. We assume students are sufficiently serious and will show up at every class, barring unforeseen circumstances. Attending class is likely to increase your grades for the exam and the paper, so attending classes is in your own interest. Importantly, we all bring different expertise, experiences, questions and perspectives to the online classroom, so please share and learn – this will make the course more interesting and fun.

Indicative division of course hours:

- Class attendance: 33 hours
- Mandatory readings: 35 hours
- Writing paper and preparing presentation: 68 hours
- Preparing and taking the exam: 32 hours
- TOTAL: 168 hours (Standard total time allocated for 6 ECTS).

Method of Assessment

The overall mark for this course is based on three separate grades, i.e.:

- Closed-book written exam (Individual; 40%);
- Short Scientific Paper (50%);
- Presentation (10%).

A necessary condition to pass this course is that all three grades must be 5.5 or higher. The exam has a resit. If a student obtains a grade lower than 5.5 for the paper or the presentation, an assessment will be made to determine how the student can improve their grade.

Further details about the assignments and examination will be provided in the course manual.

Literature

See the course manual on Canvas. The compulsory reading list will contain about 15 academic papers which will be closely related to the topics of the guest lecturers (to be read in advance).

Additional Information Target Audience

MSc students Environmental Resource Management, MSc students Earth-Sciences/Global Environmental Change and Policy track (doctoral students or other MSc students are also welcome: please contact the course coordinator).

Recommended background knowledge

Students are encouraged to take the courses ‘Value of Ecosystem Services and biodiversity’ (AM_468024) and 'Environmental Policy' (AM_468021).

Groundwater Processes

Course Code	AM_1164
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. Y. van der Velde
Examiner	dr. ir. Y. van der Velde
Teaching Staff	dr. ir. Y. van der Velde
Teaching method(s)	Seminar

Course Objective

The objective of the course on Groundwater Processes is to gain knowledge and insight in the terminology and theory of groundwater hydraulics, including its mathematical notation and physical meaning.

Course Content

The movement of groundwater through the subsurface is a fundamental part of the hydrological cycle. In this course, you will get acquainted with the fundamental hydraulics of groundwater flow. You will get profound insight into fundamental hydrological concepts related to groundwater flow and its mathematical notation and solutions. The application of this knowledge will be illustrated using some basic groundwater flow modelling exercises.

Additional Information Teaching Methods

The course consists of a set of lectures supplemented with practicals.

Method of Assessment

Written examination

Additional Information Target Audience

Hydrology MSc students and other earth sciences related MSc programs

Human Geography II

Course Code	AM_1051
Credits	12
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Science
Course Coordinator	dr. I. Pauw
Examiner	dr. I. Pauw
Teaching Staff	dr. I. Pauw
Teaching method(s)	Seminar

Course Objective

Dit studieonderdeel is bedoeld voor studenten die aardrijkskundedocent willen worden. Via literatuurstudie, eigen onderzoek en ontwerpactiviteiten verdiepen studenten hun geografische kennis en geografische vaardigheden op het gebied van sociale geografie.

Doelen:

1. Studenten verdiepen hun geografisch wereldbeeld en inzicht in geografische vraagstukken aan de hand van literatuur
2. Studenten combineren geografische kennis en geografische vaardigheden in het ontwerpen van een regionaal- geografische excursie.
3. Studenten combineren geografische kennis en geografische vaardigheden in het schrijven van een vakpublicatie die gebaseerd is op eigen geografisch onderzoek.

Course Content

Verdieping van het vak sociale geografie op een wijze die voorsorteert voor het eerstegraads leraarschap aardrijkskunde. De verdieping is doelgericht maar kent een bredere relevantie voor beroepen waarbinnen educatie en communicatie over mondiale vraagstukken een rol speelt.

Er is aandacht voor geografische concepten, benaderingen en subdisciplines. Ook het structureren van geografische kennis is een belangrijk onderdeel. Naast kennisverwerving wordt er geoefend met het schrijven van een geografische publicatie en het verrichten van een doelgroep gericht geografisch onderwijsproduct in de vorm van een excursie.

Additional Information Teaching Methods

Het vak vereist een grote zelfstandigheid van de student. Het omvat: Literatuurstudie in zelfstudie 110 uur Feedback van de docent Een tentamen: 4 uur Een regionaal-geografische excursie ontwerpen: 100 uur Een vakpublicatie over regionaal geografisch onderzoek: 100 uur Begeleiding vindt plaats via feedback en gerichte werkbesprekingen Totaal 336 uur (12 EC)

Method of Assessment

- Mondeling/schriftelijk (open boek tentamen) na literatuurstudie (40%);
- Een zelf ontworpen regionaal-geografische excursie (30%)
- Het schrijven van een vakpublicatie over een regionaal-geografisch onderzoek (30%).

Entry Requirements

Sociale geografie I afgerond

Literature

Docent verstrekt zes artikelen/hoofdstukken t.b.v. sociaal-geografische studie.

De student levert zelf vier aanvullende artikelen. Deze worden, na goedkeuring door de docent, studiemateriaal voor het tentamen.

Ook voor de twee producten, excursie en artikel, wordt literatuur aangeleverd en van de student gevraagd. Nadere

toelichting is te vinden op canvas, in de studiehandleiding.

Additional Information Target Audience

Master studenten Aardwetenschappen. Verplicht voor studenten die eerstegraads docent aardrijkskunde willen worden. Andere geïnteresseerden kunnen hun belangstelling kenbaar maken, waarna toegang tot het vak beoordeeld wordt (o.b.v. van relevantie en beschikbaarheid) .

Additional Information

Nadere informatie bij Iris Pauw- i.pauw@vu.nl

Explanation Canvas

Canvas biedt informatie (studiehandleiding) en literatuur.

Recommended background knowledge

Het vakblad Geografie lezen.

Humanity and Climate

Course Code	AM_1057
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S.J. Kluiving
Examiner	dr. S.J. Kluiving
Teaching Staff	dr. H.J.L. van der Lubbe MSc, dr. L.M. Kootker, dr. S.J. Kluiving
Teaching method(s)	Seminar, Lecture

Course Objective

Aim is to enable students to assess the best method(s) for any kind of climate research dealing with both longer and shorter time scales. This also implies obtaining knowledge and insight in past, present and future climate change which is earth sciences, archaeology, and social sciences based. Students also learn about strategies of adaptation and resilience to climate change developed by past societies in both marginal and none marginal places. Finally students learn to put into perspective present and future climate research through study of climate research of the past half a century (e.g. IPCC-reports).

Course Content

How did climate during the Quaternary shape the development of Human ancestors during this time period? How have people adapted (or failed to adapt) to marginal and non-marginal environments and to climate change?

How can we distinguish between natural versus anthropogenic climate change, and what are Future perspectives regarding climate change?

These questions lead to looking at climate change on a longer time scale, focusing on the early hominids. It also implies looking at a variety of climate research methods ranging from the natural sciences focusing on the longer time scale to historical climatology focusing on the shorter time scale. To be able to distinguish between natural and anthropogenic climate change, there is also a need to investigate strategies of adaptation of past communities to climate change, gaining insight in and understanding of their resilience and even of their perception of past climate and weather conditions.

The focus of the course is on the following thematic questions:

1. How did large and abrupt climate change events interact with the development of humans and/or hominids from the Pleistocene to the Holocene? Assess the complex reciprocal relationship between human societies and climate. Analyse relation of ‘spikes’ in climate records with archaeological and historical events.
2. How can we distinguish between natural and anthropogenic climate change? What are future perspectives regarding climate change? Investigate adaptation strategies of past communities to climate change. Gain insight in understanding of human’s resilience and perception of past climate.

3. What is the Anthropocene and when does it start? Explore the Anthropocene concept: adaptation or resilience of past societies to climate change? Analyse major transition phases from archaeological and historical sources in relation to trends of geological proxies.
4. How can we reconstruct climate variations from man's records retrieved in archaeology, ancient studies and historical geography? Apply combined earth and social science methodologies in climate research. Analyse different spatial and temporal examples in European and Middle-Eastern landscapes.

Additional Information Teaching Methods

Seminars and lectures, excursion, reading literature, presenting a scientific research paper, creating a poster presentation. Attendance is mandatory.

Method of Assessment

Student presentations (40%) of research case, poster presentation of research project (60%).

Entry Requirements

Affinity with multidisciplinary research within the realms of Earth Science and Archaeology related to climate change research.

Literature

Will be made available on Canvas.

Additional Information

This course forms an opportunity to study climate and human impact in a multidisciplinary student group.

Imaging and Assessing Landscapes

Course Code	AM_1183
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E. Koomen
Examiner	dr. E. Koomen
Teaching Staff	L.J. Bouwknegt MSc, dr. E. Koomen
Teaching method(s)	Excursion, Computer lab, Lecture

Course Objective

The landscape is the visible result of human interaction with its physical surroundings and this course explores various ways to map the results of this process and discusses its impact on the degree people value their surroundings. The course aims to introduce methods, techniques and new developments in imaging (mapping) and assessing (evaluating) the landscape.

The course starts by introducing the concept of landscape, some common ways to classify different types of landscapes, and the many components it is comprised of. We then continue with an overview of GIS-based methods to map the surface of the earth. It introduces classic and novel approaches to collect spatially explicit data that help describe landscape components and capture the way humans interact with their physical surroundings. Special attention is paid to new ways to map landscape appreciation. This introductory part of the course enables you to understand the quality issues involved in collecting and using spatial data from a variety of traditional and novel sources. In addition you are able set up and execute a mapping exercise while applying relevant spatial analysis techniques.

The second part of the course elaborates on different approaches to value landscapes. The analysis of landscape values relates to issues such as openness, cultural history, ecology, physical geography and perception. Such valuation efforts will be applied in impact assessments of various types of spatial plans. This part of the course provides you with knowledge on different valuation methods and allows you to independently form and underpin an opinion on the value of landscapes.

Course Content

The following topics are included in this course:

- Introduction to the landscape concept: differences between landscape, land use and land cover; examples of well-known landscapes; landscape classification attempts in the Netherlands and abroad.
- Describing the main landscape components (distinguishing descriptive and interpretative components) and showing how these can be implemented in spatial analysis;
- Geodata capture and practical applications of novel data sources in imaging the landscape: data sources, classic cartography and novel approaches using LiDAR data, volunteered geographical information, social media data etc.
- Data quality (error, accuracy and consistency, quality aspects of novel data sources in relation to their applicabilty).
- Valuing landscapes: indicating differences in perspectives between, for example, experts and general public.
- Economic valuation of landscape values: introducing stated and revealed preference methods and applying these to find the value of open space.
- Impact assessments: what threatens landscapes and how can we assess impacts of, for example, road infrastructure, land consolidation and urbanisation? Showing examples of existing GIS-based applications.
- Landscape and planning: how are landscapes protected in the Netherlands and abroad?

Additional Information Teaching Methods

The course consists of eight lectures (of two hours) and several non-supervised practical assignments. To finalise the assignments students will have to spend a substantial amount of time in addition to the scheduled lectures and practicals. All assignments will be evaluated as part of the final assessment. In addition a one-day field trip is organised to a location near Amsterdam to visit a landscape threatened by development, discuss its values and evaluate the role of policy in protecting it. Active participation to the excursion is required.

Method of Assessment

The assessment will be based on a written final examination (40%) and the average mark for the practical assignments (60%). For each of these components students should obtain a mark of 5.5 or higher.

Entry Requirements

This course assumes that students have a working knowledge of GIS as is, for example, obtained in the Digital spatial data course offered as part of the Earth Sciences and Earth Sciences, Economics and Sustainability Bachelor programmes of VU. A catch-up opportunity based on distance learning can be provided for students lacking this knowledge. Please consult teaching staff prior to the course when this applies to you. You have to ensure that your GIS-knowledge is up to date before the course starts.

Literature

The relevant literature will contain scientific papers in English that will be listed on Canvas at start of the course. These papers will be provided through (links on) Canvas.

Additional Information Target Audience

The course is an elective module in the Earth Sciences, Environment and Resource Management, Heritage Studies, and Digital Archaeology Master programmes. In addition, it is open to all others with an interest in the mapping and valuation of landscapes provided they possess the required knowledge listed in this course description. The course is interdisciplinary in nature and expects students to be interested beyond the borders of their own discipline. It may be relatively soft (social-sciences oriented) for earth scientists and somewhat quantitative (embedded in natural sciences) for those with a humanities background. We, of course, try to explain the concepts you are unfamiliar with, but you should be prepared to explore unknown territory. Note that the course is only taught with sufficient attendance (see registration procedure).

Custom Course Registration

The course will only be taught when at least seven students are willing to actively participate. So make sure you register and show up! For smaller groups an alternative educational form may be considered.

Recommended background knowledge

Basic knowledge about the processes that shape landscapes is expected, as is some familiarity with the peculiarities of the origin of Dutch landscapes. For those lacking this, reference is made to the following book chapter:

- De Mulder, E.F.J. (2019). Landscapes. In: The Netherlands and the Dutch. World Regional Geography Book Series. Springer, Cham. https://doi.org/10.1007/978-3-319-75073-6_3

Inclusieve Digitale Didactiek

Course Code	O_MIDD
Credits	3
Period	P2, P5
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. S.M. Calor
Examiner	dr. S.M. Calor
Teaching Staff	dr. S.M. Calor
Teaching method(s)	Seminar, Lecture

Course Objective

De doordachte afstemming van complementaire face-to-face en online activiteiten moet duidelijk tot uiting komen in het ontwerp dat studenten leveren voor hun (nieuwe of bestaande cursus). Hiervoor gebruiken zij de VU Amsterdam Blended Learning (her) ontwerpsjabloon. Een belangrijk onderdeel hiervan betreft de motivatie die studenten gebruiken. Daarnaast lichten ze hun keuze voor online en face-to-face activiteiten toe en de volgorde hiervan, waarbij ze ook aandacht besteden aan constructive alignment; leerdoelen, leeractiviteiten en (formatieve en summatieve) testactiviteiten zijn op elkaar afgestemd. Inclusiviteit is een van de kernwaarden van de VU. Studenten gebruiken de VU Amsterdam Inclusiviteitskijkkader om de mate van inclusiviteit van het ontwerp in kaart te brengen. Ten slotte heeft de introductie van Generatieve Artificial Intelligence (AI) een enorm potentieel voor het onderwijs. Studenten gebruiken een VU Amsterdam doordachte didactiek bij de integratie van Generatieve AI in hun ontwerp.

Course Content

De coronaperiode heeft de inzet van digitale onderwijsvormen zoals online, blended en hybride onderwijs in een stroomversnelling gebracht. Naar aanleiding van het rapport Ruimte voor Onderwijs tijdens Corona en de motie Kwint is de wens ontstaan om meer inzicht te krijgen in de manier waarop digitale onderwijsvormen bijdragen aan het inhalen van coronavertragingen en het verbeteren van het welzijn van studenten. Daarnaast heeft de introductie van generatieve kunstmatige intelligentie (AI) een enorm potentieel voor het onderwijs. In het keuzevak “Inclusieve Digitale Didactiek” wordt aandacht besteed aan blended learning, inclusieve digitale didactiek en de integratie van generatieve AI in het onderwijs. Recente onderzoeksresultaten uit VU onderzoek worden in het keuzevak verwerkt.

De studenten van vandaag zijn opgegroeid in een wereld waar internet altijd aanwezig was, evenals mobiele telefoons en instant messaging. Ondersteunende leertechnologieën bieden een nieuwe manier om universitair onderwijs te ontwerpen, ontwikkelen en verzorgen en vormen een aanvulling op traditionele onderwijs- en leermethoden (Orton-Johnson, 2009). Als gevolg hiervan passen universiteiten en hogescholen hun financiering aan deze veelbelovende en nieuwe technologieën aan. De mogelijkheden voor communicatie, interactie en samenwerking worden uitgebreid, waardoor het hoger onderwijs naar het kantelpunt van een belangrijke transformatie wordt geduwd, waarvoor nieuwe vaardigheden van zowel studenten als docenten nodig zijn. Leertechnologieën kunnen ongekende onderwijsmogelijkheden bieden. Het vereist echter vakmanschap om effectieve blended learning, inclusieve lessen waarin generatieve AI in is geïntegreerd, te ontwerpen en ontwikkelen. Leraren moeten kennis hebben van inhoud, didactiek en technologie, evenals het vermogen om deze kennis zodanig toe te passen dat technologie het leren verbetert.

Additional Information Teaching Methods

Dit vak is opgebouwd uit een aantal colleges. Tussentijds bestuderen studenten relevante wetenschappelijke literatuur en werken zij aan hun onderwijs ontwerp voor een vak dat zij zelf geven. Tussendoor is er gelegenheid gebruik te maken van individuele coaching. Studenten presenteren hun tussentijds ontwerp aan elkaar en ontvangen van iedereen feedback. Deze peer feedback kan gebruikt worden ter voltooiing van het onderwijsontwerp. Tijdens de presentatie besteden studenten niet alleen aandacht aan hun ontwerp, zij lichten tevens hun analyse toe alsmede hoe het blended onderwijs, inclusief onderwijs waarin generatieve AI is geïntegreerd, te ontwikkelen, te implementeren en te evalueren. Zowel presentatie als ontwerp worden beoordeeld, resulterend in één eindcijfer.

Method of Assessment

De cijfers van studenten voor deze cursus worden berekend op basis van het blended learning ontwerp en het inclusief ontwerp met daarin generatieve AI geïntegreerd en hun presentatie hiervan.

Entry Requirements

Didactiek 1, Praktijk 1 (of vergelijkbaar)

Literature

Wordt bekend gemaakt via Canvas.

Additional Information Target Audience

Dit keuzevak is bedoeld voor studenten van de VU lerarenopleidingen (VU Lerarenacademie). Ook studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten.

Custom Course Registration

Alleen studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten.

Integrated Modeling in Hydrology

Course Code	AM_1165
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. H. de Moel
Examiner	dr. H. de Moel
Teaching Staff	dr. S. Muis, dr. ir. D.M. Eilander, dr. H. de Moel
Teaching method(s)	Seminar, Computer lab

Course Objective

The objectives of this course are for the students to:

- get acquainted with the wide range of modeling tools used in integrated hydrological studies
- understand the type of questions that can be answered using such tools
- acquire hands-on modelling skills and gain experience with some commonly used analytical programs (Python, GIS, Excel)

Course Content

The course is set up with a limited amount of lectures and two large exercises. In these exercises the students will set up and apply their own models from start to finish. The first exercise concerns a spatial rainfall-runoff model using publicly available data sources. This model will be used to simulate river discharge under current and future climatic conditions, and results will be related to water resources availability and associated measures. The second exercise is a flood risk assessment where hazard, exposure and vulnerability will be
combined to estimate flood risks. Using the model, measures will be evaluated on their risk reducing effect.

Additional Information Teaching Methods

There will be a limited amount of lectures (in mornings) as the focus is on the development of the two exercises. Practical sessions are scheduled throughout the period where staff will be available to help with technical questions and to help start up the model development. The rest of the time students will work on their models, analyses and reporting themselves.

Method of Assessment

Assessment will be done in the form of a report (50%) and a poster presentation (50%) related to the assignments. Both grades need a minimum of 5.0 in order to pass the course.

Literature

Relevant course reading material (papers, reports) will be provided via Canvas.

Additional Information Target Audience

Hydrology MSc students and other earth sciences related MSc programs

Land Use Change and Ecosystems

Course Code	AM_1235
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. T.A. Pupo West
Examiner	dr. T.A. Pupo West
Teaching Staff	prof. dr. ir. P.H. Verburg, dr. ir. J. van Vliet
Teaching method(s)	Computer lab, Lecture

Course Objective

To achieve a high level of understanding of global change processes related to terrestrial ecosystems and human land use.
Objectives of this course include:

- being able to conceptualize the role of land use both as driver and as impact of global environmental change;
- achieve understanding of human-environment interactions and feedbacks between ecological, physical and social (including economic) processes;
- become familiar with research methods to study land use change and ecosystem dynamics, including the analysis of driving factors, the simulation of land use dynamics, biodiversity impacts and the assessment of human-environment interactions through global value chains;
- becoming able to match interdisciplinary research questions with appropriate scientific tools, simulation models and other methods;
- understand potential options and policies for sustainable land use.

Course Content

Land use is key to sustainability. Achieving sufficient food, mitigating climate change and conserving biodiversity all require land, so competing claims on land emerge.
In the first part students will be trained in understanding the extent, the driving factors and the impacts of land use change. We combine global and local perspectives in studying land use changes and impacts of such changes on ecosystems. In the second part of the course the students will be presented with a portfolio of research methods, theoretical models and simulation models to analyze dynamics of land use. Practicals and computer exercises will provide a hands-on experience with these methods. Methods and models discussed will originate from different disciplines and include econometric models, spatial land use models, and participatory approaches. During practicals and assignments students will match research questions with appropriate (combinations) of methods. In the third part of the course transitions to sustainable land use will be discussed and the different pathways to achieving sustainable land use are introduced.Both global perspectives, governance aspects as well as implementation at the local level (i.e. sustainable intensification of agriculture) will be discussed.

Additional Information Teaching Methods

Lectures, Practical (computer) exercises. Self-study, including literature study, group and individual assignments.

Method of Assessment

Written exam (40%) and practical and assignment results (60%).

Literature

Selected articles and practical instructions will be made available on Canvas.

Additional Information Target Audience

- Master students with an interest in sustainable development and interdisciplinary science (with some

background on quantitative analysis and preferably GIS).
- First year students of MSc Earth Sciences, specialization Global Environmental Change and Policy who meet the above entry requirements.
- Students of the MSc Econometrics, track Climate Econometrics

Recommended background knowledge

Basic knowledge of GIS is recommended.Those that are not having introductory experience with GIS are requested to do an online tutorial in advance of the course (costing a bit of extra time)

Landscape Dynamics

Course Code	AM_450331
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C.J. Beets
Examiner	dr. C.J. Beets
Teaching Staff	dr. C.J. Beets, N. van Leeuwen, prof. dr. R.T. van Balen
Teaching method(s)	Excursion, Seminar, Computer lab, Lecture

Course Objective

- To learn and understand how environmental and climate change are recorded in (terrestrial-related) depositional environments, and to understand the recording process as a function of the dynamics of this environments.
- To unravel tectonic, climatic and human impacts in fluvial sedimentary records on different regional and temporal scales (excursion).
- To understand global climate change at Milankovitch to decadal time scales.
- To appreciate the different climate sensitivity of low and high latitudes.
- To gain an incipient overview over future climate studies and their dilemmas

At the end of this course, students should be able to:

- Interpret, and operate commonly used climate proxy datasets;
- Critically read and question professional climate change publications;
- Have (state-of-the-art) knowledge of the research field of high resolution climate archive studies, and the potential challenges that lie ahead.

Course Content

This course deals with the sedimentology, geochemistry and stratigraphy of fluvial, lacustrine, coral, cave, coastal and eolian palaeoclimate records. The focus is on those processes relevant for understanding how climate and environmental change is recorded in the terrestrial archives. In addition, the susceptibility of key aspects of those environments to climate- change impacts will be addressed.

The course will target seasonal-decadal to Milankovitch climate archive studies, including the climate phenomena and fluctuations that act onsuch time-scales. Special emphasis will be on the various (terrestrial) climate archives that record on (sub)decadal time scales and the validation of applied proxies. Using the latest research studies in discussion meetings (3x), several (paleo-)climate data-sets to perform practical exercises (3x), and a field excursion to Limburg as an instruction of the ways climate records form and can be read.

Typical climate archives to be studied are:

- (Varved) sediments-for instance, lacustrine, fluvial
- Ice cores
- Speleothems
- Corals
- bivalves

Additional Information Teaching Methods

Lectures, literature study and group discussions, computer practicals, and a field excursion to Limburg.

Method of Assessment

Literature discussion Essay; Report on computer practical exercises, and report of the field excursion to southern Limburg.

Entry Requirements

Students are expected to have bachelor-level knowledge of:
Climate science, Stable isotope geochemistry and Sedimentary environments

Recommended background knowledge Bachelor courses:
Sedimentary environments (AB_1093), Climate Science (450240); Master courses: Climate Systems (450185) (this is an obligatory course scheduled parallel to Landscape dynamics),

Literature

Course notes and selected peer-reviewed research papers (because we aim at including state-of-the-art research papers, these will be selected by teaching staff at the start of the course), Paleoclimatology-Reconstructing Climates of the Quaternary- by Bradley.

Magmatic Processes

Course Code	AM_450189
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Koornneef
Examiner	dr. J.M. Koornneef
Teaching Staff	dr. P.Z. Vroon, dr. J.M. Koornneef, prof. dr. W. van Westrenen
Teaching method(s)	Seminar

Course Objective

The main aim of the course is to provide an overview of the geochemical structure and evolution of the Earth's interior. After this course you are able to (1) understand the interaction between physical and chemical processes in the Earth's interior,(2) select geochemical tools to solve problems regarding melting and chemical evolution of Earth's interior, (3) understand why and how trace elements are fractionated between mantle minerals and melt,(4) describe how different mantle components evolved isotopically over time. An emphasis will be placed on improving data handling using Excel, scientific writing, oral presentation, and critical assessment.

Course Content

Distribution of major and trace elements between solid and liquid phases; geochemical modeling of magmatic differentiation processes. Radiogenic and stable isotopes as tracers of magmatic processes: geochemical and temporal evolution of crust and mantle. The physics of magmatic processes: source, transport, emplacement/eruption. Characteristics of the principal geodynamic environments and their effects on magmatic processes.

Additional Information Teaching Methods

Lectures with associated class and home work exercises; preparation of a student paper and its oral presentation, including critical interaction between staff and students. The course counts for 6 ECTS = 160 SBU which are divided between the different components of this course in the following way (1) 12 * 3 hour lectures = 108 SBU, Presentation and essay = 28 SBU, Homework exercises = 24 SBU, Total 160 SBU = 6 ECTS

It is vital that homework exercises are undertaken as these will be the basis of some exam questions and provide the experience to understand quantitative aspects of the course.

Method of Assessment

The mark you obtain for this course consists of the following components:

Homework exercises (20%)

Paper and its presentation (40%)
Written exam (40%)

Entry Requirements

The Mantle Properties course (code AM_450156) is mandatory for the Magmatic Processes course.

Literature

Selected specialist literature papers include

- Blundy J, Wood B (1994) Nature 372, 452-454.
- Blundy J, Wood B (2003) Earth and Planetary Science Letters 210, 383- 397.

A full list of literature required for the preparation of essays and presentation will be provided at the start of the course.

Additional Information Target Audience

First year MSc students in Earth Sciences

Recommended background knowledge

The BSc Earth Science course "Inleiding in de Anorganische Geochemie" (AB_450336) is not required, but contains a broad overview of many of the basic isotopic systems which will be discussed in the Magmatic Processes course. If you are not familiar with the contents of the BSc course "Inleiding in de Anorganische geochemie", then you should read the handouts provided on Canvas.

Mantle Properties

Course Code	AM_1211
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. W. van Westrenen
Examiner	prof. dr. W. van Westrenen
Teaching Staff	prof. dr. W. van Westrenen, dr. P.Z. Vroon, prof. dr. W.P. Schellart
Teaching method(s)	Seminar, Lecture

Course Objective

The main aim of this course is to make students appreciate that heat and its transport provide a fundamental link between processes near the Earth’s surface (subduction, orogenesis, rifting, dynamic topography), and processes in the deeper parts of the Earth (including initial differentiation, convection and partial melting). An additional practical aim of this course is to start improving student communication and writing skills.

Course Content

In this course we will (1) Provide you with an overview of what seismology, petrology, and mineral physics tell us about the properties of and processes in the Earth’s interior; (2) Clarify the links between heat, pressure, mineral properties, density variations, and observations including the composition of the Earth’s core, seismic structure of the mantle, dynamic topography, and melt compositions; (3) Improve your presentation skills by having you prepare an oral presentation and (4) attend a dedicated feedback session.

Additional Information Teaching Methods

The course consists of eight half-day lectures that may include exercises, one separate exercise session, one seminar session, and one feedback session

Method of Assessment

The final mark you are given for this course consists of the following components:
(1) Presentation (25%);

1-page abstract (15%);
Assignments (30%);
Written exam (30%)

Entry Requirements

Students should have a basic understanding of global geophysics, mineralogy and petrology.

Literature

Literature references that are required background reading will be provided on Canvas during the course.

Additional Information Target Audience

All Geology and Geochemistry specialization Year 1 students

Marine Geology and Paleoclimatology

Course Code	AM_450330
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. F.J.C. Peeters
Examiner	dr. F.J.C. Peeters
Teaching Staff	prof. dr. J.B.W. Stuut, dr. F.J.C. Peeters
Teaching method(s)	Seminar, Excursion, Lecture

Course Objective

Marine sediments provide valuable information on past changes in climate- and ocean circulation. The course Marine Geology and Palaeoclimatology aims to provide insight into the processes and mechanisms which have ultimately resulted in the present day distribution of various types of marine deposits. In this course you will learn how modern and past change in environmental, ocean and climate conditions are recorded in marine environments and how marine deposits vary - in geographic distribution, shape and composition - as a function of environmental and paleoclimate changes at millennial to Milankovitch scales. At the end of this course, the student will:

1. have state-of-the-art knowledge on marine sampling techniques and various types of marine deposits and records;
2. be able to interpret and understand various types of marine paleoclimate proxy data and time-series;
3. be able to critically evaluate and present scientific literature related to the field of paleo-oceanography and paleoclimatology, including links to ice-core records.

Teachers: Dr. Frank Peeters and Dr. Jan-Berend Stuut & guest lectures researchers.

Course Content

We will start the course with an excursion to the Royal Netherlands Institute for Sea Research on Texel. Various lectures and practical exercises will deal with with sedimentology, geochemistry and stratigraphy of predominantly Quaternary marine deposits. The focus is on those processes relevant for understanding how climate/environmental change is recorded in the different paleoclimate archives. The course targets seasonal to Milankovitch climate archive studies, including the climate phenomena and processes that act on such time scales.

Additional Information Teaching Methods

Lectures and guest-lectures, paper presentation, literature study, in-class assignments, group discussions, excursion to the Royal Netherlands Institute for Sea Research.

Method of Assessment

A marine research proposal will comprise 80% of the final grade and the remaining 20% will be based upon a research article presentation. There is no written exam for this course.

Entry Requirements

Although there are no entry requirements for this master course (level 400), some basic understanding of earth sciences is an added value. In case of doubt or questions contact the coordinator (Dr Frank Peeters, f.j.c.peeters@vu.nl)

Literature

Lecture notes, selected research papers and or book chapters.

Additional Information Target Audience

This course is embedded in the VU MSc Earth Sciences program and intended for VU Earth Sciences MSc students. The course, however, is open to other master students with basic background in the earth- and or marine sciences. For further information please contact the coordinator Dr. Frank Peeters (f.j.c.peeters@vu.nl).

Explanation Canvas

Course material, announcements etc will be made available via de VU Canvas course website.

Recommended background knowledge

There are no entry requirements for this course.

Master Thesis GEC&P

Course Code	AM_1286
Credits	30
Period	P4+5+6
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. M.A. Poblete Cazenave
Examiner	dr. J.F. Brusselaers MA
Teaching Staff	dr. J.F. Brusselaers MA, dr. ir. M.A. Poblete Cazenave, dr. J.F. Brusselaers MA
Teaching method(s)	Study Group

Course Objective

The main objectives of the thesis are:

- a) To further develop and apply the key concepts, theories and methodologies taught in previous GECP courses.
- b) To independently develop and execute scientific research of societal relevance.
- c) To reflect on matters and issues in the chosen domain of global environmental change and policy, and to become able to confidently form an opinion and contribute to both scientific and practitioner’s discussions.
- d) To present and discuss research findings in a meeting with peers.
- e) To independently write a thesis on the research and its findings in an academic style.

Course Content

The MSc thesis represents the finalization of the MSc Earth Sciences - Global Environmental Change & Policy at the Vrije Universiteit Amsterdam. In the Master Thesis course of the GECP programme, the student will individually carry out a research project at one of the research groups at the VU University Amsterdam or externally at another university or (inter-)governmental organization, non-governmental organization, or company. The thesis, also in case of an external project, must have a strong research component, fit within the aims of the Global Environmental Change & Policy track and must have both an earth sciences and policy component.

Additional Information Teaching Methods

A thesis is carried out individually and independently under supervision of an appointed supervisor and after approval of the thesis Research plan by the supervisor. The workload of the thesis will be equivalent to 30 EC, which corresponds to about 20 weeks of full-time work.

The student is ultimately responsible for finding a thesis project and/or placement before the start of the master thesis course, but a number of potential projects (with supervisors) will be offered from which the student can choose. During the first week of the course, a number of on-campus sessions will be organised to support the

development of the Research Plan.

In June, all students give a presentation of their findings so far to the other GECP students and their supervisors. The presentation and attendance of students during the specific session in which they present is compulsory and part of the total assessment.

Method of Assessment

Grading of your thesis is composed of the following elements:

1. Report (60%), which is a 50/50 average of the marks of the supervisor and 2nd assessor
2. Execution (30%)
3. Presentation (10%)
4. Attitude (insufficient / sufficient / good)

Both the thesis and the presentation have to be in English.

A collaboration between a student and supervisor is formalized in an Agreement Form. The students write their research plan during the first months of the thesis project. In the research plan they further detail the research topic and research activities. The research plan details the main objective, research questions and research methodology that will be applied in the thesis. It also includes a time schedule, which allows the student and supervisor to assess during the execution phase how the student is progressing in their thesis. The research plan must be approved by the supervisor based on a “GO-NO GO” evaluation, before the student can start the thesis. One of the criteria for a positive “GO” evaluation is that the research has both an earth sciences and policy component and fits the aims of the Global Environmental Change & Policy track. Further guidance is provided in the course manual.

The presentation should not take longer than ten minutes, and is followed by a discussion of maximum five minutes. The presentation gives students the opportunity to highlight their work in front of an audience and at the same time receive feedback from supervisors and fellow students which they are able to use to finalize their thesis.

Generally, towards the end of the project the student hands in a full draft with the supervisor, on which they will then receive feedback. After incorporating this feedback, the final thesis is ready for grading by both the supervisor and the 2nd assessor.

The final thesis presents the results found during the thesis research. Research for a thesis involves a research cycle: from the problem and research questions, the way these research questions were investigated based on sound scientific research methods, the answers to the questions, to the conclusions and recommendations to solve the problem. All these steps have to be described in the thesis.

The final thesis is an original, scientifically written report consisting of 10,000 – 12,000 words including title, abstract, and table of contents, but excluding tables, figures, references and annexes or appendices. It includes proper referencing to relevant academic literature, which is included in a separate list at the end of the report.

Additional Information Target Audience

MSc students Earth Sciences, specialisation Global Environmental Change & Policy

Explanation Canvas

The course manual and lecture slides, as well as announcements regarding the thesis topic selection and allocation procedures, will be shared through Canvas.

Master Thesis Geology and Geochemistry

Course Code	AM_1186
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Koornneef
Examiner	dr. J.M. Koornneef
Teaching Staff	
Teaching method(s)	Seminar

Course Objective

Prepare and independently conduct an individual research project, or successfully fulfil a work placement/traineeship in industry, government or otherwise; and write the results in a scientific report at the academic Master's level. In their conduct, working methods, attitude, collaboration, writing and other communication, the student should demonstrate their:

- advanced knowledge and understanding of the field, thus providing a basis or opportunity for originality in developing and applying ideas;
- problem solving abilities in new or unfamiliar environments within the broader research context;
- ability to integrate knowledge and to handle complexity;
- ability to clearly communicate conclusions, and the knowledge and rationale underpinning these, to specialist and non-specialist audiences.

Information on Master thesis projects is provided by department staff in December each year and is made available on the MSc Geology & Geochemistry Communivty Canvas site.

Course Content

Individual and independent research project or work placement (traineeship) in the Master Earth Sciences specialisation Geology and Geochemistry, culminating in a written master thesis report and an oral presentation.

Additional Information Teaching Methods

Research project or work placement/traineeship.

Method of Assessment

Written report, oral presentation, attitude, and execution/methodology are assessed.

Entry Requirements

The Research Project Geology & Geochemistry (AM_1267) should be completed before the start of the thesis project. Otherwise, admission is possible only when granted by the Examination Board.

The Master thesis is subject to the school's Work placement and thesis regulations (stage- en scriptieregeling). These regulations require detailed written agreements between supervisor and student that specify the conditions for the Master thesis work placement or research project. This agreement form, filled out electronically by student and supervisors, should be checked and approved by the master specialization coordinator (Msc.GG@vu.nl) before the start of the work placement or research project. The approved agreement form outlines the goals of the research project and the project timeline that need to be followed to complete the project. In case of deviations from the timeline, a revised agreement form needs to be completed and approved before finalizing the project.

Additional Information

Either the Research Project Geology and Geochemistry (AM_1187) or Master Thesis Geology and Geochemistry (AM_1186) may be extended by 12 EC (AM_450149 Extension Master Thesis Earth Sciences). An extension must be approved before the start of one of these projects by the MSc Geology and Geochemistry coordinator. The extension can represent either (1) as a separate chapter of the MSc research/thesis report, or (2) as a stand-alone report. Like for the MSc research project, a separate agreement form needs to be completed and approved beforehand for the extension of the thesis project.

Custom Course Registration

The Master thesis is subject to the school's Work placement and thesis regulations (stage- en scriptieregeling). These regulations require detailed written agreements between supervisor and student that specify the conditions for the Master thesis work placement or research project. This agreement form, filled out electronically by student and supervisors, should be checked and approved by the master specialization coordinator (Dr. Duna Roda-Boluda) before the start of the work placement or research project. The approved agreement form outlines the goals of the research project and the project timeline that need to be followed to complete the project. In case of deviations from the timeline, a revised agreement form needs to be completed and approved before finalizing the project.

Methods of Global Environmental Change

Course Code	AM_1236

Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.C. de Ruiter
Examiner	dr. M.C. de Ruiter
Teaching Staff	dr. M.C. de Ruiter, dr. T.S. Busker, mr. G. Garcia Alvarez, dr. M.P. Bokhorst
Teaching method(s)	Seminar, Computer lab, Fieldwork, Lecture

Course Objective

The overarching aim of this course is to familiarize students with measurement methods and techniques from different disciplines that are commonly used to identify, simulate and evaluate the impact of a multi-hazard event. The following are expected outcomes of active participation and collaboration in this course:

1. The student understands the principle of risk, as well as the dynamics between risk components (Hazard x Exposure x Vulnerability).
2. The student is able to model a likely and realistic multi-hazard event for a predefined area using methods such as area analysis in GIS and in-field observations collected through various techniques.
3. The student is able to analyze potential multi-hazard scenarios and exposure, using GIS and Remote Sensing techniques.
4. The student is able to estimate the change in vulnerability to different sectors and identify feedback loops that result from the sequential occurrence of multiple consecutive hazards as well as the implementation of Disaster Risk Reduction (DRR) measures.
5. The student is able to critically evaluate the vulnerability of objects and assets at risk during a series of disasters by comparing modeled vulnerability with observed reality.
6. The student is able to evaluate the risk perceptions and adaptation dynamics in different sectors (e.g. households), using questionnaires and interviews.

Based on an overarching analysis of the data collected and resulting findings in points 1-6, the student is able to provide overall conclusions on susceptibility to multi-hazard risk and provide clear and concise policy recommendations.

Course Content

This course is a follow up of the course Earth Observation Research Methods AM_1285, and is meant only for students in the master Earth Sciences, specialization Global Environmental Change and Policy. Methods taught in the Earth Observation Research Methods course will be applied here, while other Earth Scientific, Environmental, ecological, and socio-economic methods will be introduced. Application of these methods will take place in the context of a fieldwork.

The field context is the analysis of multi-risk in a given location in Tenerife island (ES). The fieldwork is a contribution to the MYRIAD-EU project and will be presented at the Universidad de La Laguna. Both preparations and fieldwork contain hazard, exposure and vulnerability analysis of two disasters happing after each other. The impact of Disaster Risk Reduction measures on the second disaster is evaluated, while suggestions for DRR improvement is suggested.

Additional Information Teaching Methods

The students develop their own research question and related methodology. Both must: contribute to the general research question be a cooperation with other groups in terms of shared methodology and resulting data be a mixture of physical and socio-economics. The combination of techniques must answer the research question contribute to one choice experiment carried out by all groups together. (This may be the only socio-economic technique) Schedule and assessments The subject is divided in three parts with related end products: Preparation: a field plan. The first 7 days are used to conduct a desk study and to write a field plan. The first days start with lectures on the general assignment and background information on the natural conditions and economic situation in Tenerife. Inspiration lectures for your methodology follow on the second day. A few times each group of two people have a meeting with the teachers to discuss the progress of their field plan. The final field plan is presented on the day before the travel to Tenerife. Fieldwork: data collection and analyzed data. The field work starts with an introduction to the area. The field work will be carried out and data is analyzed according to the field plan in 5 days, or adjusted with reason. Every morning the students discuss their plans with the staff, when also field appointments are made. The last field day an excursion will be organized by the students. In the afternoon we have one general presentation of the whole field work as one group at the University of La Laguna, Faculty of Economics, Tenerife (ES). Poster presentation: the results have to be discussed and the subquestion per group has to be answered, followed by suggestions for policy and the students of next year. This has to be presented on an attractive and eay

to understand poster in size A1. The poster is presented on the last afternoon.

Method of Assessment

Fieldplan report, (analysed) fieldwork results, scientific poster, presentations

Entry Requirements

Experience in GIS

Literature

Several mandatory and suggested to read papers are provided in the course syllabus

Additional Information Target Audience

First year students in Master Earth Sciences, specialization Global Environmental Change and Policy

Additional Information

Accommodation with kitchen will be organised and payed by VU. The students have to take care of transport to Tenerife and on Tenerife (a car per groups of four) and food.

Custom Course Registration

A month before the start the participants list is final. No later adjustments may be made, also no withdrawals. When students do, an invoice for the non-used accomodation will follow and is charged to the student.

Explanation Canvas

Course syllabus, practical info, assignments and rubrics will be provided via Canvas

Recommended background knowledge

All subjects in the first semester of GEC&P year 1. The subject cannot be done without knowledge of and experience in GIS.

Metropole Ecology

Course Code	AM_1199
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ing. P.G.B. Chopin
Examiner	dr. ing. P.G.B. Chopin
Teaching Staff	dr. ir. J. van Vliet, prof. dr. J. Ellers, prof. dr. P.J. Ward, dr. W.H. Halfwerk
Teaching method(s)	Excursion, Seminar, Computer lab, Practical, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130164

Additional Information Teaching Methods

-

Method of Assessment

-

Additional Information Target Audience

-

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Multidisciplinary Uncertainties and Risks

Course Code	AM_1249
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. W. van Westrenen
Examiner	prof. dr. W. van Westrenen
Teaching Staff	prof. dr. W. van Westrenen
Teaching method(s)	Study Group, Lecture

Course Objective

Overall aim is to understand the main principles of Multidisciplinary Uncertainty handling and Risk Mitigation, relevant for students in Earth Sciences in light of their future positions in multidisciplinary teams.

After the course, the student will

- have an understanding of types of uncertainty (know what you know, know what you don't know; don't know what you know; don't know what you don't know)
- Understand the difference between the statistical uncertainty of data (or variability) and uncertainty of a prediction
- know about different types of bias
- know about (the role of) serendipity
- have familiarized with examples of multidisciplinary projects with unexpected outcome
- have familiarized with examples of (own) estimation errors
- have learned the importance of sharing uncertainties related to one's (monodisciplinary) results and
- have learned to overcome hesitation to discuss them
- have learned how to be prepared for unlikely scenario's
- have worked a number of practical examples

Course Content

Multidisciplinary Uncertainty Analysis is not generally covered in VU University education programmes, although it is important in science and business. Many fields of modern earth science activities require team work by multidisciplinary teams. This course is expected to contribute to your knowledge of uncertainty management when dealing with multidisciplinary teams, inside as well as outside subsurface resourcing activities.
Single discipline studies usually concentrate on the most likely scenario within that discipline. In multidisciplinary projects, it is not unusual that in one of the disciplines, a less likely scenario occurs. It is essential that all contributors to a multidisciplinary project are aware of each other's "less likely" scenario's and know how to react if one materialises. Participants need to communicate what they know and don't know. This is not easy since there is a psychological hurdle to make explicit what one does not know.
In Earth Sciences, too, you generally have to work with limited and ambiguous data from different disciplines; for example, in subsurface studies, deciding whether and where to drill a hole for geothermal energy; in studies designing a space mission to return samples from Mars, dealing with data from geology, geophysics, geochemistry and engineering. An evaluation carried out in a single discipline may arrive at a very plausible conclusion from the perspective of that discipline, which may, however, be very unlikely (or even impossible) from the perspective of a different discipline. It is therefore essential to work in multi-disciplinary teams with frequent meetings in which the findings of discipline-specific work is openly discussed and challenged. Only then is it possible to develop a technically sound integrated model that is consistent with data and findings from all disciplines, and that is as little biased as possible. In cases when unexpected outcomes were encountered, "post mortem" studies almost always

demonstrated that this resulted from under-appreciation of some data or information that was available, but with which the evaluation study team had been less familiar.

Additional Information Teaching Methods

A four-week “Multidisciplinary Uncertainties” course in 14 afternoons. The course consists of a series of flip-the-classroom exercises, complemented by some lectures by guest lecturers.

The course will be built around a number of geoscience related examples from many subfields.

Method of Assessment

Oral presentation on a multidisciplinary uncertainties topic chosen by the student. No written exam.

Entry Requirements

Bachelor degree in Earth Sciences or Applied Earth Sciences (NL: VU, UU, TUD), Applied Earth Sciences (NL: TUD), Future Planet Studies (NL: UvA), Earth Sciences and Economics (NL: VU); or advance permission of the course coordinator.

Literature

Handouts via Canvas. Reports and Literature, to be distributed.

Additional Information Target Audience

This course is of interest to students in all specialisations of the MSc Earth Sciences

Orogenesis

Course Code	AM_450190
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. W.P. Schellart
Examiner	prof. dr. W.P. Schellart
Teaching Staff	prof. dr. W.P. Schellart, dr. L. Zhou, dr. F.M. Brouwer, dr. B. Andeweg, B.A. Uunk MSc, dr. P.Z. Vroon
Teaching method(s)	Education, Seminar, Computer lab

Course Objective

This course will provide students with new knowledge and understanding about mountain building processes that occur at collision zones (e.g. Alps, Himalaya) and at subduction zones (e.g. Andes), the processes that control orogenesis (e.g. the driving forces, the rheology of the lithosphere and the tectonic setting), their consequences (e.g. brittle and ductile deformation, generation of topography, metamorphism, syn-orogenic magmatism, sedimentary basin formation, formation of mineral deposits in orogenic settings, etc.), and the methods constraining those processes such as mechanical analysis, microscopic analysis, thermochronology, and numerical modelling.

Mutual relationships and feedback relations of orogenic processes in space and time are illustrated for different segments of orogens ranging from the external zones to the core of the mountain range. Furthermore, students will develop skills (1) to differentiate between orogen types, (2) to analyse, compare and explain distinct features of orogenic structures, (3) to apply the critical taper theory to accretionary wedges and fold-and-thrust belts, (4) to apply numerical modelling as a tool to tackle orogenic processes quantitatively, and to (5) critically assess and discuss relevant literature as well as modelling results.

Course Content

Key aspects of mountain building are discussed in the context of natural examples like the Alpine mountain chain in Europe, the Himalaya and Tibet in Asia, and the Andes in South America, and in a cross-disciplinary manner.

Specific topics are:

- The anatomy, tectonic development, and thermal evolution of mountain belts at collision zones (collisional mountain belts) and subduction zones (cordilleran or accretionary mountain belts);
- Deformation, metamorphism and magmatism in axial zones;
- The critical taper theory of accretionary wedges and fold-and-thrust belts;
- The rheology of the lithosphere and its role in orogenesis;
- PTt-paths: observation, interpretation and numerical modelling;
- The internal and external zones of orogens, and the interaction between orogens and sedimentary basins;
- The late stage evolution of orogens: modes of syn- or post-orogenic extension, orogenic collapse, and exhumation mechanisms;
- Mountain building and the formation of mineral deposits;
- Real-world examples: Himalaya, European Alps, Andes.

Additional Information Teaching Methods

Lectures and practicals (9 * 3 hrs 45 min), computer practical (3 * 3 hrs 45min).

Method of Assessment

Exam (40%), reports, practicals and presentation (40%), numerical modelling practical (20%).

Entry Requirements

BSc Geology, Geochemistry or Geophysics.

Literature

The course will be based on articles from the scientific literature as well as chapters from the following books:

- "Geodynamics", second edition, D. Turcotte and G. Schubert, Cambridge University Press 2002.
- "An introduction to igneous and metamorphic petrology", 2nd ed., Winter, Prentice Hall 2010.
- "Geodynamics of the Lithosphere", 2nd ed. K. Stüwe, Springer 2007.
- "Global Tectonics", 3rd ed., P. Kearey and F. Vine, Blackwell 2008.
- "Orogenesis", 1st edition, M.R.W. Johnson, S.L. Harley, Cambridge 2012.

Selected literature for individual essay and presentation will be announced on Canvas.

Additional Information Target Audience

1st year MSc students in the Solid Earth Sciences.

Recommended background knowledge

BSc courses in tectonics, structural geology and petrology.

Pedagogische en Didactische verdieping Semester 1

Course Code	O_MPEDIVER_F
Credits	9
Period	P1+2+3
Course Level	500
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	dr. B. de Vries, D.S. Hoefakker-Laanstra, prof. dr. M. Meeter, drs. A.J.C. Monquil, dr. I. Pauw, dr. H.B. Westbroek, dr. O.P.M. van Buuren, drs. M.M. Bout, drs. J.K.W. Riksen, drs. E.D. van Noort, drs. Y.G. Meindersma, drs. N.H. Ypenburg, N. el Abdellaoui MSc, drs. H.R. Goudsmit, dr. C.D.P. van Oeveren, dr. S.M. Calor, C.L. Geraedts, M. Beukering, drs. R. Cornelisse MSc, drs. L.J. van Well-van Grootheest, drs. M.A. Huijskes MA, prof. dr. ir. F.F.J. Hermans, dr. S. Donszelmann
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de verdiepende pedagogische en (vak)didactische kennis en

vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van doorlopende lessenreeksen, en voor het handelen als docent in complexe(re) situaties.

Course Content

De cursus Pedagogische en Didactische verdieping is onderdeel van de derde fase (fase III) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 3. Tijdens Pedagogische en Didactische verdieping worden de onderdelen algemene didactiek en pedagogiek (AD), peergroup (PG) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Pedagogische en Didactische verdieping is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Pedagogische en Didactische verdieping staan de volgende kernpraktijken centraal: (1) Leerlijnen ontwerpen en evalueren, (2) De puber leren kennen, (3) Rekening houden met gedrags- en leerkenmerken, (4) Je onderwijs systematisch evalueren, (5) Omgaan met diversiteit in je klas, (6) Differentiëren, (7) Verantwoordelijkheid geven, (8) Mentor zijn, en (9) Je professionele ruimte benutten. De cursussen Didactiek 1 en 2 vormen samen het basisdeel van de ULO; de cursus Pedagogische en Didactische verdieping kan gezien worden als het verdiepingsdeel waarbij we nader ingaan op (het begeleiden van) individuele leerprocessen en het verzorgen van passend onderwijs, en het leren van leerlingen in een groepsdynamische setting. In Pedagogische en Didactische verdieping komen meer complexe kernpraktijken en bijbehorende thema's aan de orde. Het (vak)didactisch denken en handelen strekt zich uit over langere perioden, en er is aandacht voor het ontwerpen van leerlijnen. Ook wordt de lio nadrukkelijker uitgedaagd een eigen visie op onderwijs te geven en uit te dragen. Zo is de lesmethode niet langer leidend, maar wordt van lio’s in toenemende mate verwacht zelf invulling te geven aan de inhoud en didactiek van de lessen (waarbij zowel bestaand als eigen materiaal kan worden gebruikt). Tenslotte zullen de (vak)didactische overwegingen die ten grondslag liggen aan de eigen visie onderbouwd moeten worden met relevante literatuur en eigen praktijkervaringen.

Het fysieke onderwijs van Pedagogische en Didactische verdieping concentreert zich op de maandag (de instituutsdag). Tijdens de (werk)colleges AD komen genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. Tot slot zijn er peergroupbijeenkomsten o.a. gericht op intervisie en het schrijven van de meesterproef.

Het onderdeel AD in de cursus Pedagogische en Didactische verdieping is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio’s van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Pedagogische en Didactische verdieping wordt afgesloten met een (geschreven) meesterproef, bestaande uit een ontwerp voor een lessenreeks van vier tot zes lessen, waarbij rekening wordt gehouden met een specifieke casusleerling, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik. Een belangrijk onderdeel is ook de systematische evaluatie waarin één aspect van het ontwerp en het effect op het leerproces van de leerling onder de loep wordt genomen.

Entry Requirements

Voorwaardelijk voor afronding van Pedagogische en Didactische verdieping: een voldoende beoordeling (of vrijstelling) van Didactiek 2.

Literature

De volgende drie boeken zijn verplicht voor het onderdeel algemene didactiek en pedagogiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.
- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast is het volgende boek voor veel studenten een zinvolle bron:

- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Dit laatste boek is niet verplicht. Tijdens de eerste colleges zullen we kort iets over dit boek vertellen, en is er gelegenheid om het even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om het boek al dan niet aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Pedagogische en Didactische Verdieping Semester 2

Course Code	O_MPEDIVER
Credits	9
Period	P4+5+6
Course Level	500
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	drs. M.M. Bout, drs. R. Cornelisse MSc, C.L. Geraedts, dr. H.B. Westbroek, M. Beukering, dr. O.P.M. van Buuren, D.S. Hoefakker-Laanstra, dr. B. de Vries, dr. C.D.P. van Oeveren, drs. L.J. van Well-van Grootheest, drs. E.D. van Noort, dr. S.M. Calor, J. Quartel MA, N. el Abdellaoui MSc, prof. dr. M. Meeter, dr. I. Pauw, prof. dr. ir. F.F.J. Hermans, drs. Y.G. Meindersma, dr. S. Donszelmann, drs. A.J.C. Monquil, drs. N.H. Ypenburg, drs. J.K.W. Riksen, drs. H.R. Goudsmit
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de verdiepende pedagogische en (vak)didactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van doorlopende lessenreeksen, en voor het handelen als docent in complexe(re) situaties.

Course Content

De cursus Pedagogische en Didactische verdieping is onderdeel van de derde fase (fase III) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 3. Tijdens Pedagogische en Didactische verdieping worden de onderdelen algemene didactiek en pedagogiek (AD), peergroup (PG) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Pedagogische en Didactische verdieping is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Pedagogische en Didactische verdieping staan de volgende kernpraktijken centraal: (1) Leerlijnen ontwerpen en evalueren, (2) De puber leren kennen, (3) Rekening houden met gedrags- en

leerkenmerken, (4) Je onderwijs systematisch evalueren, (5) Omgaan met diversiteit in je klas, (6) Differentiëren, (7) Verantwoordelijkheid geven, (8) Mentor zijn, en (9) Je professionele ruimte benutten. De cursussen Didactiek 1 en 2 vormen samen het basisdeel van de ULO; de cursus Pedagogische en Didactische verdieping kan gezien worden als het verdiepingsdeel waarbij we nader ingaan op (het begeleiden van) individuele leerprocessen en het verzorgen van passend onderwijs, en het leren van leerlingen in een groepsdynamische setting. In Pedagogische en Didactische verdieping komen meer complexe kernpraktijken en bijbehorende thema's aan de orde. Het (vak)didactisch denken en handelen strekt zich uit over langere perioden, en er is aandacht voor het ontwerpen van leerlijnen. Ook wordt de lio nadrukkelijker uitgedaagd een eigen visie op onderwijs te geven en uit te dragen. Zo is de lesmethode niet langer leidend, maar wordt van lio’s in toenemende mate verwacht zelf invulling te geven aan de inhoud en didactiek van de lessen (waarbij zowel bestaand als eigen materiaal kan worden gebruikt). Tenslotte zullen de (vak)didactische overwegingen die ten grondslag liggen aan de eigen visie onderbouwd moeten worden met relevante literatuur en eigen praktijkervaringen.

Het fysieke onderwijs van Pedagogische en Didactische verdieping concentreert zich op de maandag (de instituutsdag). Tijdens de (werk)colleges AD komen genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. Tot slot zijn er peergroupbijeenkomsten o.a. gericht op intervisie en het schrijven van de meesterproef.

Het onderdeel AD in de cursus Pedagogische en Didactische verdieping is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio’s van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback..

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Pedagogische en Didactische verdieping wordt afgesloten met een (geschreven) meesterproef, bestaande uit een ontwerp voor een lessenreeks van vier tot zes lessen, waarbij rekening wordt gehouden met een specifieke casusleerling, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik. Een belangrijk onderdeel is ook de systematische evaluatie waarin één aspect van het ontwerp en het effect op het leerproces van de leerling onder de loep wordt genomen.

Entry Requirements

Voorwaardelijk voor afronding van Pedagogische en Didactische verdieping: een voldoende beoordeling (of vrijstelling) van Didactiek 2.

Literature

De volgende drie boeken zijn verplicht voor het onderdeel algemene didactiek en pedagogiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.
- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast is het volgende boek voor veel studenten een zinvolle bron:

- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Dit laatste boek is niet verplicht. Tijdens de eerste colleges zullen we kort iets over dit boek vertellen, en is er gelegenheid om het even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om het boek al dan niet aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Peergroup 1 Semester 1

Course Code	O_MLPEERGR_1
Credits	0
Period	P1+2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	drs. H.R. Goudsmit
Teaching method(s)	Study Group, Lecture

Course Objective

De student werkt in de sessies Peergroup (PG) aan twee leerlijnen: het onderzoekend leren kijken naar ervaringen en het leren voeren van een professionele dialoog. De bijeenkomsten maken onderdeel uit van de vakken Didactiek 1 en 2.

De student leert systematisch te reflecteren op de eigen onderwijspraktijk en hierover in gesprek te gaan. De student gebruikt hierbij feedback van derden, beeldmateriaal van de eigen praktijkuitvoering, kennis uit de school, opleiding en onderwijsonderzoek.

Subdoelen – de student kan:

- de stappen in de reflectiecirkel van Korthagen onderscheiden en doorlopen;
- leerpunten en sterke kanten van het eigen functioneren als docent benoemen;
- effectief feedback geven en ontvangen;
- tijdens de intervisie en reflectie opgedane inzichten omzetten in concrete leerpunten en organiseert de aanpak hiervan;
- verschillende typen onderwijsonderzoek onderscheiden en op kwaliteit en praktische bruikbaarheid voor zijn eigen praktijk beschouwen;
- gebruik maken van bronnen (onderzoeksliteratuur en vakbladliteratuur) voor de analyse van praktijksituaties / onderwijsontwikkelingen;
- gebruik maken van bronnen voor een onderbouwde ontwikkeling van de eigen visie en lespraktijk.

Course Content

In de peergroup staat het handelen als ‘professional’ centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

In werkgroepen op de maandagochtend (de instituutsdag) reflecteren studenten op hun handelen in de praktijk (stage/werkplek), leiden daaruit ontwikkelpunten af, formuleren acties en evalueren deze. Ze leren hierbij op een professionele wijze met elkaar in gesprek te gaan en ervaringen te delen.

Verschillende instrumenten en methodes worden gebruikt (logboek, reflectiecirkel, intervisie, videoreflectie, etc.) om de student in staat te stellen de complexiteit van de onderwijspraktijk te doorgronden en hiervan te leren. In de peergroups wordt literatuur bestudeerd over onderwijs en onderwijsonderzoek en er wordt onderzocht wat hiervan de relevantie is voor de eigen onderwijspraktijk.

De colleges PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Bij de indeling wordt getracht rekening te houden met de voorkennis en ervaring van de verschillende studenten.

Additional Information Teaching Methods

Werkgroepen. Aanwezigheid bij de peergroup is verplicht.

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de startproef en de basisproef, en komen terug in de praktijk.

Literature

Literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 1 onderdeel uitmaakt van de vakken Didactiek 1 en 2 dienen studenten apart voor dit onderdeel ingeschreven te staan.

Peergroup 1 Semester 2

Course Code	O_MLPEERG_1F
Credits	0
Period	P4+5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	N. el Abdellaoui MSc, drs. Y.G. Meindersma, drs. H.R. Goudsmit
Teaching method(s)	Study Group

Course Objective

De student werkt in de sessies Peergroup (PG) aan twee leerlijnen: het onderzoekend leren kijken naar ervaringen en het leren voeren van een professionele dialoog. De bijeenkomsten maken onderdeel uit van de vakken Didactiek 1 en 2.

De student leert systematisch te reflecteren op de eigen onderwijspraktijk en hierover in gesprek te gaan. De student gebruikt hierbij feedback van derden, beeldmateriaal van de eigen praktijkuitvoering, kennis uit de school, opleiding en onderwijsonderzoek.

Subdoelen – de student kan:

- de stappen in de reflectiecirkel van Korthagen onderscheiden en doorlopen;
- leerpunten en sterke kanten van het eigen functioneren als docent benoemen;
- effectief feedback geven en ontvangen;
- tijdens de intervisie en reflectie opgedane inzichten omzetten in concrete leerpunten en organiseert de aanpak hiervan;
- verschillende typen onderwijsonderzoek onderscheiden en op kwaliteit en praktische bruikbaarheid voor zijn eigen praktijk beschouwen;
- gebruik maken van bronnen (onderzoeksliteratuur en vakbladliteratuur) voor de analyse van praktijksituaties / onderwijsontwikkelingen;
- gebruik maken van bronnen voor een onderbouwde ontwikkeling van de eigen visie en lespraktijk.

Course Content

In de peergroup staat het handelen als 'professional' centraal. Studenten leren de regie te nemen over hun eigen

leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

In werkgroepen op de maandagochtend (de instituutsdag) reflecteren studenten op hun handelen in de praktijk (stage/werkplek), leiden daaruit ontwikkelpunten af, formuleren acties en evalueren deze. Ze leren hierbij op een professionele wijze met elkaar in gesprek te gaan en ervaringen te delen.

Verschillende instrumenten en methodes worden gebruikt (logboek, reflectiecirkel, intervisie, videoreflectie, etc.) om de student in staat te stellen de complexiteit van de onderwijspraktijk te doorgronden en hiervan te leren. In de peergroups wordt literatuur bestudeerd over onderwijs en onderwijsonderzoek en er wordt onderzocht wat hiervan de relevantie is voor de eigen onderwijspraktijk.

De colleges PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Bij de indeling wordt getracht rekening te houden met de voorkennis en ervaring van de verschillende studenten.

Additional Information Teaching Methods

Werkgroepen. Aanwezigheid bij de peergroup is verplicht.

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de startproef (Didactiek 1) en de basisproef (Didactiek 2), en komen terug in de praktijk.

Literature

Literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 1 onderdeel uitmaakt van de vakken Didactiek 1 en 2 dienen studenten apart voor het onderdeel peergroup ingeschreven te staan.

Peergroup 2 Semester 1

Course Code	O_MLPEERG_2F
Credits	0
Period	P1+2+3
Course Level	0
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	drs. H.R. Goudsmit, N. el Abdellaoui MSc, drs. Y.G. Meindersma
Teaching method(s)	Study Group

Course Objective

De student werkt in de sessies Peergroup (PG) aan het systematisch reflecteren op de eigen onderwijspraktijk en gebruikt hierbij onder meer kennis uit de school en opleiding, feedback van derden en beeldmateriaal van de eigen praktijkuitvoering. Tijdens de beenkomsten wordt ingegaan op belangrijke zaken t.a.v. de meesterproef, zoals de casusleerling en videoreflectie. De bijeenkomsten maken onderdeel uit van het vak Didactiek 3.

Course Content

In een aantal peergroupsessies op maandag (instituutsdag) staat het handelen als 'professional' centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

Additional Information Teaching Methods

Aantal werkgroepen op maandag (instituutsdag).

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de meesterproef (Didactiek 3).

Entry Requirements

Didactiek 1 en 2 (inclusief peergroup 1) afgerond (of vrijgesteld van).

Literature

Eventuele literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Peergroup 2 is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 2 onderdeel uitmaakt van het vak Didactiek 3 dienen studenten apart voor het onderdeel peergroup ingeschreven te staan.

Peergroup 2 Semester 2

Course Code	O_MLPEERGR_2
Credits	0
Period	P4+5+6
Course Level	0
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	dr. I. Pauw, drs. R. Cornelisse MSc, drs. H.R. Goudsmit
Teaching method(s)	Study Group, Lecture

Course Objective

De student werkt in de sessies Peergroup (PG) aan het systematisch reflecteren op de eigen onderwijspraktijk en gebruikt hierbij onder meer kennis uit de school en opleiding, feedback van derden en beeldmateriaal van de eigen praktijkuitvoering. Tijdens de beenkomsten wordt ingegaan op belangrijke zaken t.a.v. de meesterproef, zoals de casusleerling en videoreflectie. De bijeenkomsten maken onderdeel uit van het vak Didactiek 3.

Course Content

In een aantal peergroupsessies op maandag (instituutsdag) staat het handelen als ‘professional’ centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

Additional Information Teaching Methods

Werkgroepen.

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de meesterproef (Didactiek 3).

Entry Requirements

Didactiek 1 en 2 (inclusief peergroup 1) afgerond (of vrijgesteld van).

Literature

Eventuele literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Peergroup 2 is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 2 onderdeel uitmaakt van het vak Didactiek 3 dienen studenten apart voor het onderdeel peergroup ingeschreven te staan.

Physical volcanology

Course Code	AM_1297
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. P.Z. Vroon
Examiner	dr. P.Z. Vroon
Teaching Staff	dr. P.Z. Vroon
Teaching method(s)	Seminar

Course Objective

Modern volcanology is balanced between the descriptive and quantitative, and both of these aspects of the field will be emphasized in this course. There are three basic goals for this class:

1. We want to understand how volcanoes work: the process part;
2. We want to be able to reconstruct unseen volcanic eruptions from the deposits they leave in the field;
3. We will want to know as much as possible about the hazards volcanoes form to people.

An additional practical aim of this course is to improve your computer skills with Microsoft Excel. To this end I have designed some exercises which will show you how to use Microsoft Excel in it’s most powerful form: visual basic for applications (VBA).

Course Content

Introduction to volcanic explosions and their products; Magma properties: viscosity, density and volatiles; Non-explosive volcanic eruptions; Magmatic fragmentation and pyroclastic textures; Eruption columns and the interpretation of pyroclastic deposits; Magma chamber processes; Volcanic processes on other planets; Volcanic hazards.

Additional Information Teaching Methods

This course consists of 10 lectures in which several subjects related to volcanology will be discussed. Each lecture is accompanied by a review paper or chapter from a book that gives an overview of the topics discussed – you will get more out of the lectures if you read these papers beforehand.

In addition to following the lectures you will be asked to complete homework exercises. These should be handed in before the start of the exam. These exercises are designed to clarify aspects of the lecture topics, and are also meant to provide a link between the different lectures.

Method of Assessment

The final mark for this course consists of the following components:

homework exercises and essay (25%); written exam (75%).

Literature

Encyclopedia of Volcanoes (Sigurdsson et al., 2015). Elsevier, ISBN 978-0-12-385938-9)

https://doi.org/10.1016/C2015-0-00175-7

Additional Information Target Audience

Second year MSc students Earth Sciences, specialization Geology and Geochemistry.

Recommended background knowledge

Petrology, geochemistry, sedimentology, structural geology courses.at the BSc level.

Plastic and Chemical Pollution

Course Code	AM_1259
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S.H. Brandsma
Examiner	dr. S.H. Brandsma
Teaching Staff	dr. S.H. Brandsma, dr. R. Oldenkamp, dr. L.A. Baumann, dr. S.L. Hughes, dr. I. van der Veen
Teaching method(s)	Computer lab, Seminar, Study Group, Symposium, College case, Practical, Lecture

Course Objective

Objectives

- To evaluate key environmental pollutants, their sources, properties, distribution and fate in the environment.
- To demonstrate quantitative information on the effects of pollutants on ecological and human health.
- To demonstrate how environmental risks of pollutants are assessed.
- To relate the concept of systems thinking, and its application to studying and understanding environmental pollution problems.
- To describe the key principles of environmental science that underpin both the assessment and governance of plastic and chemical pollution.
- Formulate the environmental risk of a chemical pollutant in a report and an oral presentation.

Contents

This course provides an introduction to environmental chemistry and (eco)toxicology of plastic and chemical environmental pollution, one of the hallmarks of the hotly-debated ‘Anthropocene’, the human-dominated epoch. Is this environmental pollution just a global bundle of bad news, or can/does science and society adequately respond? If you are interested in learning about plastic pollution and other hot chemical pollution topics this course is for you.

Course Content

Topics include:

- Sources and emissions and spatial transport of plastic and chemical pollution
- Properties affecting transport and bioaccumulation of plastic and chemical pollutants
- Bioaccumulation and biotransformation in living organisms
- Ecotoxicology and human toxicology of selected plastic and chemical pollutants
- Risk assessment of plastic and chemical pollutants
- Systems thinking in environmental chemistry and one world chemistry
- Societal and policy responses to address pollution

Plastic and chemical pollution is dispersed throughout our environment, including food chains and human bodies, making it a problem of global proportions. But a solvable problem. It may be surprising but most of the >100,000 chemicals synthesized in the past 70 years have not undergone rigorous safety testing before entering the marketplace, and many of these are high-production volume chemicals, including polymers.

Approximately 80% of today’s chemicals sector produces plastic and polymers. Past lessons with pollutants like dichlorodiphenyltrichloroethane (DDT), polychlorinated biphenyls (PCBs), and chlorofluorocarbons (CFCs) showed us that chemicals with economic and societal benefits can have long-lasting serious negative consequences for public health, ecosystems, as well as the economy. After decades of research on an often chemical-by-chemical

basis, science has been revealing how certain chemicals act as developmental neurotoxicants, endocrine disruptors, genotoxicants or immunotoxicants. And this knowledge is powerful. Such research is instrumental in evidence-based, risk-based policymaking that phased out many persistent organic pollutants to date. Cutting-edge analytical environmental chemistry is critical for first signaling and then quantifying ‘emerging’ pollutants in the environment, such as microplastics, endocrine disrupting chemicals (incl. plastic additive chemicals), perfluorinated chemicals, chlorinated paraffins, etc. Without analytical data, there’s no story to tell. Understanding the toxicity of the chemicals to humans and ecosystems gives meaning to the exposure concentrations measured in the field and laboratory. These data are essential input to environmental and human health risk assessments, policy-making and legislation, and design choices during manufacturing (e.g. green chemistry substitutions and functional alternatives). The societal and policy responses pave the way for elimination of pollution sources and the stimulation of innovations that act to prevent pollution, improving the quality of the living environment for all. This course may stimulate you to think about whether or not we currently are using our best science in creating the chemical-material world in which we are living.

Additional Information Teaching Methods

Teaching method

- Lectures
- Seminars
- Presentation
- Self-study
- Working on a written project
- Supervision/feedback

Method of Assessment

The final grade is calculated from the grade of the exam (65%), assignment report (25%), and oral presentation (10%). Minimum grades needed are: exam 5.0, report and presentation 5.5. In case resubmission of the report is needed a maximum grade of 6.0 will be assigned.

Entry Requirements

BSc degree in either hydrology, ecology, environmental sciences, chemistry, biomedical sciences, or public health.

If you have already followed the BSc Analytical and Environmental Chemistry (AEC) course (51128ANM6Y) then be aware that there can be overlap between some of the lectures given in this course.

Literature

Literature

“Environmental toxicology, an open online textbook”

Other: Additional material available on the Canvas site.

Additional Information Target Audience

Is part of/may be part of/is open to students of

- MSc Ecology (elective)
- MSc Hydrology (elective)
- MSc Earth Sciences – track Global Environmental Change and Policy (elective)
- Earth Sciences’ Earth & Climate stream (elective)
- MSc Biomedical Sciences (elective)
- MSc Chemistry (elective)

Additional Information

Attendance at lectures is highly recommended and is mandatory for the presentations.

The course is given at the VU. The first 4 weeks of the course will cover most of the new material, the second half of the course will be more focused on assignment, readings, a symposium for presentations of the assignment work, and finally the exam. Ecology students may complete work on assignment in the first four weeks, with only their presentation and exam in the second half of the course schedule.

Recommended background knowledge

BSc degree in either hydrology, ecology, environmental sciences, chemistry, biomedical sciences, or public health.

If you have already followed the BSc Analytical and Environmental Chemistry (AEC) course (51128ANM6Y) then be aware that there can be overlap between some of the lectures given in this course.

Practical: Paleoclimate Change and Environmental Impacts

Course Code	AM_1144
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. W.H.J. Toonen
Examiner	dr. W.H.J. Toonen
Teaching Staff	dr. N.N.L. van der Putten, dr. H.J.L. van der Lubbe MSc, dr. W.H.J. Toonen
Teaching method(s)	Excursion, Study Group, Reading, Practical

Course Objective

- To provide hands-on experience with the most relevant methods used in paleoclimate and paleoenvironmental research.
- To assess new data in the context of previous (published) studies.
- To introduce the basics of scientific reporting (written report and/or poster presentation)

Course Content

The practical comprises a series of lectures, lab classes, and discussion meetings. The students will collect their own field and lab data (as a group of researchers), and will present the obtained results in a Nature-style publication and a conference-style poster presentation. During the practicals a marine and a terrestrial sediment core will be investigated. During a mandatory 1-day field campaign we will teach how to collect a sample core - this core will be used for further study during the course.

The research includes core description, defining sampling strategy, basic sample processing, determination of micropaleontological, palynological, geochemical and geophysical sediment properties, and advanced data analysis (e.g. End Member Modelling, the use of the Tilia software to plot pollen and macrofossil data).

The emphasis will be on climate change at the glacial-interglacial time scale and on millennial-scale climate change records (e.g., Heinrich events, the last deglaciation and the Younger Dryas event). Participation in lectures, laboratory work, oral presentations and discussion meetings are compulsory, as detailed in the OER for a practical course.

Additional Information Teaching Methods

- An introductory lecture will highlight the outline of the practical course. - Lectures on palynology, marine micropaleontology, sedimentology and applied methods.
- A 1-day field campaign is used to collect core material for further study.
- Students will be motivated to present and discuss the obtained results during a series of discussion meetings.
- Laboratory work, data analysis, preparation of presentations.
- Managing a project: the students are working together and will have to define their goals and monitor their progress.
- Self-tuition from available literature.
- Writing in an ultra-condensed format (Nature-style); how to combine novel data with existing literature, while being able to highlight the most important findings.

Method of Assessment

- Laboratory work and attitude (20 %)
- Oral presentations and participation in discussion meetings (40 %)
- Written report and/or poster presention (40 %)

Literature

Selected literature will be made available via Canvas.

Additional Information Target Audience

Master students Earth Sciences. Students from all specialisation tracks are welcomed.

Additional Information

Learning hands-on skills is time-consuming, so there are many hours of practicals. Due to the condensed teaching roster, these could overlap with other courses in the same period. If there are any concerns about the feasibility of combining certain courses, please contact the teachers - we will find a way.

Practice 1

Course Code	O_MLPRAK_1F
Credits	6
Period	P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. M.A. Huijskes MA, prof. dr. ir. F.F.J. Hermans, D.S. Hoefakker-Laanstra, prof. dr. M. Meeter, N. el Abdellaoui MSc, drs. E.D. van Noort, M. Beukering, dr. H.B. Westbroek, C.L. Geraedts, dr. O.P.M. van Buuren, dr. S.M. Calor, drs. R. Cornelisse MSc, dr. S. Donszelmann, dr. C.D.P. van Oeveren, drs. J.K.W. Riksen, drs. Y.G. Meindersma, drs. M.M. Bout, dr. I. Pauw, drs. H.R. Goudsmit, drs. L.J. van Well-van Grootheest, drs. A.J.C. Monquil, dr. B. de Vries
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 1 doorloopt de lio het eerste deel van de praktijkstage onder begeleiding van een werkplekbegeleider (wpb). De stage voor Praktijk 1 omvat minimaal 48 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk gelden, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. Tijdens Praktijk 1 werkt de student o.a. toe naar het zelf geven van een hele les.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de gehele eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 1 voert de student tegen het eind van de eerste fase op de (opleidings-)school de opdracht ‘Presenteer jezelf’ uit. De student presenteert zichzelf als leraar in opleiding en voert daarbij bewijsmateriaal aan over zijn/haar functioneren in de praktijk gedurende de eerste fase. Vervolgens beoordeelt de werkplekbegeleider de student aan de hand van het Beoordelingsformulier Praktijk. Het functioneren van de student gedurende de stage én de inbreng tijdens de presentatie vormen input voor de beoordeling.

Additional Information Target Audience

Studenten met een baan (zij-instromers, onderwijstrainees etc) geven in dit stadium al zelfstandig les. Bij deze studenten ligt de nadruk bij de begeleiding vanuit de werkplekbegeleider op het niveau van didactisch handelen in de les.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 1

Course Code	O_MLPRAK_1
Credits	6
Period	P1
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	dr. C.D.P. van Oeveren, M. Beukering, drs. M.M. Bout, drs. A.J.C. Monquil, drs. N.H. Ypenburg, drs. H.R. Goudsmit, drs. J.K.W. Riksen, dr. H.B. Westbroek, drs. R. Cornelisse MSc, dr. S. Donszelmann, N. el Abdellaoui MSc, D.S. Hoefakker-Laanstra, dr. B. de Vries, prof. dr. ir. F.F.J. Hermans, dr. I. Pauw, drs. L.J. van Well-van Grootheest, drs. Y.G. Meindersma, dr. O.P.M. van Buuren, C.L. Geraedts, prof. dr. M. Meeter, drs. E.D. van Noort, dr. S.M. Calor, drs. M.A. Huijskes MA
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 1 doorloopt de lio het eerste deel van de praktijkstage onder begeleiding van een werkplekbegeleider (wpb). De stage voor Praktijk 1 omvat minimaal 48 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk gelden, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. Tijdens Praktijk 1 werkt de student o.a. toe naar het zelf geven van een hele les.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 1 voert de student tegen het eind van de eerste fase op de (opleidings-)school de opdracht ‘Presenteer jezelf’ uit. De student presenteert zichzelf als leraar in opleiding en voert daarbij bewijsmateriaal aan over zijn/haar functioneren in de praktijk gedurende de eerste fase. Vervolgens beoordeelt de werkplekbegeleider de student aan de hand van het Beoordelingsformulier Praktijk. Het functioneren van de student gedurende de stage én de inbreng tijdens de presentatie vormen input voor de beoordeling.

Additional Information Target Audience

Studenten met een baan (zij-instromers, onderwijstrainees etc) geven in dit stadium al zelfstandig les. Bij deze studenten ligt de nadruk bij de begeleiding vanuit de werkplekbegeleider op het niveau van didactisch handelen in de les.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 2

Course Code	O_MLPRAK_2F
Credits	9
Period	P5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	dr. I. Pauw, prof. dr. M. Meeter, dr. H.B. Westbroek, M. Beukering, dr. C.D.P. van Oeveren, C.L. Geraedts, D.S. Hoefakker-Laanstra, drs. L.J. van Well-van Grootheest, drs. Y.G. Meindersma, drs. M.M. Bout, drs. R. Cornelisse MSc, N. el Abdellaoui MSc, drs. H.R. Goudsmit, dr. B. de Vries, dr. S.M. Calor, drs. A.J.C. Monquil, drs. J.K.W. Riksen, drs. N.H. Ypenburg, prof. dr. ir. F.F.J. Hermans, drs. M.A. Huijskes MA, drs. E.D. van Noort, dr. O.P.M. van Buuren, dr. S. Donszelmann
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 2 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. 2 van de stage omvat minimaal 72 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk staan, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. De student werkt o.a. toe naar het zelf geven van een kleine lessenserie.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads

bevoegdheid). Voor studenten die de gehele eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de module geldt:

Voor de afronding van Praktijk 2 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’.

Voor de master geldt:

De werkplekbegeleider beoordeelt Praktijk 2 aan de hand van het Beoordelingsformulier Praktijk.

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 2. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 2 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 2: een voldoende beoordeling van Praktijk 1 en Didactiek 1.

Additional Information

Voorwaardelijk voor afronding van Praktijk 2: een voldoende beoordeling van Praktijk 1 en Didactiek 1.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 2

Course Code	O_MLPRAK_2
Credits	9
Period	P2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil

Teaching Staff	drs. A.J.C. Monquil, M. Beukering, drs. E.D. van Noort, dr. H.B. Westbroek, prof. dr. ir. F.F.J. Hermans, D.S. Hoefakker-Laanstra, C.L. Geraedts, dr. C.D.P. van Oeveren, drs. J.K.W. Riksen, dr. S. Donszelmann, drs. L.J. van Well-van Grootheest, drs. Y.G. Meindersma, drs. M.M. Bout, dr. B. de Vries, dr. I. Pauw, drs. H.R. Goudsmit, dr. S.M. Calor, dr. O.P.M. van Buuren, drs. R. Cornelisse MSc, N. el Abdellaoui MSc, prof. dr. M. Meeter, drs. M.A. Huijskes MA, drs. N.H. Ypenburg
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 2 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. Deel 2 van de stage omvat minimaal 72 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk staan, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. De student werkt o.a. toe naar het zelf geven van een kleine lessenserie.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de gehele eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor een modulestudent geldt het volgende:

Voor de afronding van Praktijk 2 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’.

Voor een masterstudent geldt het volgende:

Praktijk 2 wordt beoordeeld door de werkplekbegeleider.

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 2. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 2 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 2: een voldoende beoordeling van Praktijk 1 en Didactiek 1.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 3

Course Code	O_MLPRAK_3F
Credits	15
Period	P1+2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. H.R. Goudsmit, dr. B. de Vries, D.S. Hoefakker-Laanstra, C.L. Geraedts, drs. R. Cornelisse MSc, drs. Y.G. Meindersma, R.B. van der Vos, drs. N.H. Ypenburg, M. Beukering, drs. A.J.C. Monquil, drs. E.D. van Noort, dr. S.M. Calor, drs. L.J. van Well-van Grootheest, dr. O.P.M. van Buuren, dr. I. Pauw, dr. C.D.P. van Oeveren, R.M.C. Canisius, drs. M.M. Bout, drs. J.K.W. Riksen, prof. dr. ir. F.F.J. Hermans, prof. dr. M. Meeter, dr. H.B. Westbroek, dr. S. Donszelmann
Teaching method(s)	Seminar

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 3 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. Deel 3 van de stage omvat minimaal 120 klassencontacturen (van 50 minuten). Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. De student gaat meer en meer zelf(standig) lesgeven. De voorbereiding en evaluatie worden samen met de werkplekbegeleider gedaan. Ten minste 60 van de 120 lesuren worden door de student zelf verzorgd.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, dienen minimaal 120 lesuren door de student zelf te worden verzorgd, waarvan 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo). Het moet in ieder geval mogelijk zijn om de lessenreeks die ten tijde van Didactiek 3 wordt ontworpen (dat parallel loopt aan Praktijk 3) uit te voeren in een bovenbouwklas. Binnen Praktijk 3 zal daarom de bovenbouw veelal centraal staan.

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. Ook met onderwerpen die bij keuzevakken op het instituut ter sprake komen zal in de praktijk worden geëxperimenteerd. De werkplekbegeleider wordt in ieder geval per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur ook met de wpb en neemt vragen, ervaringen en inzichten mee naar de instituutsdag.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 3 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’. De VU-mentor legt ook een lesbezoek af (dit kan dezelfde ‘examenles’ zijn, maar hoeft niet).

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 3. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 3 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 3: een voldoende beoordeling van Praktijk 2 en Didactiek 2.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 3

Course Code	O_MLPRAK_3
Credits	15
Period	P4+5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. E.D. van Noort, drs. N.H. Ypenburg, prof. dr. ir. F.F.J. Hermans, drs. Y.G. Meindersma, dr. I. Pauw, dr. S. Donszelmann, D.S. Hoefakker-Laanstra, dr. S.M. Calor, dr. H.B. Westbroek, dr. O.P.M. van Buuren, M. Beukering, N. el Abdellaoui MSc, dr. B. de Vries, drs. M.M. Bout, drs. R. Cornelisse MSc, drs. A.J.C. Monquil, dr. C.D.P. van Oeveren, prof. dr. M. Meeter, drs. H.R. Goudsmit, drs. L.J. van Well-van Grootheest, C.L. Geraedts, drs. J.K.W. Riksen
Teaching method(s)	Lecture

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 3 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. Praktijk 3 van de stage omvat minimaal 120 klassencontacturen (van 50 minuten). Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. De student gaat meer en meer zelf(standig) lesgeven. De voorbereiding en evaluatie worden samen met de werkplekbegeleider gedaan. Ten minste 60 van de 120 lesuren worden door de student zelf verzorgd.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, dienen minimaal 120 lesuren door de student zelf te worden verzorgd, waarvan 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo). Het moet in ieder geval mogelijk zijn om de lessenreeks die ten tijde van Didactiek 3 wordt ontworpen (dat parallel loopt aan Praktijk 3) uit te voeren in een bovenbouwklas. Binnen Praktijk 3 zal daarom de bovenbouw veelal centraal staan.

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op

het instituut aan bod komen. De werkplekbegeleider wordt in ieder geval per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur ook met de wpb en neemt vragen, ervaringen en inzichten mee naar de instituutsdag.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 3 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’. De VU-mentor legt ook een lesbezoek af (dit kan dezelfde ‘examenles’ zijn, maar hoeft niet).

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 3. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 3 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 3: een voldoende beoordeling van Praktijk 2 en Didactiek 2.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Praktijkonderzoek

Course Code	O_MLPROZ
Credits	6
Period	P4+5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. H.B. Westbroek
Examiner	dr. H.B. Westbroek
Teaching Staff	dr. H.B. Westbroek
Teaching method(s)	Study Group, Lecture

Course Objective

Aan het eind van deze cursus heeft de student:

1. Kennis van en inzicht in het opzetten, uitvoeren en presenteren van kleinschalig actieonderzoek;
2. Vaardigheid in het in samenwerking met experts op het instituut en op school opzetten, uitvoeren en presenteren van kleinschalig actieonderzoek.

Concreet houdt dit in:

- Het vertalen van een praktijkprobleem naar een onderzoeksvraag;
- Het ontwerpen en uitvoeren van een kleinschalige dataverzameling;
- Het maken van een analyseplan en uitvoeren van (kwalitatieve en/of kwantitatieve) analyse;

- Het rapporteren van de gebruikte methode, de resultaten, conclusie en discussie in een presentatie.

Course Content

In het keuzevak Praktijkonderzoek staat het leren toepassen van onderzoeksvaardigheden in de onderwijspraktijk centraal. In dit keuzevak leer je een actieonderzoek opzetten en uitvoeren ten behoeve van je eigen professionalisering. Vaak wordt een professionaliseringsvraag breder gedeeld in je sectie/school. Actieonderzoek is een vorm van onderzoekend leren. Je gebruikt theorie en bevindingen uit onderwijsonderzoek om jouw praktijkprobleem te analyseren en om ontwerp/voornemens te formuleren. Je leert hoe je in je klassen data kan verzamelen over de invloed van jouw ontwerp, op de leerprocessen van je leerlingen.

Additional Information Teaching Methods

introductiecollege (1,5 uur), werkgroepen/intervisiesessies (7,5 uur), zelfwerkzaamheid en individuele begeleidingsgesprekken (157,5 uur), presentaties (1,5 uur).

Method of Assessment

Praktijkonderzoek wordt afgesloten met een presentatie. Tussentijds dient ook een onderzoeksplan te worden ingeleverd dat wordt beoordeeld (go/no go).

Entry Requirements

Vereiste voorkennis: Didactiek 2, Praktijk 1 en onderzoekservaring op bachelor/masterniveau in het eigen domeinvak.

Literature

- Literatuur uit Didactiek 1 en Didactiek 2 vormt een basis.
- Literatuur wordt aangeboden in de vorm van wetenschappelijke artikelen (CANVAS).
- Van studenten wordt verwacht dat ze aanvullend zelf literatuur zoeken en gebruiken om hun theoretisch kader en onderzoeksmethode te kunnen uitwerken en onderbouwen.

Additional Information Target Audience

Praktijkonderzoek is een van de specialisaties binnen de ULO lerarenopleiding.

Explanation Canvas

Alle informatie over de opzet van het vak, de bijeenkomsten, toetsing en literatuur wordt aangeboden via CANVAS. Communicatie gebeurt via CANVAS (discussieomgeving; mail).

Reflection Seismic for Geologists

Course Code	AM_450170
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. K.F. Kuiper
Examiner	prof. dr. K.F. Kuiper
Teaching Staff	prof. dr. K.F. Kuiper, dr. A. Beniest, dr. T.S. Gunatilake
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

Understanding the fundamentals of the application of reflection seismic to describe geology in the subsurface at depths of 200 m - 6 km. This includes:

- Seismic Sequence Stratigraphic interpretation
- Seismic Structural interpretation
- Seismic Acquisition and processing principles
- Calibration with petrophysical information/synthetic seismograms

- Time Depth Conversion methods
- Modern Techniques, Volume interpretation, Use of Attributes
- Examples/Special Case studies (4D time lapse, rock-property prediction)

During an extensive workstation Exercise (A/B), you will gain hands-on experience with subsurface interpretation. Here you will apply the theory to a real 3D seismic dataset, using an industry interpretation software package. The Exercise part of the course is done in teams of 2 or 3 persons. Teamwork is essential in this type of work and will also be gauged during the assessment.

Course Content

This course is part of the MSc Geology and Geochemistry stream "Subsurface Resourcing" which contains a series of elective courses that prepares for activities in the subsurface (geothermal energy, CO2 sequestration (CCS), hydrocarbon exploration and production). Many examples of successful techniques, originally developed for the oil and gas industry, are discussed. These techniques are nowadays often used in geothermal energy industry. Here you will learn how to use reflection seismic to interpret the subsurface and to make estimates of uncertainties that should be taken into account for a safe use of the subsurface.

Assuming a basic knowledge of the principles of reflection seismology, this course provides a modular program with principles and interpretation of reflection seismic data and geology, integrating data from wells and gaining hands-on experience in interpreting 3D seismic data. Special attention will be paid to pitfalls in data acquisition, processing and interpretation. During the course standard methodologies used in the industry will be applied.

The course consists of:

- Introduction to seismology. The introduction will cover the technical and methodological principles of reflection seismology. Also attention will be paid to the recognition and possible treatment of issues in seismic data quality and reliability.
- Structural interpretation. Students will learn how to interpret basic geological features, such as strata relationships, faults and folds as well as the reliability of seismic interpretation at various scales;
- Seismic Sequence Stratigraphy entails the extraction of stratigraphic, sedimentological and basin evolution information from seismic data. This information can be used in exploration and basin analysis to derive regional analysis of sedimentary basin-fills in order to construct models for gross lithology prediction (Students should review the principles of sequence stratigraphy, acquired during their BSc courses);
- Interpretation on a workstation. Students work on a case study using standard workstation methodologies and learn how to handle, visualize and interpret 2D and 3D seismic data using a standard industrial software package;
- Advanced seismic interpretation. This section will show examples of recently developed techniques in 3D seismic information analysis.

Additional Information Teaching Methods

The course uses two different methods:

- Oral lectures, where the lecturer presents various topics. Students must be aware that the content of this course is difficult to find in textbooks. Therefore, understanding the handouts is essential. Attendance to all oral lectures is compulsory.
- Workstation Exercises A and B. The practicals are done in teams of 2 or 3 students. Attendance of the practicals is compulsory. By the end of the course, students present the results of their team work in an oral presentation.

Method of Assessment

The final mark is made up by 50% from the result of the exam which will cover the topics presented during course. The oral presentation of the workstation exercise contributes another 50 % to the final score.

Literature

All materials will be digitally provided through Canvas.

Additional Information

Teaching staff: Guest lecturers.

Research Project Earth & Climate

Course Code	AM_1265

Credits	24
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. H.J.L. van der Lubbe MSc
Examiner	dr. H.J.L. van der Lubbe MSc
Teaching Staff	dr. H.J.L. van der Lubbe MSc, dr. J.E. Vonk
Teaching method(s)	Seminar

Course Objective

Prepare and conduct a research project and write a scientific report at the academic Master's level. In both practical conduct and writing the student should demonstrate his or her:

- advanced knowledge and understanding of the field, thus providing a basis or opportunity for originality in developing and applying ideas;
- problem solving abilities in new or unfamiliar environments within broader contexts;
- ability to integrate knowledge and to handle complexity;
- ability to clearly communicate conclusions, and the knowledge and rationale underpinning these, to specialist and non- specialist audiences.

Course Content

Research project in the master's specialisation Earth & Climate: circa 1 weeks preparation, circa 8 weeks research, circa 7 weeks report.

Additional Information Teaching Methods

Fieldwork and/or lab analysis or meta data analysis. The student must present and document his/her results in a report.

Method of Assessment

Written report. Grading is based on the report, methodology, research attitude and presentation

Entry Requirements

At least 18 EC should be successfully acquired including the mandatory course: "Designing Interdisciplinary Research (AM_1264)

Additional Information Target Audience

The Research Project is subject to the school's Work placement and thesis regulations (stageregeling). These regulations require detailed written agreements between supervisors and student that specify the conditions for the Master thesis work placement or research project. This agreement should be put forward to the secretary of the examination board before the start of the work placement or research project.
Admission to this Research project is only granted to students with a BSc degree Earth Sciences and other students who have been admitted to the MSc degree programme by the Examination Board. Admission requirements are checked by the Examination Board on April 15th. Participants should register in time (preferably before March 1st) with the staff member/department in question and via the regular way.

Research Project Geology and Geochemistry

Course Code	AM_1267
Credits	24
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Koornneef
Examiner	dr. J.M. Koornneef

Teaching Staff
Teaching method(s)

Course Objective

Learn to perform and conduct an individual research project culminating in a written report and presentation at the academic Master's level. In their conduct, working methods, attitude, collaboration, writing and other communications, the student should demonstrate their:

- advanced knowledge and understanding of the field, thus providing a basis or opportunity for originality in developing and applying ideas;
- problem solving abilities within broader contexts;
- ability to integrate knowledge and to handle complexity;
- ability to clearly communicate conclusions, and the knowledge and rationale underpinning these, to specialist and non-specialist audiences.

Course Content

The student individually performs a guided research project of the master’s specialization Geology and Geochemistry. The project has a volume of 24 EC, which equals 16 fulltime student working weeks (42 hours/week); a total of 672 working hours.

Additional Information Teaching Methods

Research may take place in one or more of the VU Earth Science laboratories, may involve fieldwork, and could involve visits to external partners or an internship.

Method of Assessment

Final grade includes marks for the written report, execution of the project, and oral presentation.

Entry Requirements

Admission to this research project is only granted to students with a BSc degree Earth Sciences and other students who have been admitted to the MSc Earth Sciences degree program by the Examination Board (any specialization). Admission requirements are checked by the Examination Board on April 15th. Participants should register in time with the staff member/department in question and via the regular way. The student must have completed the Designing Interdisciplinary Research course (AM 1264), which serves as an introduction to the research project.

Additional Information Target Audience

First year Geology and Geochemistry specialisation students.

Additional Information

Students need to fill out an online form (Agreement Form) detailing contents and supervision of their projects, which needs to be approved by the MSc Geology and Geochemistry coordinator before the start of the project. The approved agreement form outlines the goals of the research project and the project timeline that need to be followed to complete the project. In case of deviations from the timeline, a revised agreement form needs to be completed and approved before finalizing the project.

Either the Research Project Geology and Geochemistry (AM_1187) or Master Thesis Geology and Geochemistry (AM_1186) may be extended by 12 EC (AM_450149 Extension Master Thesis Earth Sciences). An extension must be approved before the start of one of these projects by the MSc Geology and Geochemistry coordinator. The extension represent either (1) as a separate chapter of the MSc research/thesis report, or (2) as a stand-alone report. Like for the MSc research and thesis projects, a separate agreement form needs to be completed beforehand and approved for the extension.

Research Proposal Master Earth Sciences

Course Code	AM_1321
Credits	15
Period	P5+6

Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. I.M. Lehmann
Examiner	dr. I.M. Lehmann
Teaching Staff	A.H. Levering, dr. I.M. Lehmann
Teaching method(s)	Study Group

Course Objective

The main objective of this course is for students to develop a hands-on experience on designing a solution-oriented interdisciplinary research proposal. The course thereby builds directly on the course ‘Designing Interdisciplinary Research.’ Students will work on the different phases and critical aspects of proposal development, from idea to submission.

By the end of this course students should be able to:

- Evaluate and assess the quality of research proposals
- Develop creative and innovative interdisciplinary research ideas
- Translate cutting-edge research ideas into solution-oriented research plans
- Synthesize and communicate ideas orally and in written form
- Work in teams over a prolonged period of time

Course Content

An important skill in the academic world and beyond is to be able to acquire funding by putting forward a feasible and relevant research or project proposal. Learning to design, write and submit research and, or, project proposals is becoming a key skill in the professional development of academic scholars and other experts in the public sector, NGOs and consultancy arenas. The course teaches students the requisite knowledge as well as cognitive and interpersonal skills to write an interdisciplinary research proposal in the field of sustainability studies. Moreover, this course will increase students’ understanding of selected environmental problems, the state-of-the-art of research on these problems and potential solutions as well as organizing themselves in a group (e.g., sharing the workload between them).

Additional Information Teaching Methods

We will mix lectures, discussions and group activities. Key aspects of this course are related to presenting work in progress, giving and receiving feedback, and exchanging ideas and critical opinions.

Method of Assessment

The final assignment will be a written team-based research proposal with an individual component. Leading up to the final assignment, there will be presentations and written assignments. Some of these will be group, some will be individual assignments.

Science and Communication

Course Code	AM_470587
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C. Boersma MA MSc.
Examiner	dr. C. Boersma MA MSc.
Teaching Staff	dr. C. Boersma MA MSc., T.F.L. Roedema MSc, L.B. Clauss MSc, K.E. De Long MSc
Teaching method(s)	Study Group, Lecture

Course Objective

- Gain theoretical insight in the nature of science,

- Gain theoretical insight in the nature of communication,
- Gain theoretical insight in the relationship between science and society,
- Gain insight in the role of science communication in this relationship,
- Acquire knowledge of different theories and models of science communication,
- Acquire knowledge of different strategies, media and activities for science communication,
- Learn how to practically apply theoretical concepts from the field of science communication in communicating science,
- Develop practical skills for science communication (especially writing and giving oral presentations).
- Reflect on your own knowledge and competencies pertinent to your projected (ideal) role as science communicator.

Course Content

Science is all around us and shapes our lives in many different ways. From the vaccines you need to get when traveling abroad to the smartphone you use on a daily basis, and from the public transportation you use to get to the university to the ingredients of your toothpaste: scientific knowledge is elemental to all of these. Simultaneously, society shapes the ways in which science and technology develop too. Science, technology and society influence each other continuously—or, to put it differently, they ‘communicate’.

Students of the Science Communication specialization are expected to become experts in understanding and designing interactions between science and society. In order to make this interaction fruitful and valuable for both science and society, it is first of all important to gain theoretical knowledge about science, about communication and about science communication. Science and Communication provides students with the theoretical and conceptual foundations of the discipline of science communication. Thus, you will develop an in-depth understanding of communication processes at the core of several interfaces, including those between scientists from different disciplines, between different sciences and their stakeholders, and between science and the public.

Additional Information Teaching Methods

Expected contact hours:
Lectures (20 h)
Workgroups (25 h)
Home-study for group assignments (15 h)
Home-study for individual assignments/exam (100 h)

Method of Assessment

a) Participation. (pass/fail)
This consists of the following:
- the 'CARQ' presentations and summaries;
- Active participation in the work groups.
All these are assessed as pass or fail. For each one you fail, you have to do an alternative assignment.
Nota bene: if you fail your participation, this cannot be compensated with an alternative assignment!

b) An interview with a science communicator of your own choosing, including a reflection report. (20%)
d) "TED-talk" in which you present the research you did (e.g. for your Bsc thesis or (first) Msc internship) and reflect on it. (20%)
e) Exam. (60%)

To pass, your grades for the assignments and exam all need to be 5.5 or higher.

Resit: In case your grade is not sufficient, you have to take a resit. This can either consist of a second attempt at (b), (c) or (d), or a re-exam.

Literature

Academic articles. Direct links to articles will be provided on Canvas.

Additional Information Target Audience

The course Science and Communication is a compulsory course for students of the Master specialisation Science Communication (Wetenschapscommunicatie) and is a prerequisite for the internship. Science and Communication is an optional course for students from other master programs in the health and life sciences.

Science Journalism

Course Code	AM_471014

Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C. Boersma MA MSc.
Examiner	dr. C. Boersma MA MSc.
Teaching Staff	A.M. Aris, K.E. De Long MSc, dr. C. Boersma MA MSc., T.F.L. Roedema MSc
Teaching method(s)	Computer lab, Study Group, Lecture

Course Objective

After following this course, you are able to:

- Review the key changes in science journalism due to changes in the science-society relationship and digitalization of the media
- Examine contemporary issues of science journalism including the emergence of issues and publics, framing, values and emotions, expertise and evidence, uncertainty and ignorance.
- Apply an interdisciplinary toolkit to think about the interactions between science, media and the public and the role of science journalism
- Useyour skills in different science communication media, such as writing, video, and documentary to generate meaningful exchange between science and society
- Describe what the professional practice of science journalism entails

Course Content

Science journalism is needed now more than ever. Science heavily influences our society, not only through the application of its knowledge but also through its worldview. Although science and technology have shown the potential to improve our lives and living conditions significantly, they also contribute to new problems, risks and concerns. This awareness has changed the science and society relationship and put science and technology under increasing public scrutiny. In the context of the complex challenges of today, such as climate change or food security, ‘the facts’ are regularly called into question. Public trust in the solutions presented by science is under pressure. There is clear demand for the scientific world to communicate effectively with the rest of society. But how to shape the public conversation about science in society? And how do the media shape our perception and appreciation of science? What is the role of science journalism? Especially with respect to the larger transformations that are taking place in the relationship between science, media and public. Digitalization has revolutionized the media landscape. Everyone now has the means to either find or generate any information themselves. This provides new opportunities, but may also spread low-quality, sensationalized and sometimes intentionally misleading information. The challenge for science journalism today is to reinvent its practice and help shape a meaningful exchange between science and society.

In this course we review the key changes in science journalism. We examine how changes in both the science-society relationship and digitalization of the media landscape have changed the science journalism practice. We look at different roles of journalism such as reporters, storytellers, watchdogs and educators and discuss what new roles might need to be developed in light of the abovementioned changes. We analyze the communication and interaction dynamics in the current media landscape according to a set of core questions: How do issues and publics emerge? What is the role of framing? What is the role of values and emotions? What is the role of expertise, evidence? What is the role of uncertainty and ignorance? Finally, we train specific skills that you need as a science journalist, such as popular writing, making popular science videos, interviewing, and program design. Throughout the course, we work together closely with professionals from the field. Our guest speakers work as freelancer, editor or producer at diverse science media, such as newspapers, broadcasting media, online magazines and documentary film. They contribute to our course via the lectures, assignments and discussions.

Additional Information Teaching Methods

Lectures and seminars on theory and practice of science journalism and skill training (36h). Considerable time is set aside for performing science journalism in assignments (108h). All sessions indicated as training sessions in the schedule are mandatory. The assignments are assessed by lecturers and fellow students (peer-review process). Self-study (remaining hours).

Method of Assessment

Several individual assignments (60%), several small group assignments (40%).
All assignments must be passed (grade > 6).

Literature

Announced on Canvas one month before start of the course

Additional Information Target Audience

All Master students with a Beta-Bachelor degree. Students taking this course as part of their VU Science Communication Specialisation or UvA Major Science Communication will have precedence over other students. Students from other faculties and or universities need to get formal consent from the course coördinator (Frank Kupper) before enrolment.

Additional Information

The course is taught in English. Because language proficiency is crucial, there is the option to hand-in assigments in Dutch. More information: t.f.l.roedema@vu.nl.

Source to Sink

Course Code	AM_1302
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.C. Roda Boluda MSc
Examiner	dr. D.C. Roda Boluda MSc
Teaching Staff	dr. D.C. Roda Boluda MSc, dr. B. Campforts
Teaching method(s)	Computer lab, Lecture

Course Objective

This course combines Tectonic Geomorphology (i.e. “the source”) and the sedimentology of the Sedimentary Basins (i.e. “the sink”).

The learning objectives of the course are to:

1. Analyse and quantify how tectonics and climate influence the evolution of sediment routing systems, from source to sink: (a) landscape evolution and associated sediment production in mountain landscapes (sources), (b) sediment transfer along fluvial systems, and (c) the infill of sedimentary basins (sinks).
2. Learn how to extract information from landscapes and sedimentary basins to reconstruct past tectonic and climatic conditions, and characterize active tectonics.

To achieve these objectives, the course involves training in:

- Understanding the interplay of (physical) mechanisms responsible for the evolution of sediment routing systems, with a focus on tectonics, climate and their interaction.
- Integrating different data sets and geochronological techniques to understand the dates & rates of processes controlling landscape evolution and sedimentary infills, and understanding the strength and limitations of these methods.

As well as in transferable skills such as:

- Formulating hypotheses, and critically evaluating data to test them.
- Combining class material and independent, critical reading of literature.
- Working in small interdisciplinary groups.
- Communicating effectively in oral and written form.

Course Content

The main topics addressed in this course are:

- Key concepts in sediment routing systems from source to sink, and the interplay of tectonics, climate and surface processes, for 10^0-10^6 year timescales
- The methods that can be used to quantify dates, rates and sediment provenance, and test hypotheses on the evolution of Earth's surface
- The main processes controlling the morphology and evolution of mountain topography, hillslopes and rivers, as well as their associated sediment production, and the information we can extract from them.
- The main controls on erosion, sediment fluxes and sedimentary basins infill, stratigraphic architecture, and

preservation, and the information we can extract from these archives.
- Sediment routing system's transient responses, signal propagation and preservation, and autogenic processes, as well as how to interpret these on the stratigraphic record.
- Current debates and open questions on how sediment routing systems responded to and record past tectonic and climatic changes.

Additional Information Teaching Methods

Lectures, seminar-style paper discussions, quizzes, presentations, computer practicals, and home assignments involving exercises, reading and report writing.

Method of Assessment

- Individual presentation on a paper: 15%
- Querying and leading the Q&A on another paper presented by a colleague: 10%
- Assignnements on landscape evolution modelling sessions: 15%
- Group presentation on roundtable debate session: 15%
- Report on computer practical: 15%
- Final written exam: 30%. A minimum grade of 5.5 is required to make an average with the other assessed components and pass the course

Literature

The course content will mainly be based on these books:

- Tectonic Geomorphology, D.W. Burbank and R.S. Anderson, 2nd edition, 2012. John Wiley & Sons
- Geomorphology: the mechanics and chemistry of landscapes, R.S. Anderson and S.P. Anderson, 1st edition, 2010. Cambridge University Press.
- Key Concepts in Geomorphology, P.R. Bierman and D.R. Montgomery, 2nd edition, 2020. McMillan Learning.
- Allen, P.A. and Allen, J.R. (2004). Basin Analysis. Blackwell Publishing. ISBN: 978-0-470-67376-847
- Allen, P.A. (2017) Sediment Routing Systems. Cambridge University Press. ISBN: 9781107091993

These readings are not mandatory but would provide you with extra information that can help you getting a deeper and broader understanding on the subject.

A reading list of published papers for each session, some for mandatory and some for optional reading, will also be provided on Canvas.

Additional Information Target Audience

Master students of the Earth Sciences Geology & Geochemistry program, Master students of the Earth & Climate program

Sustainable Energy Systems

Course Code	AM_468018
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. M.A. Poblete Cazenave
Examiner	
Teaching Staff	dr. M. Werthschulte, dr. ir. M.A. Poblete Cazenave
Teaching method(s)	Study Group, Lecture

Course Objective

The goal of the course "Sustainable Energy Systems" is to introduce students to key aspects of the energy transition from a systemic socio-economic perspective. Besides a brief introduction to the technological aspects of the energy transition, the lectures will focus on the economic, societal, and political conditions for sustainable energy generation, distribution, and use. The applied component of the course will stimulate the students to critically reflect on the positions of various stakeholders in the energy transition and to develop their own ideas for business, community-oriented projects or public policies that could accelerate the energy transition.

After following this course, students should be able to:

1. Understand the basic technical and economic principles underlying today's energy generation, distribution, and consumption around the globe and have an idea of the possible pathways into a sustainable energy future.
2. Understand the economic, social, and environmental challenges related to the energy transition and critically discuss possible solutions and strategies for the decarbonization challenge.
3. Know about a diverse set of policy instruments and strategies that can be implemented to support the energy transition.
4. Think critically about efficiency, effectiveness, fairness, and distributional consequences of different principles, policy instruments, and approaches to the energy transition. Learn to recognise trade-offs and potentially competing objectives.
5. Identify niches for sustainable energy innovations, understand the main components of a successful business model or community project and apply this in developing a business plan or community project for a clean energy innovation.
6. Understand stakeholders' positions and how to constructively debate to find synergistic solutions.
7. Demonstrate the capacity to operate in a multidisciplinary team and constructively contribute to the development of a shared project.
8. Synthesize information into a pitch for a non-academic audience.

Course Content

Environmental externalities like CO2 emissions leading to global climate change and air pollution urge the world to transition towards a more sustainable and less-carbon intensive renewables-based energy system. Energy is central to nearly every major challenge and opportunity the world faces today, such as health, employment, education, or food production. Furthermore, access to clean and sustainable energy for all is essential and part of the UN Sustainable Development Goals (UN, 2015). The transition towards an affordable, reliable, and sustainable energy system can be accelerated by investing in renewable energy sources, prioritizing energy efficient technologies and practices, and by building a clean energy infrastructure, but changes to the physical infrastructure need to go hand in hand with economic, societal, and behavioural changes.

The course "Sustainable Energy Systems" is built on a series of lectures and seminars by VU lecturers and selected guest speakers to introduce students to key aspects of the energy transition from a socio-economic and socio-technical perspective. Every lecture will be a combination of theory and evidence-based discussions, relating theoretical arguments with recent experiences and examples in the domain of the transition to sustainable energy.

The course is built on 4 main blocks:

1. The socio-technical systems perspective of the energy transition (socio-technical pathways for greening energy production, distribution and use)
2. Sustainable energy demand innovations and deployment: from available energy efficient technologies to public programmes and policies to international agreements
3. Energy markets and policies (economics and policy instruments to support the energy transition, future electricity markets)
4. Consumer behaviour and social aspects (sustainable energy consumer behaviour, energy poverty, energy access)

Additional Information Teaching Methods

The course is worth 6 ECTS credits which corresponds to 168 hours of work per student. The course comprises two sets of activities. The first takes place in class, where information is presented through lectures, presentations, debates, seminars, etc. The second includes assessments where the student's ability to achieve the course's objectives is tested. Assessments involve group assignments (presentation and written assignments), and the exam. Feedback opportunities are included through meetings as well as assessments. Approximate time allocation: • Class: 28 hours • Reading and exam preparation: 70 hours • Assignment: 60 hours • Presentations: 10 hours.

Method of Assessment

Type of assessment:

- Two group written assignments (A) worth 30% of the final grade.
- One group project involving a written assignment (A) and presentation (Pres) worth 40% of the final grade.
- Written exam (E) worth 30% of the final grade.

Minimum grade to pass the course: 5.5 in each assessment component. It is not possible to compensate one component with another. It is compulsory to attend the group presentation session.

Sustainable Mineral Resources

Course Code	AM_1289
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. L. Zhou
Examiner	dr. L. Zhou
Teaching Staff	dr. L. Zhou
Teaching method(s)	Excursion, Seminar, Computer lab

Course Objective

Mineral resources, the source of primary metals, play a key role in enabling the green technologies for energy transition. This 6 EC elective course aims to provide students with a comprehensive view of the critical role of mineral resources in contemporary society, and allow students to better define their role as geoscientists in the energy transition.

Students will study the complexity and vulnerability of metals supply chain, the criticality of metals gauged by their economic importance and supply risks, the fundamental geology of critical mineral resources, and the sustainable practices taken by both minerals industry and academia.

Furthermore, the course comprises a series of guest lectures and a unique assessment method. Students will be able to develop skills to effectively communicate with professionals, and to write a detailed research proposal that demonstrates their academic competence.

Course Content

Class 1: Mineral resources and society
Class 2: Metals supply chain
Class 3: Metal criticality
Class 4: Responsible sourcing for metals
Class 5: Mineral deposit genesis and types
Class 6: Geology of rare earth element mineral deposits
Class 7: Geology of cobalt and nickel mineral deposits
Class 8: Geology of lithium mineral deposits
Class 9: Unconventional sources for metals
Class 10: Mineral processing and metal extraction methods
Class 11: Mineral exploration methods
Class 12: Looking into the future (students' view and presentations)

Excursion; lab practical; a series of guest lectures.

Additional Information Teaching Methods

- Lectures with in-class exercises and assignments
- Practical: ore hand specimen identification
- An oral presentation by individual students on a selected subject throughout the course
- A site visit to an industrial site

Method of Assessment

- Research proposal (50%)
- Oral presentation (20%)
- Assignments (20%)
- Engagement during the course (10%)

Entry Requirements

BSc Earth sciences

Literature

Students are recommended to read selected chapters from the following textbooks to gain a preliminary understanding of ore deposit geology.

- Ridley John. Ore Deposit Geology. Cambridge University Press, 2013.
- Laurence Robb. Introduction to Ore-Forming Processes, 2nd Edition. Wiley-Blackwell, 2020.
- Friedrich-W. Wellmer, Peter Buchholz, Jens Gutzmer, Christian Hagelüken, Peter Herzig, Ralf Littke, Rudolf K. Thauer. Raw Materials for Future Energy Supply. Springer, 2018.

A selection of papers and government and industry reports will be listed on Canvas.

Additional Information Target Audience

The course is open to all MSc students in Earth Sciences, as well as other groups of MSc students with research interests in energy critical metals and metal supply chain.

Thesis Project Earth & Climate

Course Code	AM_1228
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. H.J.L. van der Lubbe MSc
Examiner	dr. H.J.L. van der Lubbe MSc
Teaching Staff	dr. J.E. Vonk, dr. H.J.L. van der Lubbe MSc
Teaching method(s)	Seminar

Course Objective

Prepare and conduct an independent research project, or successfully fulfil a work placement/traineeship in trade, industry, government or otherwise; and to write a scientific report at the academic Master's level. In practical conduct, working methods, attitude, collaboration, writing and other communication the student should demonstrate his or her:

- advanced knowledge and understanding of the field, thus providing a basis or opportunity for originality in developing and applying ideas;
- problem solving abilities in new or unfamiliar environments within broader contexts;
- ability to integrate knowledge and to handle complexity;
- ability to clearly communicate conclusions, and the knowledge and rationale underpinning these, to specialist and non-specialist audiences.

Course Content

Research project or work placement (traineeship) in the master's specialization Earth & Climate; with a volume of 27 EC and related master thesis report.

Additional Information Teaching Methods

Research project or work placement/traineeship.

Method of Assessment

Written report/thesis, research methodology, attitude and presentation

Entry Requirements

The Research project Earth & Climate should be completed before the start of the thesis.

The Master thesis is subject to the school's Work placement and thesis regulations (stage- en scriptieregeling). These regulations require detailed written agreements between supervisor and student that specify the conditions for the Master thesis work placement or research project.
This agreement should be put forward to secretary of the Examination Board before the start of the work placement or research project.
The master thesis work placement or research project may be extended by a volume of 12 EC using the optional subject 'Extension of master thesis
in Earth Sciences' (450149). Information on Master thesis projects is provided by departmental lecturers and is made available on the departmental pages of the Faculty website.

Tweetalig Onderwijs

Course Code	O_MTTO
Credits	3
Period	P2
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. S. Donszelmann
Examiner	dr. S. Donszelmann
Teaching Staff	dr. S. Donszelmann
Teaching method(s)	Seminar

Course Objective

Na afloop van het vak heb je een beeld van het Tweetalig Onderwijs (TTO) binnen het Nederlandse voortgezet onderwijs en beschik je over basale vaardigheden om de Engelse taal op een zinvolle manier als instructietaal én als leertaal in te zetten.

Course Content

Op hoe langer hoe meer scholen wordt tweetalig onderwijs aangeboden (TTO). Het doel van dit type onderwijs is dat leerlingen tegelijk met vakinhoud ook Engels leren. Soms pakt dit goed uit, maar het kan ook zijn doel voorbij schieten. Hoe zit dit?

Het gebruik van de doeltaal (Engels in dit geval) leidt niet bij voorbaat tot het leren van Engels. Leerzaam inzetten van de doeltaal is gebonden aan allerlei didactische voorwaarden waarvan de docent zich vaak maar beperkt bewust is. Het gevolg kan heel goed zijn dat leerlingen bij veel vakken nauwelijks Engels bijleren ten opzichte van het Engels dat zij al beheersten, en bovendien minder vakinhoud meekrijgen dan in het Nederlands het geval zou zijn omdat de docent zich omslachtig in een vreemde taal uitdrukt. Doeltaal-Leertaal onderzoek (Dönszelmann, 2019) geeft inzicht in de vraag welk didactisch instrumentarium er nodig is om leerlingen op leerzame manier onder te dompelen in de vreemde taal, terwijl er tegelijkertijd ook vakinhoud wordt overgebracht.

Binnen het keuzevak onderzoeken we wat dit betekent voor de TTO-praktijk binnen Nederlandse VO-scholen, en hoe goed TTO-onderwijs eruit moet zien.

Additional Information Teaching Methods

Werkcolleges

Method of Assessment

Kort literatuurverslag, of onderzoek binnen de eigen praktijk (als hier TTO-onderwijs voorkomt).

Entry Requirements

Didactiek 2, Praktijk 1

Literature

Wordt via Canvas bekend gemaakt.

Additional Information Target Audience

Dit keuzevak is bedoeld voor studenten van de VU lerarenopleidingen (VU Lerarenacademie). Ook studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten

Custom Course Registration

Er geldt een maximum van 5 voor bijvakstudenten van andere lerarenopleidingen.

Vernieuwende onderwijsconcepten

Course Code	O_MLVNON
Credits	3
Period	P2, P5
Course Level	500
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	prof. dr. M. Meeter
Examiner	prof. dr. M. Meeter
Teaching Staff	prof. dr. M. Meeter, dr. H.B. Westbroek
Teaching method(s)	Seminar, Lecture

Course Objective

Na het volgen van dit keuzevak, kun je:

1. beschrijven hoe het onderwijs in Nederland de afgelopen eeuwen vorm gekregen heeft en hoe met het onderwijs antwoord gegeven is op de vragen: wat willen we onze kinderen leren; waartoe dient onderwijs en hoe geven we onderwijs vorm?
2. een aantal vernieuwende vormen van VO-onderwijs opsommen (traditionele vernieuwende onderwijsconcepten, zoals Montessori, Dalton, Vrije School en hedendaagse vernieuwende onderwijsconcepten zoals gepersonaliseerd leren/Kunskapsskolan, Agora) en beschrijven wat de ideeën achter de vernieuwingen zijn.
3. een eigen leervraag over vernieuwende onderwijsconcepten en/of processen die daarbij horen formuleren en vragen bedenken die je tijdens het schoolbezoek aan docenten en leerlingen wilt stellen om de leervraag te kunnen beantwoorden.
4. de eigen leervraag beantwoorden met gebruikmaking van theorie, gesprekken met docenten en leerlingen, de bijeenkomsten en het schoolbezoek en een onderbouwd antwoord geven op de vraag: Wat vind ik belangrijk bij het onderwijs dat ik verzorg? Wat spreekt mij aan?
5. een begrippenkader hanteren waarmee je kunt nadenken en praten over vernieuwingsprocessen en vernieuwende onderwijsconcepten.
6. een beredeneerd oordeel vormen en een professionele dialoog voeren over voor- en nadelen van vernieuwingen en ongeteste aannames en kritische implementatiedetails benoemen.

Course Content

In het Nederlandse onderwijs, zeker in competitieve markten als de Amsterdamse, wordt continu vernieuwd. Scholen sturen bijvoorbeeld aan op het implementeren van gepersonaliseerd leren of projectmatig werken of onderwijs in gemengde stamgroepen. Bij dit vak bezoeken we een aantal vernieuwingsscholen en analyseren we de beweegredenen en de proceskant van onderwijsvernieuwing (de implementatie).

Additional Information Teaching Methods

Werkcolleges (waarin de dialoog centraal staat en waarin je collega's uit het de scholen interviewt) en schoolbezoeken.

Method of Assessment

Analyse van een schoolvernieuwing in de vorm van een academisch essay of in de vorm van een podcast of educlip (keuze) aan de hand van wetenschappelijke- en vakliteratuur, interviews, observaties en schoolbezoeken.

Entry Requirements

Didactiek 1 en 2, Praktijk 1

Literature

Informatie over te lezen artikelen wordt gedeeld op Canvas.

Additional Information Target Audience

Dit keuzevak is bedoeld voor studenten van de VU lerarenopleidingen (VU Lerarenacademie). Ook studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten.

Water Politics and Governance

Course Code	AM_1192
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J. Dell'Angelo
Examiner	dr. J. Dell'Angelo
Teaching Staff	dr. J. Dell'Angelo
Teaching method(s)	Lecture

Course Objective

Water is central for ecosystems, human wellbeing and development. The relationship between society and water resources is critical and complex. Water issues have been historically treated as a technical problem, something to be resolved through hydraulic and engineering solutions. However, the increasing and exacerbating competition over water resources due to expanding global population, climate change and conflicting production activities raises new political and democratic challenges. Understanding water governance implies developing multidisciplinary knowledge about the different problems, approaches, perspectives and controversies related to how societies cooperate and compete over water resources. This course aims at building students' knowledge, critical thinking and analytical skills on the governance of freshwater resources with particular emphasis on the institutional, political and organizational dimensions.

Course objectives:

The main objective of this course is for you to develop an expert understanding of the multidimensional challenges of governance of water resources. Collectively as a class, and individually, you will develop expertise in the water governance world by engaging with the core scientific literature, discussing and tackling real world problems and independent research activities. By the end of the course, you will have gained an understanding of the complexity of water governance: the various models of governance, its structures and institutions at multiple scales, the different values and uses of water, and the main frames and debates from multiple perspectives.

By the end of this course you will be able to:

1. Describe various principles and theories for water governance and □management
2. Apply these to analyze different water policy and governance interventions, by analyzing them from diverse and diverging perspectives
3. Formulate your own analysis and suggestions regarding water-governance interventions on multiple scales: local, national and international
4. Synthesize and communicate your ideas and expertise orally and in written form

Course Content

Tentative program which can be revised, main themes are:

- Water politics and critical perspectives
- Neoliberalism and privatization
- Commons Theory and Community Irrigation Schemes
- Water Law and Water Rights
- NEXUS, Virtual Water, Water & Energy, Water and Mining
- International cooperation and water conflicts
- Global virtual water trade, water grabbing, water war and peace

Additional Information Teaching Methods

This course will be intensive, during three weeks you will be "immersed" in water governance studies. We will spend an important amount of the course in the classroom but you will be primarily required to study the readings to prepare for lectures and discussions. We will have three weekly meetings and each day we will mix: frontal lectures, discussions and group activities.

The structure of each session, generally, consists of lectures at the start, where we will first highlight concepts and key questions related to the topics we are working on. We will then engage in-group discussions associated with the key issues addressed in each session. The group activities will include simulations and team assignments and presentations. Readings will be assigned for each session and it will be essential to be well prepared in order to successfully participate and address key questions that we will tackle in the discussions. Finally, writing exercises, group activities, and debates will be an integral part of the course too.

Participation and preparation: Due to the complexity of the issues addressed and the concentrated length of the course I highly recommend attending every session. Of course, if you will have serious family or health reasons you can count on our help for making up for your absence and catching up, but otherwise it will really not be in your interest to miss classes. You are kindly invited inform the teaching assistant in case you cannot attend one session. The amount of the material that we will go through and the very short time to prepare between the end of classes and the final exam will not allow you to be successful in this course if you procrastinate.

We will have group discussions on key questions related to the readings lead by myself and other teaching assistants. In the discussions and lectures we will call on students directly to address specific question during the group discussion. Referring to the “Socratic method” you might be challenged to defend your informed opinion and perspective on specific subjects. (This is a method that might seem a little intimidating, but is particularly useful to develop the capacity to debate publicly and defend your opinions – in line with the learning goals of the course).

Preparation advice: As MSc students you are very qualified, motivated and brilliant therefore some of this advice might be unnecessary. However, I can’t stress enough the importance to hit the ground and start running from day one, actively participate and not fall behind...keep in mind it’s only 3 weeks of classwork. A critical aspect is that you make an effort to study the assigned readings before class. If you don’t you will see pages adding up quickly and you not be as able to engage productively in discussions and have less information and tools to understand the lectures.

Method of Assessment

- Class participation, public debated, discussion and presentations.
- Water documentary review: This is a team project that includes a. search and selection of water documentaries; b. a written review of one selected documentary; c. a presentation to the class.
- In class final exam: there will be a closed-book 2.15h digital exam prevalently with open ended questions.

Literature

Selected articles and book chapters available in Canvas, organized around the above topics. There is no textbook for this class. For the most part, we will use scientific peer reviewed articles and, in certain cases, policy reports or other kind of documents. As we will be using scientific articles I will provide instructions on what I expect you should be focusing.

There will also be an additional number of suggested readings for the people interested in learning more about specific topics. Also, please note that the reading list available here in some rare cases might change during the course but you will all be clearly notified. You will find the final readings list every week and all the required readings in weekly folders on Canvas.

Additional Information Target Audience

Master level students with an interest in understanding the role of governance in the emergence of water issues, and their potential resolution

Additional Information

The course is coordinated and taught by Dr. Jampel Dell'Angelo (Associate Professor of Water Governance and Politics) with invited guest speakers. Language of teaching is English.

Custom Course Registration

Through the regular VU channels

Recommended background knowledge

no specific background required, just an eager motivation to engage, debate, learn and READ (a lot)

Water Risks

Course Code	AM_1210
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. ir. A.F. van Loon

Examiner	prof. dr. ir. A.F. van Loon
Teaching Staff	prof. dr. ir. A.F. van Loon, dr. J.A. de Bruijn
Teaching method(s)	Seminar, Lecture

Course Objective

The objective of this course is to provide students with a thorough understanding of concepts, datasets, methods, and applications in flood and drought risk assessments and management.

By the end of the course, students should be able to:

1. Review the scientific literature in order to discuss consecutive / co-occurring / compound floods and droughts, including hazard, exposure, vulnerability, and adaptation.
2. Discuss flood and drought risk reduction measures and strategies in case studies and propose improvements to current disaster risk assessment and management.
3. Describe the main datasets and methods (e.g. conceptual, statistical, computer simulation models) used to represent flood and drought hazard, exposure, and vulnerability, and explain their pros and cons.
4. Apply the main datasets and methods used to assess flood and drought hazard, exposure, and vulnerability.
5. Integrate data and information on hazard, exposure, and vulnerability to analyse flood and drought risk.

Course Content

Risk is defined as the probability of a natural hazard multiplied by its consequences, and can be seen as the product of the hazard, exposure, and vulnerability. In this course, students will learn the main concepts of water risk, how these water risks can be managed, and how these fit into international agreements on disaster risk reduction and recovery. They will also learn about different indicators that can be used to express flood and drought risk, and how flood and drought risk have changed over the past half century and are projected to change in the 21st century. Students will learn about the main datasets and methods that can be used to assess hazard, exposure, and vulnerability, and apply several of these to assess flood and drought risk at different spatial scales. Finally, students will learn about different measures that can be taken to reduce flood and drought risk, and the pros and cons of these measures.

Additional Information Teaching Methods

In weeks 1-6, the course will be taught as a mixture of (guest) lectures, practicals, discussions, and serious games. Sessions are interactive, so active participation of all students is required. We aim to develop an open and respectful atmosphere, in which diverse views can be expressed. A sit-in exam is taken in week 8.

Several sessions are mandatory (e.g., guest lectures) and for the other sessions attendance is strongly recommended. The sessions require preparation and/or work after the session of around three hours (per session). This consists of reading scientific papers, looking up information and working on practicals.

In order to participate in the course, students need to bring their own laptop computers and will need to be able to install and run the following programs: Python, SFINCS, Microsoft Office, QGIS. For Python and SFINCS a guide will be distributed before the course and assistance will be provided during the first lecture if needed.

Method of Assessment

The course will be assessed through 2 components: the results report of computer practicals (50%) and a sit-in exam (50%). The final grade is a weighted mean of the grades for the 2 components, whereby students must achieve a grade of at least 5.5 for each component. Resit opportunities will be provided in the form of: a re-take of the sit-in exam and/or improving the results report of the computer practicals.

Entry Requirements

Knowledge of GIS, basic programming and hydrological processes.

Literature

A reading list will be provided on CANVAS.

Additional Information Target Audience

MSc Hydrology, MSc Earth Sciences (Global Environmental Change & Policy, Earth & Climate), MSc Climate Econometrics, and others as long as they have knowledge of GIS, basic programming and hydrological processes.

Recommended background knowledge

In preparation for the course, students can find some guidance on learning Python and several mini-tutorials in last year's syllabus: https://vu-ivm.github.io/Water-Risks/mini_tutorials/learning-python.html. Note that the practical exercises may be updated before the start of the course.

To brush up on hydrological process knowledge, students can have a look at a hydrology textbook, such as "Hydrology: an introduction" by Brutsaert or "Principles of hydrology" by Ward & Robinson (both are available in the VU library).

Wereldburgerschap deel 1

Course Code	O_MWRLDBS_1
Credits	3
Period	P1, P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	D.S. Hoefakker-Laanstra, dr. I. Pauw, drs. H.R. Goudsmit
Teaching method(s)	Study Group

Course Objective

Na het volgen van dit keuzevak:

- kan je het belang van wereldburgerschap voor leerlingen, docenten en scholen onderbouwd over het voetlicht brengen.
- ken je de verschillende definities van wereldburgerschap en kan je op grond daarvan een onderbouwde eigen formulering geven.
- kan je de verschillende vormen van wereldburgerschap noemen en kan je bij de verschillende vormen leerlijnen en concrete activiteiten noemen.
- kan je aan de hand van de gekozen definitie en de methodologie van wereldburgerschap een onderbouwde leerlijn wereldburgerschap voor een school ontwerpen
- kan je in de klas in het kader van wereldburgerschap een maatschappelijke controverse les ontwerpen en uitvoeren .
- kan je in je klas met leerlingen met behulp van de dialoog maatschappelijke thema's bespreken

Course Content

Leerlingen van nu groeien op in een gepolariseerde multiculturele omgeving in een globaliserende wereld, waarin belangrijke thema's als klimaat, de energietransitie en ook oorlog en vrede de landsgrenzen overstijgen. De samenleving van nu vraagt nieuwe en andere competenties van leerlingen. Zo is het belangrijk, dat leerlingen in dialoog met elkaar kunnen gaan in plaats van in debat. En moeten leerlingen zich een aanpak eigen maken om de grote maatschappelijke thema's voor zichzelf behapbaar te maken. Ook vraagt het van leerlingen om een open en onderzoekende houding ten opzichte van andere meningen en ideeën. Anders gezegd, de maatschappij vraagt om leerlingen, die zich zowel nu in hun eigen omgeving als later in de maatschappij kunnen ontwikkelen tot wereldburgers . En het is de taak van het onderwijs hen handvatten mee te geven om wereldburgers te worden. Hoe je als docent en als school dat kan doen is het onderwerp van deze specialisatie.

Additional Information Teaching Methods

Werkcolleges

Method of Assessment

Verschillende toetsvormen: leerlijn Wereldburgerschap ontwerpen, controverse les ontwerpen en uitvoeren

Entry Requirements

Didactiek 1, Praktijk 1

Literature

- Wereldgericht Onderwijzen.Biesta in praktijk, B.Eidhof e.a, 2021
- Artikelen via Canvas

Wereldburgerschap Deel 2

Course Code	O_MWRLDBS_2
Credits	3
Period	P2, P5
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	dr. I. Pauw, drs. H.R. Goudsmit, D.S. Hoefakker-Laanstra
Teaching method(s)	Seminar, Lecture

Course Objective

Doelen.

- Studenten kennen de verschillende internationale onderzoeken naar de diverse schoolsystemen, kunnen de indicatoren van de verschillende onderzoeken aangeven en zijn in staat tot het geven van een onderbouwde kijk op deze onderzoeken.
- Studenten kennen de kritische momenten in een schoolsysteem en zijn in staat deze momenten te waarderen
- Studenten zijn in staat andere onderwijssystemen dan het Nederlandse in kaart te brengen en deze te relateren aan het eigen onderwijsstelsel
- Studenten kunnen het begrip Internationalisering definieren, de mate van Internationalisering op hun eigen school in kaart brengen en een leerlijn Internationalisering voor een school ontwerpen
- Studenten kunnen hun visie op goed onderwijs verdiepen naar aanleiding van analyseren van andere schoolsystemen. Daarnaast zijn ze ook in staat een meer onderbouwde kijk op het Nederlands onderwijssysteem te geven

Course Content

Onlangs sprak een bekende Nederlandse publicist naar aanleiding van hevige discussies rond de klimaatcrisis de gevleugelde woorden: "Velen hebben een mening, slechts weinigen komt een oordeel toe." Nederlanders staan er om bekend overal een mening over te hebben en die te pas en te onpas te ventileren.

Je kan je afvragen of dat ook niet voor het onderwijs geldt. Er wordt immers veel gezegd en geschreven over ons onderwijs zonder dat het altijd duidelijk waarop het geventileerde is gebaseerd.

En hoewel wij er dagelijks deel van uit maken is het maar de vraag hoe onderbouwd onze eigen visie is op het Nederlands onderwijssysteem. Wordt onze eigen kijk ook niet teveel ingegeven door een beperkte hoeveelheid aan eigen ervaringen en missen wij niet het overzicht of de helikopter view om te komen tot een gefundeerd oordeel? Missen wij ook niet voldoende vergelijkingsmateriaal om tot een werkelijk gefundeerd oordeel te komen? Deze keuzemodule wil dan ook in de eerste plaats een stimulans zijn om tot een dergelijk verdiepte visie te komen.

Overal in de wereld tonen regeringen en politici interesse in de uitkomsten van de benchmarks, die internationaal worden georganiseerd door instellingen als de OESO (PISA; Eduaction at Glance). De uitkomsten van deze rapporten zorgen in de deelnemende landen vaak voor onderwijshervormingen. Maar het doen van internationaal vergelijkend onderzoek roept ook veel discussie op. Kan je onderwijssystemen wel vergelijken op grond van cijfers, zijn de onderzoeken wel betrouwbaar en valide, kun je onderwijssystemen uberhaupt wel met elkaar vergelijken? In dit keuzevak zullen we dan ook ingaan op de verschillende onderzoeken, onderzoeksmethodes en kritieken. Maar ook gaan we zelf (een klein) onderzoek doen naar de verschillende onderwijssystemen om te kijken of we ons een beeld kunnen vormen van andere schoolsystemen.

Dit keuzevak is enerzijds een aanvulling op deel 1 van Wereldburgerschap maar is ook als "los"keuzevak te volgen

Additional Information Teaching Methods

Werkcollege

Method of Assessment

Het schrijven van een essay

Entry Requirements

Didactiek 1 en praktijk 1

Literature

Artikelen uitgereikt op college

Recommended background knowledge

Didactiek 1 en Praktijk 1